MAT 3004 - Abstract Algebra I

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Chapter 1

Preliminaries

In this Chapter, we will recall some knowledge which will be assumed throughout the course.

1.1 Divisibility of Integers

We begin our studies on the integers $\mathbb{Z}$. Let $t, s \in \mathbb{Z}$ with $t \neq 0$. We write $t \mid s$ if and only if

$$\exists\, u \in \mathbb{Z}\ , \text{such that } s = tu.$$

If u does not exist, write $t \nmid s$.

Remark 1.1. Every integer divides 0 whereas 0 is a divisor of only 0.

Theorem 1.2 (Division Algorithm)**.** *Let $a, b \in \mathbb{Z}$ with $b > 0$. Then there exists unique integers q, r such that $a = bq + r$, where $0 \leq r < b$.*

Proof. (Existence) Let $a, b \in \mathbb{Z}$ with $b > 0$ Consider the set:

$$S = \{a - bk \mid k \in \mathbb{Z},\ a - bk \geq 0\}.$$

We first show that S is nonempty:
If $a \geq 0$, then $a = a - b \cdot 0 \in S$.
If $a < 0$, since $b > 0$, we may choose $k = 2a$, then we can get: $a - b(2a) = a(1 - 2b) \in S$.

By the well-ordering property of $\mathbb{Z}_{\geq 0}$, the set S has the has a minimal element r.
By the definition of S, there exists $q \in \mathbb{Z}$ such that:

$$r = a - bq > 0.$$

Thus, $a = bq + r$.

Then we need to show that $r < b$. Suppose, to the contrary, that $r \geq b$. Then:

$$a - b(q + 1) = r - b \geq 0,$$

So $a - b(q + 1) \in S$. But this element satisfies:

$$a - b(q + 1) < a - bq = r,$$

contradicting the minimality of r. Therefore $0 \leq r < b$, completing the proof of existence.

The proof of uniqueness of q and r is left as an exercise. □

Definition 1.3. The **greatest common divisor (gcd)** of two nonzero integers a and b is the largest of all common divisors of a and b. We denote this integer by $\gcd(a, b)$.

Definition 1.4. Two nonzero integers a and b are **coprime** or **relatively prime** if $\gcd(a, b) = 1$.

Remark 1.5. One can also define the gcd of $a_1, a_2, \ldots, a_k$ as the largest of common divisors of all a_i's. Then one has

$$\gcd(a_1, \ldots, a_k) = \gcd(\gcd(a_1, a_2), a_3, \ldots, a_k) = \gcd(\gcd(\gcd(a_1, a_2), a_3), \ldots, a_k).$$

Similarly, if $\gcd(a_1, \ldots, a_k) = 1$, we say $a_1, \ldots, a_k$ are **relatively prime**. Note that $a_1, \ldots, a_k$ are relatively prime does not necessarily mean $\gcd(a_i, a_j) = 1$ for any $i \neq j$. For instance:

$$\gcd(6, 10, 15) = 1$$

but $\gcd(6, 10) = 2$, $\gcd(6, 15) = 3$, $\gcd(10, 15) = 5$.

Definition 1.6. An integer p is **irreducible** if $p \neq \pm 1$ and the only divisors of p are ± 1 and $\pm p$ (except for zero). A nonzero integer, except for ± 1, is **reducible**(or **composite**) if it is not irreducible.

Definition 1.7. A nonzero integer p is **prime** if and only if it is irreducible.

1.2 Euclidean algorithm and Bézout's Theorem

To find the greatest common divisor of two integers, one can apply Euclidean algorithm:

Example 1.8. To find $\gcd(374, 221)$, one can keep dividing the larger integer by the smaller integer:

$$\begin{aligned}
374 &= 1 \cdot 221 + 153 \\
221 &= 1 \cdot 153 + 68 \\
153 &= 2 \cdot 68 + 17 \\
68 &= 4 \cdot 17 + 0
\end{aligned}$$

when the remainder is zero, the last remainder to it (colored in red) is the greatest common divisor. Therefore,

$$\gcd(374, 221) = 17$$

Theorem 1.9 (Bézout's Theorem)**.** *Suppose* $\gcd(p, q) = m$*. Then there exist integers* $a, b \in \mathbb{Z}$ *such that*

$$m = ap + bq.$$

Example 1.10. By Bezout's Theorem, there exists a, b integers such that:

$$374a + 221b = 17$$

We now determine such integers explicitly:

To find these integers, one do backward induction from the bottom equality to the top:
From the bottom equality:

$$17 = 153 - 2 \cdot 68$$

Substitute the second equality ($68 = 221 - 1 \cdot 153$) to the above equation:

$$17 = 153 - 2(221 - 1 \cdot 153) = 3 \cdot 153 - 2 \cdot 221$$

Now substitute the first equality ($153 = 374 - 1 \cdot 221$) to the above equation:

$$\begin{aligned} 17 &= 3(374 - 221) - 2 \cdot 221 \\ &= 3 \cdot 374 - 3 \cdot 221 - 2 \cdot 221 \\ &= 3 \cdot 374 - 5 \cdot 221 \end{aligned}$$

Thus:

$$a = 3, \quad b = -5$$

In the special case of when a and b are relatively prime, there exist integers s and t such that

$$as + bt = 1.$$

This plays an important role in algebra. More generally, for $a_1, \dots, a_k \in \mathbb{Z}$, one has

$$a_1 s_1 + \cdots + a_k s_k = \gcd(a_1, \dots, a_k)$$

for some integers $s_1, \dots, s_k$. This can be checked by using the fact that $\gcd(a_1, a_2, a_3 \dots, a_k) = \gcd(\cdots \gcd(\gcd(a_1, a_2), a_3), \dots, a_k)$ and apply the above algorithms repeatedly.

Finally, we state a standard yet important theorem for natural numbers, and is slightly extended to all integers:

Theorem 1.11 (Fundamental Theorem of Airthmetic). *Every nonzero integer $n \in \mathbb{Z}$ can be factorized into a product of primes up to ± 1, i.e. $n = p_1 p_2 \cdots p_r$ if $n > 0$ or $n = -p_1 p_2 \cdots p_r$ if $n < 0$. Moreover,this product is unique up to ± 1 and the ordering of factors. That is, if*

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s,$$

where the p's and q's are primes, then $r = s$ and, after renumbering the q's, we have $p_i = \pm q_i$ for all i.

1.3 Modular Arithmetic $\mathbb{Z}_n$

Sometimes, we would like to study integers that has a certain 'cycle'. For instance, days in a week has a cycle of 7, months in a year has a cycle of 12, hours in a day has a cycle of 24 and so on. We define the *congruence numbers*:

$$\mathbb{Z}_n := \{0 \pmod{n},\ 1 \pmod{n},\ \ldots, (n-1) \pmod{n}\} = \{[0]_n, [1]_n, \ldots, [n-1]_n\}$$

For example, when $n = 7$, it corresponds to days in a week, and when $n = 12$ it corresponds to months in a year.

The addition and multiplication rule in $\mathbb{Z}_n$ is given by:

$$\begin{aligned} [a] + [b] &= [\text{remainder of } a+b \text{ in } n] \\ [a] \cdot [b] &= [\text{remainder of } ab \text{ in } n] \end{aligned}$$

(we may sometimes omit the subscript $[\ ]_n$ if it is clear from the context). For instance, when $n = 7$:

$$[4]_7 \cdot [5]_7 = [4 \cdot 5]_7 = [20]_7 = [6]_7$$

There are many applications on modular arithmetic - for instance, RSA cryptography, fast Fourier transform, etc. It can also be used to verify the validity of statements about divisibility regarding all positive integers by checking only finitely many cases.

Example 1.12. To see:

$$[n^3 + (n+1)^3 + (n+2)^3]_9 \equiv [0]_9$$

for all integers $n \in \mathbb{Z}$, one only needs to check this holds for $n = 0,\ 1, \ldots, 8$.

1.4 Equivalence Relations

One has encountered the notion of 'equivalence' in different contexts within or outside of mathematics. This allows us to relate different objects that are not necessarily identical to each other.

For instance, one studies 'similar triangles' in middle school mathematics, where two non-identical triangles $\Delta_1 \sim \Delta_2$ are similar if one can resize one triangle to get another. Another example is the congruence numbers we studied in the previous relation, where $1 \neq 8 \neq 15 \neq 22 \neq 29$ but $[1] = [8] = [15] = [22] = [29]$ in $\mathbb{Z}_7$.

Here is a precise definition of what it means for two elements to be equivalent:

Definition 1.13 (Equivalence Relation)**.** An equivalence relation on a set X is a binary relation R on X that has the following three properties:

1. $(x, x) \in R$ for all $x \in X$ (reflexivity).
2. $(x, y) \in R$ implies $(y, x) \in R$ (symmetry).
3. $(x, y) \in R$ and $(y, z) \in R$ imply $(x, z) \in R$ (transitivity).

We will write $x \sim y$ instead of $(x, y) \in R$.
If $\sim$ is an equivalence relation on a set X and $x \in X$, then the **equivalence class containing/with representative** x is the subset:

$$[x] = \{y \in X \mid x \sim y\}.$$

The collection of equivalence classes of X is denoted as:

$$X/\sim := \{[x] \mid x \in X\}$$

Example 1.14.

1. Let S be the set of all triangles in a plane. If $a, b \in S$, define $a \sim b$ if a and b are similar-that is, if a and b have corresponding angles that are the same. Then $\sim$ is an equivalence relation on S.

2. Let $S = \mathbb{Z}$ be the set of integers. We define $a \sim b$ if $a \equiv b \bmod 7$, or equivalently $[a]_7 = [b]_7$. Then one can show that $\sim$ is an equivalence relation, and the equivalence class containing 1 is

$$[1] = \{\cdots, -13, -6, 1, 8, 15, \cdots\} = [8] = [15]$$

This justifies our notation of using $[\ ]_n$ for the elements in $\mathbb{Z}_7$.

One important aspect of equivalence relation on S is that it gives a **partition** of S. Namely, a partition of S is a disjoint union of nonempty subsets $P_i \subseteq S$ such that

$$\bigsqcup_{i \in I} P_i = S$$

Theorem 1.15. *Let $\sim$ be an equivalence relation on a set S. Then the collection of equivalence classes constitute a partition of S.*

Conversely, for any partition P_i of S, there is an equivalence relation on S whose equivalence classes are precisely the elements of P_i.

Proof. Let $\sim$ be an equivalence relation on a set S. For any $a \in S$, $a \in [a]$ since $a \sim a$. Therefore, $[a]$ is nonempty, and $\bigcup_{a \in S} [a] = S$.

There are repetitions in the union above. One therefore has to show that for $a, b \in S$, one either has $[a] = [b]$ or $[a] \cap [b] = \emptyset$ is disjoint.

To do so, suppose $c \in [a] \cap [b]$, so that $c \sim a$ and $c \sim b$. By symmetry and transitivity, one therefore has $a \sim b$. Then for any $x \in [a]$, one has $x \sim a$ (and $a \sim b$). So $x \sim b$ and hence $x \in [b]$. In other words,

$$[a] \subseteq [b] .$$

The above argument can be reversed, so that one has $[b] \subseteq [a]$ as well. Thus, $[a] = [b]$.

To prove the converse, let $\{P_i \mid i \in I\}$ be a partition of S. Define $a \sim b$ if $a, b \in P_i$ for some $i \in I$. One then can easily check that $\sim$ is an equivalence relation on S. $\square$

1.5 Functions

Definition 1.16. A **function** $\phi : A \to B$ from a set A to a set B is a rule that assigns to each element $a \in A$ exactly one element $\phi(a) \in B$. The set A is called the **domain** of ϕ, and B is called the **codomain** of ϕ.

When we say ϕ is **well-defined**, we need to show the following:

- For all $a \in A$, $\phi(a) \in B$; and
- If $a = a' \in A$, then $\phi(a) = \phi(a') \in B$.

The above definition of well-definedness may look trivial at the first sight, but this may become an issue when one has two or more 'representatives' of the same element in A. For instance, a map $\phi : \mathbb{Z}_7 \to \mathbb{Z}$ given by $\phi([a]_7) = a$ is *not* well-defined, since $[1]_7 = [8]_7$ but $\phi([1]_7) = 1 \neq 8 = \phi([8]_7)$.

Definition 1.17. A function $\phi : A \to B$ is called

- injective (or one-to-one) if for every $a, a' \in A$, $\phi(a) = \phi(a')$ implies $a = a'$.
- surjective (or onto) if for any $b \in B$, there exists $a \in A$ such that $\phi(a) = b$.
- bijective if it is both injective and surjective.

In the case when $\phi : A \to B$ is bijective, one says that ϕ is **invertible**, and has an inverse $\psi : B \to A$ (often denoted as ϕ^{-1}) such that

$$\phi^{-1} \circ \phi = \mathrm{id}_A, \qquad \phi \circ \phi^{-1} = \mathrm{id}_B.$$

where $\mathrm{id}_S : S \to S$ is the identity map $\mathrm{id}_S(s) := s$ for all $s \in S$.

1.6 Polynomials

In this section, we will introduce some basic aspects of polynomials over a field $\mathbb{F}$. For beginners, it is safe to assume $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Definition 1.18.

1. A polynomial over $\mathbb{F}$ has the form

$$p(z) = a_m z^m + \cdots + a_1 z + a_0, \quad (a_m \neq 0).$$

   Here $a_m z^m$ is called the **leading term** of $p(z)$; m is called the degree; a_m is called the **leading coefficient**; $a_m, \ldots, a_0$ are called the coefficients of this polynomial.

2. A polynomial over $\mathbb{F}$ is **monic** if its leading coefficient is $1_{\mathbb{F}}$.

3. A polynomial $p(z) \in \mathbb{F}[z]$ is **irreducible** if for any $a(z), b(z) \in \mathbb{F}[z]$,

$$p(z) = a(z)b(z) \implies \text{either } a(z) \text{ or } b(z) \text{ is a constant polynomial.}$$

   Otherwise $p(z)$ is **reducible**.

Example 1.19. The polynomial $p(x) = x^2 + 1$ is irreducible over $\mathbb{R}[x]$; but $p(x) = (x - i)(x + i)$ is **reducible** in $\mathbb{C}[x]$.

Theorem 1.20. *Division Algorithm For all $p, q \in \mathbb{F}[z]$ such that $p \neq 0$, there exists unique $s, r \in \mathbb{F}[x]$ satisfying $\deg(r) < \deg(q)$, such that*

$$p(z) = s(z) \cdot q(z) + r(z).$$

Here $r(z)$ is called the ***remainder****.*

Theorem 1.21 (Root Theorem)**.** *For $p(x) \in \mathbb{F}[x]$, and $\lambda \in \mathbb{F}$, $x - \lambda$ divides $p(x)$ if and only if $p(\lambda) = 0$.*

Proof. 1. If $(x - \lambda)$ divides p, then $p(x) = (x - \lambda)q(x)$ for some $q(x) \in \mathbb{F}[x]$. Thus clearly $p(\lambda) = 0$.

2. For the other direction, suppose that $p(\lambda) = 0$. By division theorem, there exists $q(x), r(x) \in \mathbb{F}[x]$ such that

$$p(x) = (x - \lambda)q(x) + r(x) \quad \text{with } \deg(r(x)) < \deg(x - \lambda) = 1. \tag{1.1}$$

Therefore, $r(x) = r$ must be a constant polynomial. Substituting λ into both sides in the above equation, we have

$$0 = p(\lambda) = 0 \cdot q(x) + r \implies r = 0.$$

Therefore, $p = (x - \lambda) \cdot q(x)$, i.e., $(x - \lambda)$ divides $p(x)$. $\square$

Corollary 1.22. *A polynomial with degree n has at most n roots counting multiplicity.*

Definition 1.23 (Algebraically Closed)**.** A field $\mathbb{F}$ is called **algebraically closed** if every non-constant polynomial $p(x) \in \mathbb{F}[x]$ has a root $\lambda \in \mathbb{F}$, or equivalently, all polynomials in $p(x) \in \mathbb{F}[x]$ can be factorized into linear terms:

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_n)$$

for $c, \lambda_1, \ldots, \lambda_n \in \mathbb{F}$.

We have seen before that $\mathbb{R}$ is not algebraically closed. Nevertheless, we have:

Theorem 1.24 (Fundamental Theorem of Algebra)**.** $\mathbb{C}$ *is algebraically closed.*

We will skip the proof of the theorem. This can be proved using complex **analysis** (MAT 3253); or **topology** of S^1 (MAT 4002); or **algebra**ic number theory (MAT 5210).

In general, $\mathbb{F}$ may not necessarily be factorized into linear terms. But the factorization is still unique. This can be seen as an analogue of the fundamental theory of arithmetic in $\mathbb{Z}$:

Theorem 1.25 (Unique Factorization)**.** *Every $f(x) = a_n x^n + \cdots + a_0$ in $\mathbb{F}[x]$ can be factorized as*

$$f(x) = a_n[p_1(x)]^{e_1} \cdots [p_k(x)]^{e_k}$$

*where p_i's are **monic, irreducible, distinct**. Furthermore, this expression is unique up to the permutation of factors.*

This will be proved in the chapter of Ring Theory. Assuming its validity for the moment, we can now define:

Definition 1.26 (Factor)**.** If $p(x) = q(x)s(x)$ with $p, q, s \in \mathbb{F}[x]$, then we say

- $p(x)$ is **divisible** by $s(x)$;
- $s(x)$ is a **factor** of $p(x)$;
- $s(x)|p(x)$;
- $s(x)$ **divides** $p(x)$;
- $p(x)$ is **multiple** of $s(x)$.

Definition 1.27 (Common Factor)**.**

1. The polynomial $g(x)$ is said to be a **common factor** of $f_1, \ldots, f_k \in \mathbb{F}[x]$ if

$$g|f_i, \quad i = 1, \ldots, k$$

2. The polynomial $g(x)$ is said to be a **greatest common divisor** of $f_1, \ldots, f_k$ if
    - g is **monic**.
    - g is common factor of $f_1, \ldots, f_k$

- g is of largest possible (maximal) degree.

$\gcd(f_1, f_2)$ is easy to compute for factorized polynomials. For example, let $f_1(x) = (x^2 + x + 1)^3(x-3)^2x^4$ and $f_2(x) = (x^2+1)(x-3)^4x^2$ in $\mathbb{R}[x]$, then

$$\gcd(f_1, f_2) = (x-3)^2x^2.$$

As for general polynomials, the gcd can be computed using Euclidean algorithm as in the case of integers: For example, given $x^3 + 6x + 7$ and $x^2 + 3x + 2$, we imply

$$\begin{aligned} x^3 + 6x + 7 &= (x-3)(x^2 + 3x + 2) + (13x + 13) \\ x^2 + 3x + 2 &= \frac{x+2}{13}(13x + 13) + 0 \end{aligned}$$

Therefore, $\gcd(x^2 + 3x + 2, 13x + 13)$ is equal to a scalar multiple of $13x + 13$ such that it is monic, namely

$$\gcd(x^3 + 6x + 7, x^2 + 3x + 2) = x + 1.$$

Similarly, one has Bezout's theorem for polynomials:

Theorem 1.28 (Bezout)**.** *Let $g = \gcd(f_1, f_2)$, then there exists $r_1, r_2 \in \mathbb{F}[x]$ such that*

$$g(x) = r_1(x)f_1(x) + r_2(x)f_2(x)$$

More generally, $g = \gcd(f_1, \ldots, f_k)$ implies there exists $r_1, \ldots, r_k$ such that

$$g = r_1f_1 + \cdots + r_kf_k$$

1.7 Introduction to Abstract Algebra

We now give a brief introduction of abstract algebra - in a nutshell, abstract algebra is about generalization of number systems we studied in kindergarten, such as:

$$\begin{aligned} &\mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\} \\ &\mathbb{Q} = \left\{ \frac{m}{n} \,\middle|\, m \in \mathbb{Z},\ n \in \mathbb{Z} \setminus \{0\} \right\} \\ &\mathbb{R} = \text{real numbers (limits of Cauchy sequences in } \mathbb{Q}) \\ &\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\} \end{aligned}$$

All these number systems have addition and multiplication, e.g. in $\mathbb{Q}$:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

And they all possess nice properties, e.g.,

$$(a+b)+c = a+(b+c) \tag{A1}$$

$$(ab)c = a(bc) \tag{A2}$$

$$ab = ba \tag{C1}$$

$$a+b = b+a \tag{C2}$$

$$a(b+c) = ab+ac \tag{D1}$$

$$(a+b)c = ac+bc \tag{D2}$$

One goal in abstract algebra is to study different number systems, and to find out their common features and obtain theorems that hold for all such generalized number systems.

Example 1.29.

$$M_{2\times 2}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \; a,b,c,d \in \mathbb{R} \right\}$$

We can still do $+$ and $\times$ on $M_{2\times 2}(\mathbb{R})$, but we no longer have (C1) - in other words,

$$AB \neq BA \quad \text{in general for matrices.}$$

Chapter 2

Groups

2.1 Basic Definitions

Definition 2.1. Let S be a set. A *binary operation* on S is a map:

$$*: S \times S \to S$$

Let $T \subseteq S$ be a subset. We say that the binary operation is *closed* in T if:

$$\forall a, b \in T, \quad a * b \in T$$

Example 2.2. Let $S = \mathbb{Z}$. Then the following are binary operations:

- $S = \mathbb{Z}, \quad * = +$ (addition)
- $S = \mathbb{Z}, \quad * = \times$ (multiplication)

Let

$$T = \{\text{all even integers}\} \subset \mathbb{Z}.$$

For $a, b \in T$, $a + b \in T$ (the sum of two even numbers is even). Hence $(T, +)$ is closed in $(S, +)$. Also, if $a, b \in T$, $a \cdot b \in T$ (the product of two even numbers is even). Therefore, $(T, \times)$ is also closed in $(S, \times)$.

However, let

$$T' = \{\text{all odd integers}\}$$

Then $(T', +)$ is **not** closed in $(S, +)$. But for any $(2p + 1), (2q + 1) \in T'$:

$$(2p + 1) \cdot (2q + 1) = 4pq + 2p + 2q + 1 = 2(2pq + p + q) + 1 \in T'$$

Therefore, $(T', \times)$ is closed in $(S, \times)$.

Example 2.3. 1. Let $S = \mathbb{R}^n$, and $* =$ be the vector addition operation. If $W \leq \mathbb{R}^n$ is a vector subspace, then $(W, +)$ is closed in $(\mathbb{R}^n, +)$.

2. Let $S = M_{n\times n}(\mathbb{R})$, and $* =$ multiplication of the ne matrix. Suppose $T = GL_n(\mathbb{R})$ is the subset of all invertible real matrices. To check if T is closed in $(S, \times)$: If $A, B \in T$ (invertible matrices), then $A \cdot B$ is invertible since

$$(AB)^{-1} = B^{-1}A^{-1}$$

or alternatively:

$$\det(AB) = \det(A)\det(B) \neq 0$$

Hence, T is closed in $(S, \times)$.

Definition 2.4 (Group)**.** A **group** G is a set along with a binary operation $* : G \times G \to G$ satisfying:

1. $(a * b) * c = a * (b * c) \quad \forall a, b, c \in G \qquad$ (associativity)
2. $\exists\ e \in G$ such that $e * a = a * e = a,\ \forall a \in G \qquad$ (identity)
3. $\forall a \in G,\ \exists b \in G$ s.t. $a * b = b * a = e$ (b is often written as a^{-1}, called the inverse of a)

Example 2.5. (a) $(\mathbb{Z}, +)$ is a group. To see this, note that

$$\begin{cases} (a+b)+c = a+(b+c) \\ 0+a = a+0 = a \quad \forall a \in \mathbb{Z} \quad \text{(i.e., } e = 0) \\ a+(-a) = (-a)+a = 0 \quad \text{(i.e., } a^{-1} = -a) \end{cases}$$

(b) Similarly $(\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{C}, +)$ as well as $(\mathbb{Z}_n, +)$ are groups.

(c) $(\mathbb{Z}, -)$ is **not** a group, since $(a-b)-c \neq a-(b-c)$ and $0-a \neq a-0 = a$ in general, i.e. it is not associative.

(d) $(\mathbb{Z}, \times)$ is also **not** a group. Although $(a\times b)\times c = a\times(b\times c)$ is associative, and $a\times 1 = 1\times a = a$ (so that the identity can be taken as $e = 1$), but there is *no* $a \in \mathbb{Z}$ such that $2 \times a = 1 = e$.)

(e) $(\mathbb{Q}, \times)$ is **not** a group. Although now one can take $a = \frac{1}{2}$ such that $2 \times \frac{1}{2} = 1$, yet the element $0 \in \mathbb{Q}$ does not have a multiplicative inverse.

(f) To resolve the issue, let $\mathbb{Q}^* = \{q \in \mathbb{Q} \mid q \neq 0\}$. Then $(\mathbb{Q}^*, \times)$ is a group.

(g) Similarly, $(\mathbb{R}^*, \times)$ and $(\mathbb{C}^*, \times)$ are groups.

(h) The set of 2×2 real matrices, denoted $M_2(\mathbb{R})$, under matrix multiplication $(M_2(\mathbb{R}), \times)$ is **not** a group, because not all matrices have a multiplicative inverse (namely, singular matrices do not have a multiplicative inverse).

(i) Let $GL(2, \mathbb{R}) = \{A \in M_2(\mathbb{R}) \mid \det(A) \neq 0\}$ be the set of 2×2 invertible matrices with real entries. Then $(GL(2, \mathbb{R}), \times)$ is a group, known as the general linear group of degree 2 over $\mathbb{R}$.

Question 2.6. *Given $n \in \mathbb{N}$, for which $a \in \mathbb{Z}$ does $[a]_n$ (the congruence class of a modulo n) have a multiplicative inverse in $\mathbb{Z}_n$?*

Answer: An element $[a]_n \in \mathbb{Z}_n$ has a multiplicative inverse if and only if $\gcd(a, n) = 1$. For instance, when $n = 6$, then:

- For $a = 3$, $\gcd(3, 6) = 3 \neq 1$. Therefore, $[3]_6$ does not have a multiplicative inverse in $\mathbb{Z}_6$.
- For $a = 5$, $\gcd(5, 6) = 1$. Therefore, $[5]_6$ has a multiplicative inverse in $\mathbb{Z}_6$. Indeed, $[5]_6 \cdot [5]_6 = [25]_6 \equiv [1]_6 \pmod 6$, so $[5]_6^{-1} = [5]_6$.

Remark 2.7. When there is no ambiguity, we often write ab instead of $a * b$. Also, for all $m \in \mathbb{Z}$, we denote

$$a^m := \begin{cases} \overbrace{a * a * \cdots * a}^{m \text{ times}} & \text{if } m > 0 \\ e & \text{if } m = 0 \\ \overbrace{a^{-1} * a^{-1} * \cdots * a^{-1}}^{-m \text{ times}} & \text{if } m < 0 \end{cases}$$

In the case when the operation $*$ is the 'usual' addition $+$, we call $(G, +)$ **additive group**, and write

$$0 := e, \quad -a := a^{-1}, \quad m \cdot a := a^m.$$

The order of multiplication matters, namely for $b \in G$, multiplying $a \in G$ on the left or on the right may result in different elements. In the case when they are the same, we have:

Definition 2.8 (Abelian Group)**.** A group $(G, *)$ is called *abelian* if its operation is commutative; that is, for all $a, b \in G$, we have $a * b = b * a$.

Example 2.9.

- Examples of abelian groups include: $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, and $(\mathbb{Z}_n, +)$.
- The group $(GL(2, \mathbb{R}), \times)$ is NOT abelian, as matrix multiplication is generally not commutative.

Proposition 2.10. *Let $(G, *)$ be a group.*

1. *(Uniqueness of Identity) The identity element e in G is unique.*
2. *(Uniqueness of Inverses) For each $a \in G$, its inverse a^{-1} is unique.*
3. *(Cancellation Laws) For $a, b, c \in G$:*
   - *If $a * b = a * c$, then $b = c$.*
   - *If $b * a = c * a$, then $b = c$.*

2.2 Cayley Table

Definition 2.11 (Order of a Group)**.** The *order* of a group G, denoted $|G|$, is the number of elements in the group. If $|G| < \infty$, the group is called a *finite group*.

Definition 2.12 (Cayley Table)**.** For a finite group $(G, *)$, a *Cayley table* (or group table) is a square table where the rows and columns are labeled by the elements of the group. The entry in the row corresponding to a and the column corresponding to b is $a * b$.

Example 2.13. The Cayley table for $\mathbb{Z}_9^*$ is:

$\times$	$[1]_9$	$[2]_9$	$[4]_9$	$[5]_9$	$[7]_9$	$[8]_9$
$[1]_9$	$[1]_9$	$[2]_9$	$[4]_9$	$[5]_9$	$[7]_9$	$[8]_9$
$[2]_9$	$[2]_9$	$[4]_9$	$[8]_9$	$[1]_9$	$[5]_9$	$[7]_9$
$[4]_9$	$[4]_9$	$[8]_9$	$[7]_9$	$[2]_9$	$[1]_9$	$[5]_9$
$[5]_9$	$[5]_9$	$[1]_9$	$[2]_9$	$[7]_9$	$[8]_9$	$[4]_9$
$[7]_9$	$[7]_9$	$[5]_9$	$[1]_9$	$[8]_9$	$[4]_9$	$[2]_9$
$[8]_9$	$[8]_9$	$[7]_9$	$[5]_9$	$[4]_9$	$[2]_9$	$[1]_9$

Proposition 2.14. *In a Cayley table for a finite group, each element of the group appears exactly once in each row and exactly once in each column. In other words, for all $a \in G$, the sets*

$$\{a * g \mid g \in G\}$$

and

$$\{g * a \mid g \in G\}$$

contains all elements of G exactly once, with no repetitions.

2.3 Subgroups

Definition 2.15 (Subgroup)**.** Let $(G, *)$ be a group and H be a non-empty subset of G. We say that H is a *subgroup* of G, denoted $H \leq G$, if $(H, *)$ is itself a group under the same operation $*$ restricted to H.

Proposition 2.16 (Subgroup Test)**.** *Let $(G, *)$ be a group and H be a non-empty subset of G. Then H is a subgroup of G if and only if:*

1. *For all $a, b \in H$, $a * b \in H$, i.e. $*$ is closed in H.*
2. *For all $a \in H$, $a^{-1} \in H$.*

Equivalently, one can check the following:

$$\textit{For all } a, b \in H,\ a * b^{-1} \in H.$$

Example 2.17. 1. $(\mathbb{Z}, +) \leq (\mathbb{Q}, +) \leq (\mathbb{R}, +) \leq (\mathbb{C}, +)$.

2. If $W \leq \mathbb{R}^n$ is a vector subspace, then $(W, +) \leq (\mathbb{R}^n, +)$ is a subgroup.

3. Let $(G, *) = (\mathbb{Z}, +)$, and $H =$ all even integers $= 2\mathbb{Z} \subseteq G$. Take $2p, 2q \in H$ $(p, q \in \mathbb{Z})$. Then $2p + 2q = 2(p + q) \in H$ and $-2p = 2(-p) \in H$. Hence, $H \leq G$ is a subgroup of G. More generally, for all $k \in \mathbb{Z}$, $(k\mathbb{Z}, +) \leq (\mathbb{Z}, +)$.

   Meanwhile, if $K =$ all odd integers $\subseteq G$, then K is **NOT** a subgroup of G, since $1, 3 \in K$ but $1 + 3 \notin K$.

4. Let $(G, *) = (GL(n, \mathbb{R}), \cdot)$, and

$$SL(n, \mathbb{R}) := \{A \in G \mid \det(A) = 1\}$$

   Then $SL(n, \mathbb{R}) \leq GL(n, \mathbb{R})$:

   - For all $A, B \in SL(n, \mathbb{R})$, $\det(AB) = \det(A) \cdot \det(B) = 1 \cdot 1 = 1$. Therefore, $AB \in SL(n, \mathbb{R})$ for all $A, B \in SL(n, \mathbb{R})$
   - For all $A \in SL(n, \mathbb{R})$, $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$. Therefore, $A^{-1} \in SL(n, \mathbb{R})$ as well.

5. Let $(G, *) = (\mathbb{Z}_8, +)$. Then $H = \{0, 4\}$ is a subgroup of G, while $K = \{0, 3\}$ is **NOT** a subgroup.

Definition 2.18. Let $(G, *)$ be a group. Then:

1. $(G, *) \leq (G, *)$ is a subgroup of G. We say all $H \leq G$ satisfying $H \neq G$ a **proper subgroup** of G.
2. $\{e\} \leq G$ is a subgroup. We say all $H \leq G$ with $H \neq \{e\}$ a **nontrivial subgroup** of G.

2.4 Cyclic Groups

Definition 2.19 (Cyclic Subgroup)**.** Let $(G, *)$ be a group. A **cyclic subgroup** generated by $g \in G$ is the subgroup

$$\langle g \rangle := \{g^m \mid m \in \mathbb{Z}\}.$$

(Exercise: Check $\langle g \rangle \leq G$ is a subgroup of G, i.e.: for any $g^a, g^b \in \langle g \rangle$ $\begin{cases} g^a * g^b \in \langle g \rangle \\ (g^a)^{-1} \in \langle g \rangle \end{cases}$.)

Example 2.20. Here are some examples of cyclic subgroups:

1. Let $(G, *) = (\mathbb{Z}, +)$, then

$$\langle 2 \rangle = \{2, 2+2, 2+2+2, \ldots, 0, (-2), (-2)+(-2), (-2)+(-2)+(-2), \ldots\} = 2\mathbb{Z}.$$

2. Let $(G, *) = (GL(2, \mathbb{R}), \cdot)$, then

$$\left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\rangle = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^m \middle| m \in \mathbb{Z} \right\} = \left\{ \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \middle| m \in \mathbb{Z} \right\}.$$

(Check: $\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+b \\ 0 & 1 \end{pmatrix}$)

3. Let $(G, *) = (\mathbb{Z}_8, +)$, then

$$\langle 2 \rangle = \{0, 2, 4, 6, \dots\} = \{0, 2, 4, 6\}$$

$$\langle 3 \rangle = \{0, 3, 6, 9, 12, 15, 18, 21, 24, \dots\} = \{0, 3, 6, 1, 4, 7, 2, 5, 0, \dots\} = G$$

Definition 2.21 (Cyclic Group)**.** Let $(G, *)$ be a group. We say G is **cyclic** if there is $g \in G$ such that $\langle g \rangle = G$.

Example 2.22.
- $(G, *) = (\mathbb{Z}, +)$ is cyclic, since $G = \langle 1 \rangle$.
- $(G, *) = (\mathbb{Z}_8, +)$ is cyclic, since $G = \langle 3 \rangle = \langle 1 \rangle$.
- $(G, *) = (GL(2, \mathbb{R}), \cdot)$ is **NOT** cyclic, since for all $H = \langle g \rangle$, H is countable but G is uncountable. Hence, H can never be equal to G.
- $(G, *) = (\mathbb{Z}_8^*, \cdot) = \{1, 3, 5, 7\}$. Then the cyclic subgroups of G are:
  $\langle 1 \rangle = \{1\}$,
  $\langle 3 \rangle = \{3^0 = 1, 3^1 = 3, 3^2 = 1, 3^3 = 3, \dots\} = \{1, 3\}$;
  $\langle 5 \rangle = \{1, 5\}$;
  $\langle 7 \rangle = \{1, 7\}$.
  Note that none of them is equal to G. So G is **not** cyclic.

Question: $(G, *)$ group, $H = \langle g \rangle$. What's the order of H? For instance, if $G = (\mathbb{Z}_8, +)$, then $|\langle [2] \rangle| = 4,\ |\langle [3] \rangle| = 8,\ |\langle [4] \rangle| = 2$.

Definition 2.23 (Order)**.** Let $g \in G$. The *order* of g is equal to the smallest positive integer k such that $g^k = e$. If no such k exists, then we say the order of g is ∞.

Proposition 2.24. *If* $\operatorname{ord}(g) = k$*, then* $|\langle g \rangle| = k$*.*

Example 2.25.
- $(G, *) = (\mathbb{Z}_8, +)$. $\operatorname{ord}([2]) = 4$ since

$$[2] = [2], [2] + [2] = [4], [2] + [2] + [2] = [6], [2] + [2] + [2] + [2] = [8] = [0].$$

Also, $\operatorname{ord}([3]) = 8$, since

$$1 \cdot [3] = [3],\ 2 \cdot [3] = [6],\ 3 \cdot [3] = [9] = [1],\ 4 \cdot [3] = [12] = [4],$$

$$5 \cdot [3] = [15] = [7],\ 6 \cdot [3] = [18] = [2],\ 7 \cdot [3] = [21] = [5],\ 8 \cdot [3] = [24] = [0].$$

- Let $(G, *) = (GL(2,\mathbb{R}), \cdot)$. Then $\operatorname{ord}\left(\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}\right) = \infty$, since

$$\begin{pmatrix}1 & m\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}^m \neq \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \quad \forall m \in \mathbb{N}.$$

Remark 2.26. If $|G| < \infty$ is finite, then $\langle g\rangle \leq G$ is a finite subgroup with $|\langle g\rangle| \leq |G|$. Indeed, all $H \leq G$ has $|H| \mid |G|$ (see Section 2.7 - Lagrange's Theorem below).
e.g.: If $|G| = 12$, then there is **NO** $H \leq G$ with $|H| = 8$, can only be $1, 2, 3, 4, 6, 12$.
Therefore, $\operatorname{ord}(g) \mid |G|$ (take $H = \langle g\rangle$).

2.5 More examples of Groups

Permutation group / Symmetric group S_n

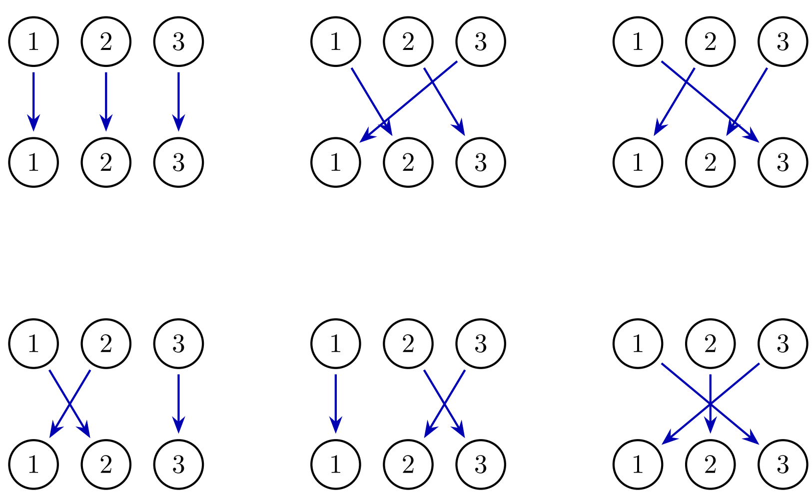
Definition 2.27. A *permutation* of $X_n = \{1, 2, \ldots, n\}$ is a bijection $\sigma : X_n \to X_n$.

For example, when $n = 3$, $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1$ is a permutation.

Definition 2.28. The **symmetric / permutation group** of n elements is the collection of all permutations $\sigma : X_n \to X_n$.

$$S_n := \{\sigma : X_n \to X_n \mid \sigma \text{ is bijective}\}$$

For instance, when $n = 3$, one has:



Hence $|S_3| = 6$. More generally,

$$|S_n| = n \cdot (n-1) \cdots 1 = n!.$$

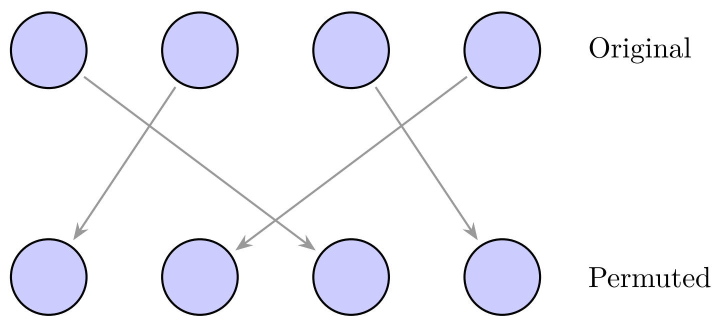
Proposition 2.29. $(S_n, \circ)$ *is a group, where* $\circ$ *is the composition of functions.*

Proof. We need to show that the three axioms of groups hold for $(S_n, \circ)$:

1. $(\sigma_1 \circ \sigma_2) \circ \sigma_3 = \sigma_1 \circ (\sigma_2 \circ \sigma_3)$ is true, since composition of functions $\circ$ is associative.
2. Take $e = \mathrm{id} : X_n \to X_n$, $e(i) := i$ for all $1 \leq i \leq n$. Then $\sigma \circ e = e \circ \sigma = \sigma$.
3. Since σ is bijective, then $\sigma^{-1} : X_n \to X_n$ exists with $\sigma \circ \sigma^{-1} = \sigma^{-1} \circ \sigma = e$.

$\square$

Remark 2.30. Groups are often used to study "symmetry" of objects. For example, S_n can be used to study symmetry of n identical objects. Here is an example of $n = 4$:



Although we have carried out a permutation $1 \mapsto 3$, $2 \mapsto 1$, $3 \mapsto 4$, $4 \mapsto 2$, one cannot tell the difference of the objects before and after permutation.

As another example, to describe (some) symmetries of $\mathbb{R}^n$, we have:

$$GL(n, \mathbb{R}) = \{A : \mathbb{R}^n \to \mathbb{R}^n \mid A \text{ is a bijective linear transformation}\}$$

which can be seen as a certain kind of 'permutation' on the points in $\mathbb{R}^n$.

Calculations on S_n

We use cycle notations to denote elements of S_n.

Definition 2.31. A permutation $\sigma \in S_n$ is called a k**-cycle** (or a cycle of length k) if there exists a set of k distinct elements $\{a_1, a_2, \ldots, a_k\} \subseteq \{1, 2, \ldots, n\}$ such that:

$$\sigma(a_1) = a_2, \quad \sigma(a_2) = a_3, \quad \ldots, \quad \sigma(a_{k-1}) = a_k, \quad \sigma(a_k) = a_1$$

and $\sigma(i) = i$ for all $i \notin \{a_1, a_2, \ldots, a_k\}$.

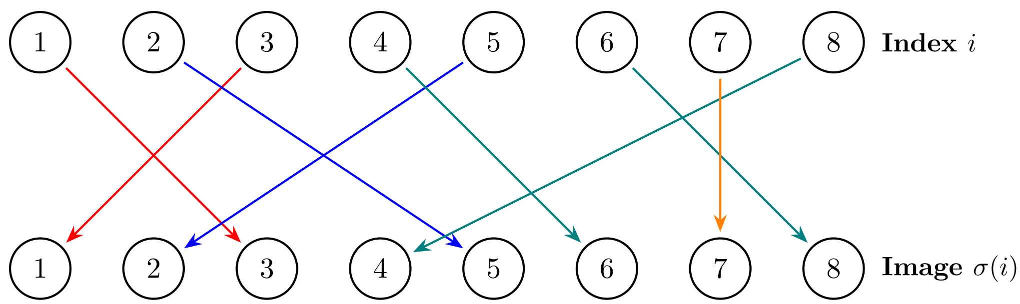
As an example, the element $\sigma \in S_4$ in Remark 2.30 is a 4-cycle $\sigma = (1\ 3\ 4\ 2)$.

Note that every $\sigma \in S_n$ can be expressed as a product of **disjoint** cycles (possibly of different lengths):

$$\sigma = (a_1 \dots a_k)(b_1 \dots b_l) \cdots (c_1 \dots c_m) := (a_1 \dots a_k) \circ (b_1 \dots b_l) \circ \cdots \circ (c_1 \dots c_m).$$

where the sets $\{a_1, \dots, a_k\}$, $\{b_1, \dots, b_l\}$, $\cdots$, $\{c_1, \dots, c_m\}$ are disjoint subsets of X_n. For instance, in S_8, the permutation

$$1 \to 3, \quad 2 \to 5, \quad 3 \to 1, \quad 4 \to 6, \quad 5 \to 2, \quad 6 \to 8, \quad 7 \to 7, \quad 8 \to 4$$



can be split into:

- $1 \to 3$ and $3 \to 1$. This gives the cycle $(1\,3)$.
- $2 \to 5$ and $5 \to 2$. This gives the cycle $(2\,5)$.
- $4 \to 6$, $6 \to 8$, and $8 \to 4$. This gives the cycle $(4\,6\,8)$.
- $7 \to 7$. This is a fixed point, giving the 1-cycle (7).

Therefore,

$$\sigma = (1\,3)(2\,5)(4\,6\,8)(7) := (1\,3) \circ (2\,5) \circ (4\,6\,8) \circ (7)$$

Here are some examples in calculating products in S_n:

Example 2.32. Let $\sigma = (1\,2\,3)$, $\tau = (1\,3\,4\,2)(7\,6)$ in S_7. Then:

- $\sigma\tau = (1\,2\,3)(1\,3\,4\,2)(7\,6) = (1)(2)(3\,4)(7\,6) = (3\,4)(6\,7)$.
- $\tau\sigma = (1\,3\,4\,2)(7\,6)(1\,2\,3) = (1)(2\,4)(3)(6\,7) = (2\,4)(6\,7)$.

Therefore,

$$\sigma \circ \tau \neq \tau \circ \sigma \textbf{ in general, i.e. } S_n \textbf{ is not abelian for } n \geq 3.$$

Remark 2.33.
- From now on, we'll express $\sigma \in S_n$ using (disjoint) product of k-cycles.
- If α and β are k-cycles with disjoint entries, then $\alpha\beta = \beta\alpha$.
  (e.g.: $\alpha = (1\,3\,2)$, $\beta = (4\,7) \Rightarrow \alpha\beta = (1\,3\,2)(4\,7) = (4\,7)(1\,3\,2) = \beta\alpha$).
  Otherwise, $\alpha\beta \neq \beta\alpha$ in general.

- The inverse of a k-cycle is given by:

$$(i_1\, i_2 \dots i_k)^{-1} = (i_1\, i_k\, i_{k-1} \dots i_2)$$

- More generally, for any $\sigma \in S_n$, $\sigma = (i_1 \dots i_k)(j_1 \dots j_\ell) \dots (m_1 \dots m_p)$ where the cycles are disjoint, then:

$$\sigma^{-1} = (i_1\, i_k \dots i_2)(j_1\, j_\ell \dots j_2) \dots (m_1\, m_p \dots m_2)$$

Alternating Group A_n

Definition 2.34 (Transposition)**.** A 2-cycle $\tau = (i\, j) \in S_n$ is called a **transposition**.

Proposition 2.35. *Every $\sigma \in S_n$ can be expressed as a product of (NOT necessarily disjoint) transpositions.*

Proof. $\sigma = (i_1 \dots i_k)(j_1 \dots j_\ell) \dots (m_1 \dots m_p)$ with each $i, j, \dots, m$ cycles are disjoint. Then for $(i_1 \dots i_k)$:

$$(i_1 \dots i_k) = (i_1\, i_k)(i_1\, i_{k-1}) \dots (i_1\, i_3)(i_1\, i_2)$$

Do the same for $(j_1 \dots j_\ell), \dots, (m_1 \dots m_p)$. Then we're done. □

Remark 2.36. The expression of $\sigma \in S_n$ into product of transpositions is **NOT** unique. For example,

$$\sigma = (2\,3) = (2\,3)(2\,3)(2\,3) = (1\,2)(2\,3)(1\,3).$$

But the number of transpositions in each expression of σ is always **even** or **odd**, as we will see below.

Lemma 2.37. *If $e = \tau_1 \cdots \tau_k$, where each τ_i is a transposition. Then $k \equiv 0 \pmod 2$ is even.*

Proof. Apply induction on k, that is,

$$\text{If } e = \tau_1 \cdots \tau_k, \text{ then } k \text{ is even.} \qquad (*)$$

The case of $k = 0$ is trivial. Also, it is obvious that the $k \neq 1$, since e cannot be equal to any single transposition τ. So (*) holds for $k = 1$.

Now suppose by induction hypothesis that (*) holds for all expressions of $e = \tau_1' \cdots \tau_{k'}'$ for $k' \leq m$. Consider

$$e = \tau_1 \cdots \tau_m \tau_{m+1},$$

an expression of e with $(m+1)$ transpositions. Let $\tau_{m+1} = (a\, b)$, Then:

Case 1: $\tau_m = (a\, b)$. In this case, $e = \tau_1 \cdots \tau_{m-1}(a\, b)(a\, b) = \tau_1 \cdots \tau_{m-1}$. By induction, $m-1 \equiv 0 \pmod 2$. Hence, $m + 1 \equiv 0 \pmod 2$ is even.

Case 2: $\tau_m \neq (a\, b)$. Then $\tau_m = \begin{cases} (a\, c) & c \neq b \\ (b\, d) & d \neq a \\ (i\, j)(a\, b) & \{i, j\} \cap \{a, b\} = \emptyset \end{cases}$. Then one has

$$\tau_m \tau_{m+1} = \begin{cases} (a\, c)(a\, b) = (a\, b)(b\, c) \\ (b\, d)(a\, b) = (a\, d)(d\, b) \\ (i\, j)(a\, b) = (a\, b)(i\, j) \end{cases}$$

(where the right hand side in the new expression ($(b\ c)$, $(d\ b)$, $(i\ j)$) have no "a"s). Therefore,

$$e = \tau_1 \cdots \tau_{m-1} \tau_m \tau_{m+1} = \tau_1 \cdots \tau_{m-1} (a\, \star)(*\, *).$$

where $* \neq a$.

Continue the same argument on $\tau_1 \ldots \tau_{m-1}(a\, *)$. If $\tau_{m-1} = (a\, *)$, then $\tau_{m-1}(a\, *)$ goes away. Then one can apply induction as in Case 1 and get the same conclusion. Otherwise, $\tau_{m-1} \neq (a\, *)$, then we use Case 2 to move one more step to the left and get:

$$e = \tau_1 \cdots \tau_{m-2} (a\, \square)(*\, *)(*\, *)$$

with $* \neq a$.

We keep moving a to the left, and claim that Case 1 must occur somewhere, so that we can apply the induction argument to conclude $m + 1$ is even. Otherwise, we can keep applying Case 2 to 'push' a to the leftmost position, and get

$$e = (a\, \triangle)(*\, *) \cdots (*\, *)$$

with $* \neq a$. But this **cannot** happen, since the expression on right-hand-side permutes $a \to \triangle$ (note that $(a\ \triangle)$ is the only transposition on the right that moves a), which contradicts that fact that it is the identity element. $\square$

Proposition 2.38. *Let $\sigma = \tau_1 \cdots \tau_k = \tau_1' \cdots \tau_\ell'$ be two expressions of σ as product of transpositions. Then $k \equiv \ell \pmod 2$.*

Proof.
$$e = \sigma^{-1}\sigma = (\tau_1 \cdots \tau_k)^{-1}(\tau_1' \cdots \tau_\ell') = \tau_k^{-1} \cdots \tau_1^{-1} \tau_1' \cdots \tau_\ell' = \tau_k \cdots \tau_1 \tau_1' \cdots \tau_\ell'.$$

Then by Lemma 2.37, $k + \ell \equiv 0 \pmod 2$. $\square$

Definition 2.39. $\sigma \in S_n$ is called an **even / odd permutation** if σ is a product of an even / odd number of transpositions.

Definition 2.40 (Alternating Group)**.** The **alternating group** A_n is the subgroup of S_n given by:

$$A_n := \{\sigma \in S_n \mid \sigma \text{ is even}\}$$

(Check: $A_n \leq S_n$. Also, is the set $\{\sigma \in S_n \mid \sigma \text{ is odd}\}$ a subgroup?)

Proposition 2.41. $|A_n| = \frac{1}{2}|S_n| = \frac{n!}{2}$

Proof. Let $\tau \in S_n$ be a transposition and define $f : A_n \to \{\sigma \in S_n \mid \sigma \text{ odd}\}$ by:

$$f(\sigma) := \sigma\tau.$$

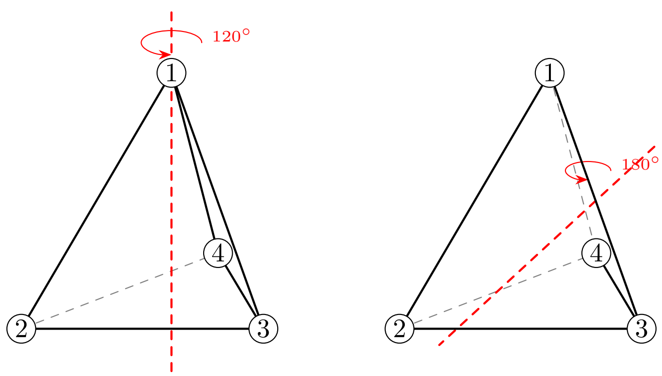
Note that f is bijective with $f^{-1} = f$. Hence, $|A_n| = |\{\sigma \in S_n \mid \sigma \text{ odd}\}|$.

Since $A_n \cap \{\sigma \in S_n \mid \sigma \text{ odd}\} = \emptyset$, and $A_n \sqcup \{\sigma \in S_n \mid \sigma \text{ odd}\} = S_n$, therefore

$$|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}.$$

□

Example 2.42. 1. A_4 is the group of "symmetries of a tetrahedron".



Rotation $(2\,3\,4)$ Rotation $(1\,3)(2\,4)$

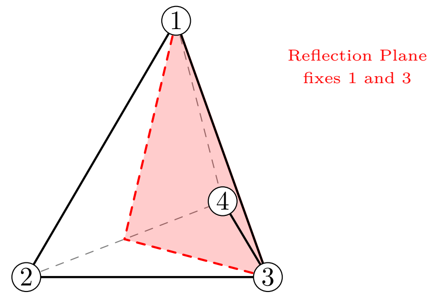
- A rotation of the tetrahedron corresponds to a permutation of its 4 vertices. For example, a 120° rotation about a vertex fixies one vertex and cyclically permutes the other three:

$$(2\,3\,4) = (2\,4)(2\,3) \in A_4$$

- A 180° rotation about the midpoints of opposite edges corresponds to:

$$(1\,3)(2\,4) \in A_4$$

2. If we allow reflections (not just rotations), we get the full symmetric group, e.g. $(2\,4) \in S_4 \setminus A_4$ comes from the reflection:



Reflection $(2\,4)$

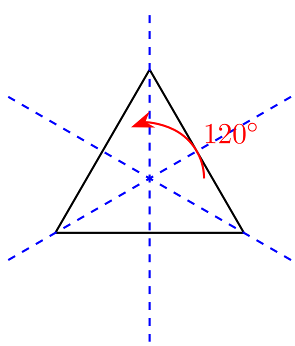
3. **Galois Theory**: A_5 is a simple group (we will define simple group later). This is a purely group theoretic observation, but it is used by Galois to showing that a general quintic (degree-5) equation **cannot** be solved using radicals, i.e. $+, \times, \frac{*}{*}, \sqrt[k]{*}$.

Dihedral Groups D_n

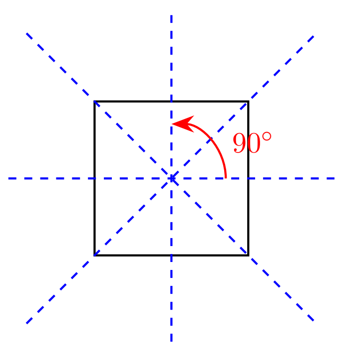
We begin by giving an informal definition of the dihedral group D_n:

The dihedral group D_n describes the rotational and reflectional symmetries of a regular n-gon. Here are the pictures for $n = 3, 4, 5$:

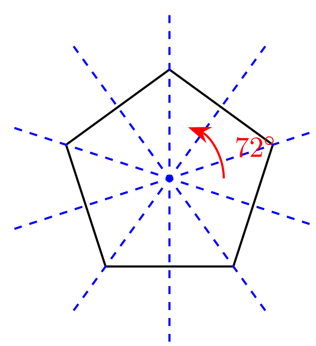
Regular 3-gon



Regular 4-gon



Regular 5-gon



For example, when $n = 3$, D_3 describes the symmetries of an equilateral triangle:

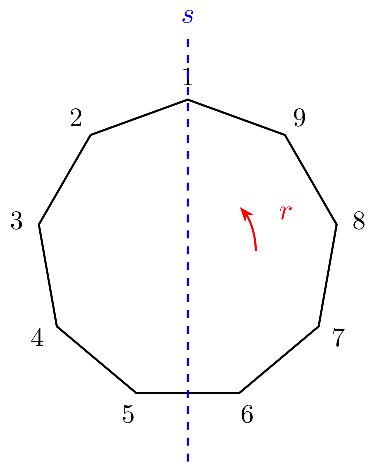
- Rotations: e, σ_{120°, σ_{240°.
- Reflections: 3 axes of symmetry, each fixing one vertex and swapping the other two.

Hence, D_3 has 6 elements. As for $n = 4$, D_4 describes the symmetries of a square, namely:

- There are 4 rotations e, σ_{90°, σ_{180°, σ_{270°
- There are also 4 reflections - 2 reflections on along the diagonals, and 2 reflections along the midpoints of the parallel sides.

Hence, $|D_4| = 8$, and more generally, $|D_n| = 2n$ for all $n \geq 3$.

We now give an algebraic representation of n-gon symmetries. Label the vertices of the n-gon by $1, 2, \ldots, n$. Let r be a rotation by $\left(\frac{360}{n}\right)^\circ$ anticlockwise, , and s be a reflection along the axis passing through vertex 1. For $n = 9$, it looks like:



Then $r^n = e$ and $s^2 = e$. Moreover, if we trace how the vertices are moved along under r and s, we have

$$r \longleftrightarrow (1\ 2\ 3 \ldots n), \qquad s \longleftrightarrow (2\ \ (n-1))(3\ \ (n-2)) \cdots$$

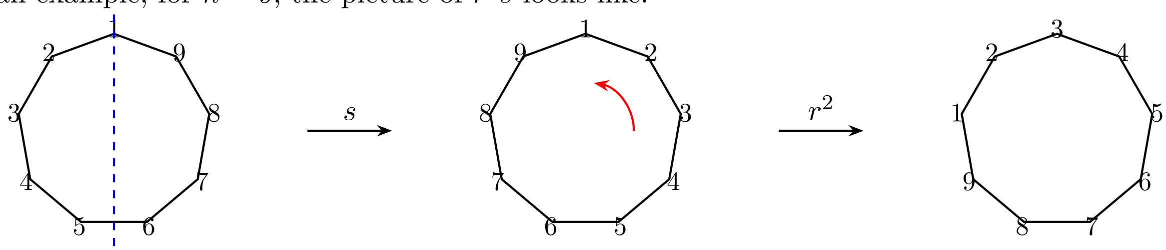
Obviously, all n rotations can be expressed as

$$r^i, \quad 0 \leq i < n.$$

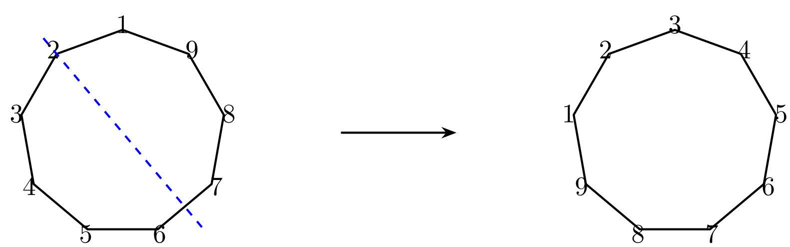
Indeed, the n reflections in D_n can be expressed as

$$r^i s \quad 0 \leq i < n.$$

As an example, for $n = 9$, the picture of $r^2 s$ looks like:



This has the same effect as doing the reflection:



In terms of movement of vertices,

$$r^2 s \longleftrightarrow (1\ 3)(4\ 9)(5\ 8)(6\ 7).$$

More generally, $r^k s$ represents a reflection along an axis obtained by rotating the "axis passing through vertex 1" by $180 \times \left(\frac{k}{n}\right)^\circ$ anticlockwise.

Definition 2.43. The dihedral group D_n is given by the set:

$$D_n = \{e, r, r^2, \ldots, r^{n-1}, s, rs, r^2 s, \ldots, r^{n-1}s\}$$

such that $r^n = e, s^2 = e$ and $sr^l = r^{n-l}s$ for all $0 \leq l \leq n$. In terms of *generators and relations* (not to be defined precisely in this course), D_n can be written as

$$D_n = \langle s, r \mid r^n = e, s^2 = e, sr^l = r^{n-l}s \text{ for all } l\rangle.$$

For example, in D_5, one can write

$$s^3 r^{14} s^5 r^{-3} s^{16} = sr^{10} r^4 sr^5 r^{-3} e = sr^4 sr^2 = s(sr^1)r^2 = s^2 r^3 = r^3.$$

Product Groups (External Direct Product)

Definition 2.44 (Product Group)**.** Let $G_1, \ldots, G_n$ be groups. The **product group** $G := \prod_{i=1}^n G_i = G_1 \times \cdots \times G_n$ is given by

$$G := \{(g_1, \ldots, g_n) \mid g_i \in G_i\}$$

whose multiplication is given by:

$$(g_1, \ldots, g_n) * (h_1, \ldots, h_n) := (g_1 h_1, g_2 h_2, \ldots, g_n h_n)$$

In particular, $|G| = |G_1| \times \cdots \times |G_n|$ and $e_G := (e_1, \ldots, e_n) \in G$, where $e_i = e_{G_i}$ is the identity element of G_i.

Example 2.45. The product group $\mathbb{Z}_2 \times \mathbb{Z}_3$ is given by

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \{([a], [b]) \mid a = 0, 1;\ b = 0, 1, 2\}$$

with $(1, 2) * (0, 1) = (1 + 0, 2 + 1) = (1, 3) \equiv (1, 0)$. (Exercise: $\mathbb{Z}_2 \times \mathbb{Z}_3 = \langle (1, 1)\rangle$ is cyclic.)

One can construct many other groups using products, for instance:

$$S_3 \times D_4; \quad \mathbb{Z} \times \mathbb{R}; \quad GL(2, \mathbb{R}) \times S_5 \times \mathbb{Z}_7; \quad \ldots$$

2.6 Homomorphism and Isomorphism

Motivation: Recall that in understanding D_4, one can look at how the vertices of the squares are permuted. This gives the following 'dictionary' between elements of D_4 and S_4:

D_4		S_4	D_4		S_4
e	$\leftrightarrow$	e	s	$\leftrightarrow$	$(2\,4)$
r	$\leftrightarrow$	$(1\,2\,3\,4)$	rs	$\leftrightarrow$	$(1\,2)(3\,4)$
r^2	$\leftrightarrow$	$(1\,3)(2\,4)$	r^2s	$\leftrightarrow$	$(1\,3)$
r^3	$\leftrightarrow$	$(1\,4\,3\,2)$	r^3s	$\leftrightarrow$	$(1\,4)(2\,3)$

To understand D_4, it is equally good to understand the 8 elements in S_4 in the above dictionary.

Definition 2.46 (Homomorphism & Isomorphism)**.** Let $(G,*), (H,\circledast)$ be groups. A *homomorphism* from G to H is a map $\phi : G \to H$ such that

$$\phi(g_1 * g_2) = \phi(g_1) \circledast \phi(g_2).$$

If ϕ is bijective, then ϕ is an *isomorphism.*

Example 2.47. Here is a long list of (non-)examples of group homomorphisms:

- $\phi : D_4 \to S_4$ given by $\phi(r) = (1\,2\,3\,4)$, $\phi(s) = (2\,4)$ and so on, as in the above dictionary. Then ϕ is an *injective* homomorphism. For instance,

$$\phi(r^2s) = (1\ 3) = (1\ 3)(2\ 4)(2\ 4) = \phi(r^2)\phi(s)$$

- $\phi : \mathbb{Z}_3 \to S_4$ given by

$$\phi([0]) = e, \quad \phi([1]) := (1\ 2\ 4), \quad \phi([2]) := (1\ 4\ 2),$$

  then ϕ is a homomorphism. For instance,

$$\phi([1]+[1]) = \phi([2]) = (1\ 4\ 2) = (1\ 2\ 4) * (1\ 2\ 4) = \phi([1]) * \phi([1])$$

- Let $(G,*) = (\mathbb{R}^n,+)$ and $(H,\circledast) = (\mathbb{R}^m,+)$. Then any linear transformation

$$T : \mathbb{R}^n \to \mathbb{R}^m$$

  is a group homomorphism, since

$$T(x_1 + x_2) = T(x_1) + T(x_2)$$

- $\phi : (\mathbb{Z}_{15}^*, \times) \to (\mathbb{Z}_{15}, +)$ given by

$$\phi([k]) := [k]$$

is **NOT** a homomorphism, since

$$\phi([2]\cdot[4]) = \phi([8]) = [8] \neq [6] = [2]+[4] = \phi(2)+\phi(4)$$

- $\exp : (\mathbb{R},+) \to (\mathbb{R}_{>0},\cdot)$ given by $\exp(a) := e^a$ is a homomorphism (here $\mathbb{R}_{>0}$ is the set of all positive real numbers), since:

$$\exp(a+b) = e^{a+b} = e^a \cdot e^b = \phi(a)\cdot\phi(b)$$

  Indeed, it is an isomorphism since $\exp : \mathbb{R} \to \mathbb{R}_{>0}$ is bijective.

- S_3 and $\mathbb{Z}_6$ both have 6 elements, but there are no bijective homomorphism $\phi : S_3 \to \mathbb{Z}_6$ (see below). Similarly, there are not bijective homomorphism from D_4 to $\mathbb{Z}_8$.
- $i : (\mathbb{Z},+) \hookrightarrow (\mathbb{R},+)$ defined by $i(a) := a$ for all $a \in \mathbb{Z}$ is a homomorphism. More generally, if $H \leq G$, then $i : (H,*) \hookrightarrow (G,*)$ is a homomorphism.
- $\pi : (\mathbb{Z},+) \to (\mathbb{Z}_n,+)$ with

$$\pi(a) := [a]_n$$

  is a homomorphism, since

$$\pi(a+b) = [a+b]_n = [a]_n + [b]_n = \pi(a)+\pi(b).$$

- $\det : GL(n,\mathbb{R}) \to (\mathbb{R}^*,\cdot)$ is a homomorpism, since

$$\det(A\cdot B) = \det(A)\cdot\det(B).$$

- Let $\phi : S_n \to (\{\pm 1\},\cdot)$ by

$$\phi(\sigma) := \begin{cases} +1, & \text{if } \sigma \text{ is even} \\ -1, & \text{if } \sigma \text{ is odd} \end{cases}$$

  Then ϕ is a surjective homomorphism, i.e.

$$\phi(\sigma\tau) = \phi(\sigma)\cdot\phi(\tau) \quad \forall \sigma,\tau \in S_n,$$

  since:

			$\phi(\sigma\tau)$		$\phi(\sigma)\phi(\tau)$
σ even,	τ even	$\Rightarrow \sigma\tau$ even	1	$=$	$1\cdot 1$
σ even,	τ odd	$\Rightarrow \sigma\tau$ odd	-1	$=$	$1\cdot(-1)$
σ odd,	τ even	$\Rightarrow \sigma\tau$ odd	-1	$=$	$(-1)\cdot 1$
σ odd,	τ odd	$\Rightarrow \sigma\tau$ even	1	$=$	$(-1)\cdot(-1)$

Proposition 2.48. *Let $(G,*),(H,\circledast)$ and $(K,\star)$ be groups. Then:*

(a) If $\phi : G \to H, \psi : H \to K$ are homomorphism, then $\psi \circ \phi : G \to K$ is homomorphism

(b) If $\phi : G \to H$ is homomorphism, then $\phi(e_G) = e_H$, $\phi(a^{-1}) = (\phi(a))^{-1}$.

(c) If $\phi : G \cong H$ is isomorphism, then $\phi^{-1} : H \to G$ satisfies

$$\phi^{-1}(h_1 \circledast h_2) = \phi^{-1}(h_1) * \phi^{-1}(h_2).$$

In other words, the inverse of isomorphism is an isomorphism.

Proof. (a) Since ϕ and ψ are homomorphisms, $\phi(g_1 * g_2) = \phi(g_1) \circledast \phi(g_2)$ and $\phi(h_1 \circledast h_2) = \phi(h_1) \star \phi(h_2)$. Hence we have:

$$\begin{aligned}(\psi \circ \phi)(g_1 * g_2) = \psi(\phi(g_1 * g_2)) &= \psi(\phi(g_1) \circledast \phi(g_2)) \\ &= \psi(\phi(g_1)) \star \psi(\phi(g_2)) \\ &= (\psi \circ \phi)(g_1) \star (\psi \circ \phi)(g_2).\end{aligned}$$

(b) Since $\phi(g) = \phi(e_G * g) = \phi(e_G) \circledast \phi(g)$, one has

$$\begin{aligned}\phi(g)(\phi(g))^{-1} &= \phi(e_G) \circledast (\phi(g) \circledast (\phi(g))^{-1}) \\ e_H &= \phi(e_G).\end{aligned}$$

Similarly, since $\phi(e_G) = \phi(a * a^{-1}) = \phi(a) \circledast \phi(a^{-1})$, therefore

$$e_H = \phi(a) \circledast \phi(a^{-1}) \Rightarrow \phi(a^{-1}) = (\phi(a))^{-1}$$

(c) Note that

$$\phi(\phi^{-1}(h_1 \circledast h_2)) = h_1 \circledast h_2 = (\phi \circ \phi^{-1}(h_1)) \circledast (\phi \circ \phi^{-1}(h_2)) = \phi(\phi^{-1}(h_1) * \phi^{-1}(h_2))$$

Since ϕ is bijective, one can apply ϕ^{-1} on both sides of the equation and get

$$\phi^{-1}(h_1 \circledast h_2) = \phi^{-1}(h_1) * \phi^{-1}(h_2).$$

□

Definition 2.49 (Kernel and Image)**.** Let $\phi : G \to H$ be a group homomorphism.

- The **kernel** of ϕ is:
$$\ker \phi := \{g \in G \mid \phi(g) = e_H\}.$$
- The **image** of ϕ is:
$$\text{im } \phi := \phi(G) = \{\phi(g) \mid g \in G\}.$$

Example 2.50.
- Let A be an $m \times n$ real matrix, which defines a linear transformation A :

$(\mathbb{R}^n, +) \to (\mathbb{R}^m, +)$. Then

$$\ker(A) = \{\mathbf{v} \in \mathbb{R}^n \mid A(\mathbf{v}) = \mathbf{0}_{\mathbb{R}^m}\} = \text{Null space of } A.$$

$$\text{im}(A) = \{A(\mathbf{v}) \in \mathbb{R}^m \mid \mathbf{v} \in \mathbb{R}^n\} = \text{Column space of } A.$$

- Let $\pi : (\mathbb{Z}, +) \to (\mathbb{Z}_n, +)$. Then

$$\ker(\pi) = \{\text{multiples of } n\} = \langle n \rangle, \quad \text{im}(\pi) = (\mathbb{Z}_n, +).$$

- Let $\det : GL(n, \mathbb{R}) \to (\mathbb{R}^*, \cdot)$. Then

$$\ker(\det) = \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\} =: SL(n, \mathbb{R}), \quad \text{im}(\det) = \mathbb{R}^*.$$

Proposition 2.51. *(a)* $\ker \phi \leq G$, *im* $\phi \leq H$ *are subgroups of* G *and* H *respectively.*

(b) ϕ *is an isomorphism if and only if* $\ker \phi = \{e_G\}$ *and im* $\phi = H$*.*

(c) If G *is cyclic / abelian, then* $\phi(G)$ *is cyclic / abelian.*

Proof. (a) $\forall g_1, g_2 \in \ker \phi$:

(i) $\phi(g_1 g_2) = \phi(g_1)\phi(g_2) = e_H \cdot e_H = e_H \Rightarrow g_1 g_2 \in \ker \phi$

(ii) $\phi(g_1^{-1}) = (\phi(g_1))^{-1} \Rightarrow g_1^{-1} \in \ker \phi$

(b) Skipped. ($\ker \phi = \{e_G\} \iff \phi$ is injective; im $\phi = H \iff \phi$ is surjective)

(c) If G is abelian, i.e.: $ab = ba\ \forall a, b \in G$. Then $\forall \phi(a), \phi(b) \in \phi(G)$:

$$\phi(a)\phi(b) = \phi(ab) = \phi(ba) = \phi(b)\phi(a)$$

If $G = \langle g \rangle$ is cyclic, then for all $\phi(a) \in \phi(G)$, $a = g^k$ for some $k \in \mathbb{Z}$, and:

$$\phi(a) = \phi(g^k) = (\phi(g))^k \in \langle \phi(g) \rangle$$

(when $k < 0$ is negative, Proposition 2.48(b) is needed to prove the second equality). Therefore, $\phi(G) \subseteq \langle \phi(g) \rangle$. Meanwhile, $\langle \phi(g) \rangle \subseteq \phi(G)$ since $\phi(g) \in \phi(G)$. Therefore, $\phi(G) = \langle \phi(g) \rangle$ is cyclic. □

Example 2.52.

- $S_3, \mathbb{Z}_6$ both have 6 elements. But $S_3 \ncong \mathbb{Z}_6$ – Suppose by contrary that

$$\phi : \mathbb{Z}_6 \xrightarrow{\cong} S_3.$$

Then $\phi(\mathbb{Z}_6) = S_3$ and hence $\phi(\mathbb{Z}_6)$ is abelian by the above proposition. However, we already know that S_3 is not abelian, so this is impossible.

- Similarly, $\mathbb{Z}_2 \times \mathbb{Z}_2 \ncong \mathbb{Z}_4$. (Hint: Suppose on contrary, $\phi : \mathbb{Z}_4 \to \mathbb{Z}_2 \times \mathbb{Z}_2$ and study $\phi(x)$)

From now on, we will classify groups up to isomorphism, i.e. we will not distinguish $A_3 \cong \mathbb{Z}_3$ but $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_4$ are different!

Theorem 2.53 (Classification of cyclic groups)**.** *Let $G = \langle g \rangle$ be a cyclic group. Then:*

(a) If $|G| = \infty$, then $G \cong (\mathbb{Z}, +)$.

(b) If $|G| = n < \infty$, then $G \cong (\mathbb{Z}_n, +)$.

Proof. (a) Consider the map $\phi : \mathbb{Z} \to G = \langle g$ by $\phi(a) := g^a$. Then

$$\phi(a + b) = g^{a+b} = g^a g^b = \phi(a)\phi(b).$$

Therefore, ϕ is a surjective homomorphism. Suppose on contrary that ϕ is **NOT** injective, i.e. there exists $m > n$ such that $\phi(m) = \phi(n)$. Then:

$$\phi(m) = \phi(n) \;\Rightarrow\; g^m = g^n \Rightarrow\; g^{m-n} = e \Rightarrow\; \mathrm{ord}(g) \leq m - n$$

(Exercise: $g^k = e \;\Leftrightarrow\; \mathrm{ord}(g) \mid k$). Therefore, $\mathrm{ord}(g) < \infty$ and hence $|G| = |\langle g \rangle| = |\mathrm{ord}(g)| < \infty$ (c.f. Proposition 2.24), contradicting $|G| = \infty$. Therefore, ϕ must be injective, and is an isomorphism between $\mathbb{Z} \cong G$.

(b) Let $|G| = |\langle g \rangle| = n < \infty$. Then $\mathrm{ord}(g) = n$ by Proposition 2.24 again, with

$$G = \{e = g^0, g^1, \ldots, g^{n-1}\}.$$

Consider $\psi : G \to \mathbb{Z}_n$ with $\psi(g^i) := i$ for $0 \leq i \leq n - 1$. Then ψ is a surjective homomorphism, since

$$\psi(g^a g^b) = \psi(g^{a+b}) = a + b = \psi(g^a) + \psi(g^b).$$

Since $|G| = |\mathbb{Z}_n| = n$, then any surjective map is also injective. Therefore, $\psi : G \xrightarrow{\cong} \mathbb{Z}_n$ is bijective. $\square$

2.7 Lagrange's Theorem

To begin this section, recall that an equivalence relation $\sim$ on a set S partitions the set into disjoint equivalence classes. That is, $S = \bigsqcup_{\alpha \in I} C_\alpha$, where C_α are the equivalence classes.

Example 2.54.

- Let $G = \mathbb{Z}$ and $H = 3\mathbb{Z}$. Define $a \sim b$ iff $a \equiv b \pmod 3$. This is an equivalence relation where the classes are

$$C_0 = 0 + 3\mathbb{Z},\ C_1 = 1 + 3\mathbb{Z}, \text{ and } C_2 = 2 + 3\mathbb{Z}.$$

- Let $G = GL(n, \mathbb{R})$ and $H = SL(n, \mathbb{R})$. Define $A \sim B$ iff $\det(A) = \det(B)$. The equivalence classes are $C_\alpha = \{B \in GL(n, \mathbb{R}) \mid \det(B) = \alpha\}$ for $\alpha \in \mathbb{R}^*$.

Definition 2.55. Let G be a group and $H \leq G$ be a subgroup. We define a relation on G by

$$a \sim b \text{ if and only if } a^{-1}b \in H.$$

Proposition 2.56. *The relation $a \sim b \iff a^{-1}b \in H$ is an equivalence relation.*

Proof. (i) **Reflexive:** $a \sim a$ since $a^{-1}a = e \in H$.

(ii) **Symmetric:** If $a \sim b$, then $a^{-1}b \in H$. Since H is a group, $(a^{-1}b)^{-1} = b^{-1}a \in H$, so $b \sim a$.

(iii) **Transitive:** If $a \sim b$ and $b \sim c$, then $a^{-1}b \in H$ and $b^{-1}c \in H$. Thus $(a^{-1}b)(b^{-1}c) = a^{-1}c \in H$, so $a \sim c$.

□

Definition 2.57 (Cosets)**.** Let G be a group and $H \leq G$.

- The **left coset** of H with representative $a \in G$ is the set:

$$aH := \{ah \mid h \in H\}$$

- The **right coset** of H with representative $a \in G$ is the set:

$$Ha := \{ha \mid h \in H\}$$

Example 2.58.

- Let $G = \mathbb{Z}, H = 3\mathbb{Z}$. The left cosets (using additive notation) are $0 + 3\mathbb{Z}$, $1 + 3\mathbb{Z}$, and $2 + 3\mathbb{Z}$.
- Let $G = S_3, H = \langle (1\,2) \rangle = \{e, (1\,2)\}$. The left cosets are:

$$eH = \{e, (1\,2)\}$$
$$(1\,2\,3)H = \{(1\,2\,3), (1\,3)\}$$
$$(1\,3\,2)H = \{(1\,3\,2), (2\,3)\}$$

- Let $G = D_4, H = \langle s \rangle = \{e, s\}$. The distinct left cosets are eH, rH, r^2H, and r^3H. Each coset contains 2 elements.

Remark 2.59. The left coset aH is precisely the equivalence classes of the equivalence relation

$$a \sim_L b \iff a^{-1}b \in H.$$

Indeed, one can easily check that

$$aH = \{ah \mid h \in H\} = \{b \in G \mid b = ah \text{ for } h \in H\} = \{b \in G \mid a^{-1}b = h \in H\} = \{b \in G \mid a \sim_L b\}.$$

Similarly, the right coset Ha is the equivalence class of the relation $\sim_R$ defined by

$$a \sim_R b \iff ab^{-1} \in H.$$

Theorem 2.60 (Lagrange's Theorem)**.** *Let G be a finite group and $H \leq G$. Then the order of H divides the order of G. Specifically:*

$$|G| = [G : H] \cdot |H|$$

where $[G : H]$ is the ***index*** *of H in G (the number of distinct left cosets).*

Proof. The left cosets of H partition G. Let $m = [G : H]$. We can write G as a disjoint union:

$$G = a_1 H \sqcup a_2 H \sqcup \cdots \sqcup a_m H$$

where $a_i H \cap a_j H = \emptyset$ for $i \neq j$. We claim that every coset has the same size, namely

$$|a_i H| = |eH| = |H|.$$

To see so, define a map $f : H \to a_i H$ by $f(h) = a_i h$. This map is surjective by the definition of a coset. It is also injective because $a_i h_1 = a_i h_2 \implies h_1 = h_2$ by left cancellation. Thus, $|a_i H| = |H|$ for all i. It follows that:

$$|G| = \sum_{i=1}^{m} |a_i H| = \sum_{i=1}^{m} |H| = m|H|.$$

In other words, $|G| = [G : H]|H|$ as required. □

Corollary 2.61. *Let G be a finite group. For any $g \in G$,* $\mathrm{ord}(g)$ *divides $|G|$.*

Proof. Let $H = \langle g \rangle$. By definition, $|H| = \mathrm{ord}(g)$. By Lagrange's Theorem, $|H|$ divides $|G|$. □

Example 2.62 (Fermat's Little Theorem)**.** Let p be a prime. For any $a \in \mathbb{Z}$ such that $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the group $G = \mathbb{Z}_p^*$, which has order $p - 1$. For any $[a] \in G$, the order d of $[a]$ divides $p - 1$. Thus $p - 1 = dk$ for some integer k. Then:

$$[a]^{p-1} = ([a]^d)^k = [1]^k = [1]$$

and hence $a^{p-1} \equiv 1 \pmod{p}$. □

Remark 2.63. More generally, for the group $G = \mathbb{Z}_n^*$, the order is given by Euler's totient function $\phi(n)$. Thus, for any a coprime to n, $a^{\phi(n)} \equiv 1 \pmod{n}$.

2.8 Group Action

In this section, we will apply Lagrange's Theorem to go deeper into studying the symmetry of any set X by defining an 'action' of a group G on X. Here is the definition:

Definition 2.64. Let G be a group and X be a set. A **(left) group action** of G on X is a function

$$\phi : G \times X \to X,$$

often denoted by $\phi(g, x) = g \cdot x$, such that the following are satisfied for all $x \in X$ and $g_1, g_2 \in G$:

1. (Identity) $e \cdot x = x$, where e is the identity element of G.
2. (Compatibility) $(g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x)$.

A more intuitive way to understand the group action of G on X is to define

$$\sigma_g : X \to X$$

by $\sigma_g(x) := g \cdot x$. In other words, each $g \in G$ defines how X is moved around.

Proposition 2.65. *Let $\phi : G \times X \to X$ be a group action. Then the map*

$$\sigma : G \to \mathrm{Aut}(X), \qquad g \mapsto \sigma_g$$

is a well-defined homomorphism.

Remark 2.66. For any set X, $\mathrm{Aut}(X)$ is defined as:

$$\mathrm{Aut}(X) := \{f : X \to X \mid f \text{ is bijective}\}.$$

If there are extra structure on X, we often impose more conditions on the bijective map in the definition of $\mathrm{Aut}(X)$. For instance, if $X = G$ is a group, $\mathrm{Aut}(G)$ is usually understood as:

$$\mathrm{Aut}(G)\ (= \mathrm{Aut}_G(G)) := \{\phi : G \to G \mid \phi \text{ is a bijective homomorphism}\}.$$

Proof. To see σ is well-defined, one needs to show $\sigma_g \in \mathrm{Aut}(X)$ is bijective. Indeed, by the identity and compatibility axioms, σ_g has an inverse $\sigma_{g^{-1}}$, making σ_g a bijection. Thus $\sigma_g \in \mathrm{Aut}(X)$. As for σ being a homomorphism, one applies compatibility axiom and get

$$\sigma_{g_1 g_2}(x) = (g_1 g_2) \cdot x = g_1 \cdot (g_2 \cdot x) = \sigma_{g_1}(\sigma_{g_2}(x))$$

for all $x \in X$. Therefore,

$$\sigma(g_1 g_2) := \sigma_{g_1 g_2} = \sigma_{g_1} \sigma_{g_2} =: \sigma(g_1)\sigma(g_2),$$

i.e. σ is a homomorphism. $\square$

From now on, we will use

$$\sigma(g)(x) \longleftrightarrow \sigma_g(x) \longleftrightarrow g \cdot x$$

interchangeably to denote the action of $g \in G$ on $x \in X$.

Example 2.67.

- By definition of symmetric group, one automatically has a bijective map

$$\sigma : S_n \to \mathrm{Aut}(X_n),$$

which is an S_n-action on $X_n := \{1, 2, \dots, n\}$.

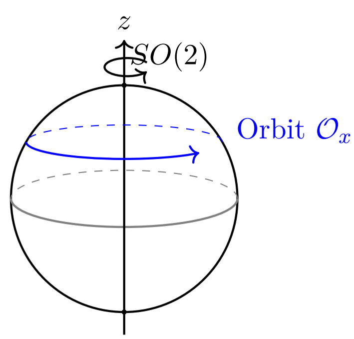
- Similarly, one also has an injective D_n-action

$$\gamma : D_n \to \mathrm{Aut}(X_n).$$

- Let $G = SO(2) := \left\{ \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \middle| \ \theta \in \mathbb{R} \right\}$ be the group of rotations on $\mathbb{R}^2$. Define a G-action on the unit sphere $X = S^2 := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ by:

$$\sigma : SO(2) \to \mathrm{Aut}(S^2), \qquad \sigma \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} := R_\theta,$$

where $R_\theta(\mathbf{x}) := \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{x}$ for $\mathbf{x} \in S^2$. So the $SO(2)$-action is precisely the rotation on the sphere along the polar axis:



Definition 2.68 (Orbit and Stabilizer)**.** Let G act on a set X.

- The **orbit** of x, denoted by $G \cdot x$ or $\mathcal{O}_x$, is the subset of X containing all elements that x can be moved to by G:

$$\mathcal{O}_x = \{g \cdot x \mid g \in G\} \subseteq X$$

- The **stabilizer** of x, denoted by G_x or $\mathrm{Stab}(x)$, is the set of elements in G that fix x:

$$G_x = \{g \in G \mid g \cdot x = x\} \subseteq G$$

Example 2.69. Note that for $x, y \in X$,

$$x \sim y \quad \Leftrightarrow \quad x, y \text{ are in the same orbit}$$

is an equivalence relation (Exercise). Therefore, the orbits form a partition of the set X. For instance, an orbit of $SO(2)$-action on S^2 in the above example is a *latitude line* on the sphere (marked in blue). Obviously, lines of different latitudes give a partition of S^2.

As for the stabilizer, one can check that G_x is always a subgroup of G (Exercise). Therefore, if G is finite, one always has

$$|G_x| \;\Big|\; |G|$$

by Lagrange's theorem. Indeed, one has a more refined result in terms of the size of the orbit:

Theorem 2.70 (Orbit-Stabilizer Theorem)**.** *Let G be a finite group acting on a set X. For any $x \in X$:*

$$|G| = |G_x| \cdot |\mathcal{O}_x|$$

Equivalently, the size of the orbit is the index of the stabilizer: $|\mathcal{O}_x| = [G : G_x]$.

Proof. To prove $|\mathcal{O}_x| = [G : G_x]$, we will construct a bijection f between the set of all left cosets of G_x in G and the elements of the orbit $\mathcal{O}_x$. To do so, let $G/G_x = \{gG_x \mid g \in G\}$ be the set of left cosets, and define:

$$f : G/G_x \to \mathcal{O}_x \qquad f(gG_x) := g \cdot x$$

1. Well-defined: We must ensure that if two cosets are equal, their images are equal. Suppose $g_1G_x = g_2G_x$. Then $g_2^{-1}g_1 \in G_x$. By the definition of the stabilizer:

$$\begin{aligned}
(g_2^{-1}g_1) \cdot x &= x \\
g_2 \cdot (g_2^{-1}g_1 \cdot x) &= g_2 \cdot x \\
(g_2g_2^{-1}g_1) \cdot x &= g_2 \cdot x \\
g_1 \cdot x &= g_2 \cdot x
\end{aligned}$$

Thus $f(g_1G_x) = f(g_2G_x)$, so f is well-defined.
2. Injective: Suppose $f(g_1G_x) = f(g_2G_x)$. Then:

$$\begin{aligned}
g_1 \cdot x &= g_2 \cdot x \\
g_2^{-1} \cdot (g_1 \cdot x) &= g_2^{-1} \cdot (g_2 \cdot x) \\
(g_2^{-1}g_1) \cdot x &= e \cdot x = x
\end{aligned}$$

This implies $g_2^{-1}g_1 \in G_x \;\Leftrightarrow\; g_1G_x = g_2G_x$. Thus f is injective.
3. Surjective: By the definition of the orbit $\mathcal{O}_x$, any element $y \in \mathcal{O}_x$ is of the form $g \cdot x$ for some $g \in G$. Then $y = f(gG_x)$, so f is surjective.

Since f is a bijection, the number of elements in the orbit is equal to the number of left cosets:

$$|\mathcal{O}_x| = |G/G_x| = [G : G_x]$$

By Lagrange's Theorem, we know that $[G : G_x] = \frac{|G|}{|G_x|}$. Therefore:

$$|\mathcal{O}_x| = \frac{|G|}{|G_x|} \implies |G| = |\mathcal{O}_x| \cdot |G_x|.$$

□

Example 2.71. Let G be the group of rotational symmetries of the tetrahedron. Then G defines an action on the 4 faces of the tetrahedron. Suppose x be a specific face of the tetrahedron, then:

- There are 4 faces, and any face can be rotated to any other face, so $|\mathcal{O}_x| = 4$.
- The rotations fixing face x are the 3 rotations (0, 120, 240 degrees) around the axis through the center of that face. So $|G_x| = 3$.

By the Orbit-Stabilizer Theorem, one has $|G| = 4 \times 3 = 12$. But of course we already knew that the rotational symmetries of a tetrahedron is isomorphic to A_4, the alternating group. So we have re-confirmed that $|A_4| = 12$.

Example 2.72. Let G be any finite group, and $X = G$. Then G act on itself by conjugation:

$$g \cdot x = gxg^{-1}.$$

In such a case:

- The orbit $\mathcal{O}_x$ is the **conjugacy class** of x.
- The stabilizer G_x is the **centralizer** $Z_G(x) := \{g \in G \mid gx = xg\}$.

By the orbit-stabilizer theorem, the size of a conjugacy class must divide the order of the group. Also, it can be computed by computing the centralizer group $Z_G(x)$ (see Homework).

We end this section with Burnside's Lemma, which provides a way to count the number of distinct orbits in X for any finite group action.

Theorem 2.73 (**Burnside's Lemma**). *Let G be a finite group acting on a finite set X. Then the number of orbits, denoted $|X/G| := |\{\mathcal{O}_x \mid x \in X\}|$, is equal to:*

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where $X^g = \{x \in X \mid g \cdot x = x\}$ be the set of points fixed by g.

Proof. Consider the set of all fixed pairs $S = \{(g, x) \in G \times X \mid g \cdot x = x\}$. We will count the number of elements in S in two different ways.
Method 1: Summing over G For a fixed $g \in G$, the number of elements x such that $(g, x) \in S$ is exactly $|X^g|$. Therefore:

$$|S| = \sum_{g \in G} |X^g|$$

Method 2: Summing over X For a fixed $x \in X$, the number of elements g such that $(g, x) \in S$ is exactly the size of the stabilizer G_x. Therefore:

$$|S| = \sum_{x \in X} |G_x|$$

Therefore, one has:

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |G_x|$$

Let $\Omega_1, \Omega_2, \ldots, \Omega_k$ be the distinct orbits of X (note that $k = |X/G|$). Then for any $x \in X$, $x \in \Omega_{i_x}$ for a unique $1 \leq i_x \leq k$. Therefore:

$$\sum_{g \in G} |X^g| = \sum_{x \in X} |G_x| = \sum_{x \in X} \frac{|G|}{|\mathcal{O}_x|} = |G| \sum_{x \in X} \frac{1}{|\mathcal{O}_x|} = |G| \sum_{i=1}^{k} \sum_{x \in \Omega_i} \frac{1}{|\Omega_i|} \qquad (*)$$

where the second $=$ comes from the Orbit-Stabilizer Theorem.

For any specific orbit Ω_i, the inner sum $\sum_{x \in \Omega_i} \frac{1}{|\Omega_i|}$ is just adding the value $\frac{1}{|\Omega_i|}$ to itself $|\Omega_i|$ times:

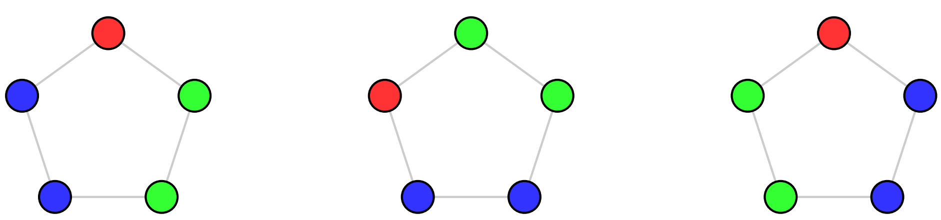
$$\sum_{x \in \Omega_i} \frac{1}{|\Omega_i|} = |\Omega_i| \cdot \frac{1}{|\Omega_i|} = 1$$

Therefore, Equation $(*)$ reduces to

$$\sum_{g \in G} |X^g| = |G| \sum_{i=1}^{k} 1 = |G| \cdot k = |G| \cdot |X/G|$$

and consequently $|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$. □

Example 2.74. To see how many distinct ways can the vertices of a regular pentagon be colored using 3 colors up rotations and reflections. For example, the following colorings are treated the same:



Original (x) **Rotated** $(r \cdot x)$ **Reflected** $(s \cdot x)$

since the second can be obtained from the first by rotation, and the third can be obtained from the first by reflection.

Consider the action of $G = D_5$ on the set X of 5 vertices each with three different possible colors. Therefore, there are

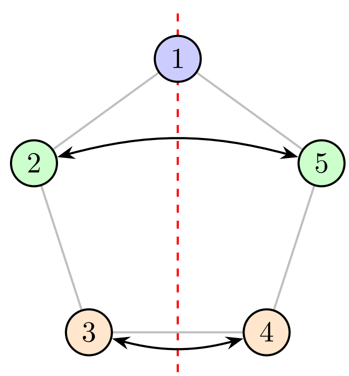
$$|X| = 3 \times 3 \times 3 \times 3 \times 3 = 3^5 = 243$$

elements in X. The three diagrams above are *in the same orbit* under our action. So we need to calculate $|X/G|$ – the *distinct* orbits of X under the action of G.

To do so, we need to calculate X^g for each $g \in D_5$. A coloring $x \in X$ is fixed by $g \in D_5$ if all vertices in the same cycle of the permutation g have the same color. If g has ℓ cycles, there are 3^ℓ fixed colorings. Therefore, one has

Element Type	**No. of Elements**	**Cycle Structure**	**No. of Cycles**	$\lvert X^g \rvert$
Identity	1	$(1)(2)(3)(4)(5)$	5	$3^5 = 243$
Rotations	4	$(1\,2\,3\,4\,5)$	1	$3^1 = 3$
Reflections	5	$(1)(2\,5)(3\,4)$	3	$3^3 = 27$

Here is an example of the number of possible colorings for a reflection $s \in D_5$ along the axis passing through vertex 1. Among all the possible 3^5 colorings of the five vertices, the colorings such that s fixes the coloring is of the form:



Cycles of the Reflection:
- **Cycle 1:** $\{1\}$
- **Cycle 2:** $\{2, 5\}$
- **Cycle 3:** $\{3, 4\}$

Total independent choices of X^s:
$3 \times 3 \times 3 = 3^3$

By Burnside Lemma, the number of distinct colorings $|X/G|$ is given by:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g| = \frac{1}{10} (1 \cdot 243 + 4 \cdot 3 + 5 \cdot 27) = \frac{390}{10} = 39$$

There are **39** distinct ways to color the vertices of a regular pentagon with 3 colors under dihedral symmetry.

2.9 Normal Subgroups

Recall that for a subgroup $H \leq G$, the group G can be partitioned into left cosets $G = \bigsqcup_i \ g_i H$ or right cosets $G = \bigsqcup_j \ H b_j$. A natural question is whether these partitions are identical—that is, whether $gH = Hg$ for all $g \in G$.

In general, the answer is no. Consider $G = S_3$ and $H = \langle(1\,2)\rangle$:

$$\begin{aligned}(1\,3)H &= \{(1\,3), (1\,3)(1\,2)\} = \{(1\,3), (1\,2\,3)\}\\ H(1\,3) &= \{(1\,3), (1\,2)(1\,3)\} = \{(1\,3), (1\,3\,2)\}\end{aligned}$$

Since $(1\,3)H \neq H(1\,3)$, we look for subgroups that satisfy this property.

Definition 2.75 (Normal Subgroup)**.** A subgroup $H \leq G$ is called a **normal subgroup** if $gH = Hg$ for all $g \in G$. We denote this by $H \lhd G$.

Example 2.76.

- $A_n \lhd S_n$ for all n. Indeed, for all $g \in G$, if $g \in A_n$ is even, then $gh, hg \in A_n$ for all $h \in A_n$ and $gA_n = A_n = A_n g$. Otherwise, if g is odd, then one can easily check that $gA_n = A_n g$ is the collection of all odd permutations.
- More generally, if $H \leq G$ and $[G : H] = 2$, then $H \lhd G$ (Homework).
- If G is abelian, then any subgroup $H \leq G$ is normal. For any $g \in G$ and $h \in H$, $gh = hg$, so $gH = Hg$.
- The trivial subgroup $\{e\}$ and the group G itself are always normal in G.

Theorem 2.77. *Let $H \leq G$ be a subgroup. The following are equivalent:*

(i) $H \lhd G$.

(ii) $ghg^{-1} \in H$ *for all* $h \in H$ *and* $g \in G$.

(iii) $gHg^{-1} = H$ *for all* $g \in G$.

Proof.

- **(i) $\Rightarrow$ (ii):** By assumption, $gH = Hg$. Thus for any $h \in H$, $gh \in Hg$, so $gh = h'g$ for some $h' \in H$. Multiplying by g^{-1} on the right gives $ghg^{-1} = h' \in H$.
- **(ii) $\Rightarrow$ (iii):** Condition (ii) implies $gHg^{-1} \subseteq H$ for all $g \in G$. Applying this to g^{-1}, we have $g^{-1}H(g^{-1})^{-1} \subseteq H \Rightarrow g^{-1}Hg \subseteq H$. Multiplying by g on the left and g^{-1} on the right gives $H \subseteq gHg^{-1}$. Thus $gHg^{-1} = H$.
- **(iii) $\Rightarrow$ (i):** If $gHg^{-1} = H$, then multiplying by g on the right gives $(gHg^{-1})g = Hg$, so $gH = Hg$.

$\square$

Example 2.78. We have $SL(n, \mathbb{R}) \lhd GL(n, \mathbb{R})$. Indeed, let $a \in GL(n, \mathbb{R})$ and $h \in SL(n, \mathbb{R})$. Then:

$$\det(aha^{-1}) = \det(a)\det(h)\det(a)^{-1} = \det(a) \cdot 1 \cdot \frac{1}{\det(a)} = 1$$

Thus $aha^{-1} \in SL(n, \mathbb{R})$, and the result follows from (ii) above.

Corollary 2.79. *Let $\phi : G \to H$ be a group homomorphism. Then* $\ker\phi \lhd G$.

Proof. Let $k \in \ker\phi$ and $g \in G$. We check the conjugation condition:

$$\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g^{-1}) = \phi(g)e_H(\phi(g))^{-1} = \phi(g)(\phi(g))^{-1} = e_H$$

Since $\phi(gkg^{-1}) = e_H$, we have $gkg^{-1} \in \ker\phi$. By the previous theorem, $\ker\phi \triangleleft G$. □

Example 2.80.

- Consider $\det : GL(n,\mathbb{R}) \to \mathbb{R}^*$. Since $\ker(\det) = SL(n,\mathbb{R})$, it follows that $SL(n,\mathbb{R}) \triangleleft GL(n,\mathbb{R})$.
- Consider $\psi : S_n \to (\{\pm 1\}, \cdot)$ given by the parity of the permutation. Since $\ker\psi = A_n$, it follows that $A_n \triangleleft S_n$.

2.10 Quotient Groups

Let $H \triangleleft G$ be a normal subgroup. We want to define a group structure on the set of left cosets $G/H := \{gH \mid g \in G\}$. To do this, we must ensure that the natural multiplication of cosets is independent of the choice of representative.

Definition 2.81 (Quotient Group)**.** Let G be a group and $H \triangleleft G$. The **quotient group** G/H is the set of left cosets $\{gH \mid g \in G\}$ equipped with the binary operation:

$$(aH) * (bH) := (ab)H$$

Remark 2.82. Since many different elements can represent the same coset (i.e., $a_1H = a_2H$ even if $a_1 \neq a_2$), we must verify that the operation is well-defined.

Proposition 2.83. *If $H \triangleleft G$, the operation $(aH) * (bH) = (ab)H$ is well-defined and makes G/H into a group.*

Proof. First, we show the operation is **well-defined**. Suppose $aH = a'H$ and $bH = b'H$. This implies $h_a := a^{-1}a' \in H$ and $h_b := b^{-1}b' \in H$. We compute:

$$a'b' = (ah_a)(bh_b) = ab(b^{-1}h_ab)h_b$$

Since $H \triangleleft G$, the element $h_3 := b^{-1}h_ab$ is in H. Thus $a'b' = ab(h_3h_b)$. Since $h_3h_b \in H$, we have $a'b' \in (ab)H$, so $(a'b')H = (ab)H$.

Next, we check the **group axioms**:

1. **Associativity:** $((aH)(bH))(cH) = (ab)H(cH) = (abc)H = aH(bc)H = (aH)((bH)(cH))$.
2. **Identity:** The coset $eH = H$ serves as the identity since $(aH)(eH) = (ae)H = aH$.
3. **Inverses:** The inverse of aH is $a^{-1}H$, since $(aH)(a^{-1}H) = (aa^{-1})H = eH$.

□

Example 2.84.

- Let $G = \mathbb{Z}$ and $H = n\mathbb{Z}$. Since G is abelian, H is normal. The quotient group is:

  $$G/H = \{0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \ldots, (n-1) + n\mathbb{Z}\}$$

  The operation is $(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}$. This group is isomorphic to $(\mathbb{Z}_n, +)$. We often write $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

- Since $A_n \triangleleft S_n$ and $|S_n/A_n| = |S_n|/|A_n| = 2$, the quotient group S_n/A_n is a group of order 2. Any group of order 2 is isomorphic to $\mathbb{Z}_2$, so $S_n/A_n \cong \mathbb{Z}_2$.

- Since $SL(n, \mathbb{R}) \triangleleft GL(n, \mathbb{R})$, we can form the quotient group. Each coset is of the form:

  $$A \cdot SL(n, \mathbb{R}) = \{B \in GL(n, \mathbb{R}) \mid \det(B) = \det(A)\}$$

  The multiplication of cosets corresponds to the multiplication of their determinants: $\det(AB) = \det(A)\det(B)$. Thus, $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong (\mathbb{R}^*, \times)$.

- Let $K = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\} \triangleleft S_4$. The quotient group S_4/K has order $24/4 = 6$. It can be shown that $S_4/K \cong S_3$.

The construction of quotient groups is a powerful tool in group theory:

- **Construction:** It allows us to construct new groups from existing ones (e.g., constructing $\mathbb{Z}_n$ from $\mathbb{Z}$).
- **Cauchy's Theorem:** If G is a finite group and p is a prime dividing $|G|$, then G contains an element of order p. Quotient groups are used in the inductive proof of this theorem.
- **Classification:** They are essential for the classification of finite groups and understanding group extensions.

Theorem 2.85 (First Isomorphism Theorem)**.** *Let $\phi : G \to H$ be a group homomorphism. Then:*

$$G/\ker\phi \cong \text{im } \phi$$

Example 2.86.

- Let $\phi : \mathbb{Z} \to \mathbb{Z}_n$ be defined by $\phi(a) = [a]_n$. Then $\ker\phi = n\mathbb{Z}$ and $\text{im } \phi = \mathbb{Z}_n$. By the First Isomorphism Theorem, $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.
- Let $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$ be the determinant map. Then $\ker(\det) = SL(n, \mathbb{R})$ and $\text{im}(\det) = \mathbb{R}^*$. Thus, $GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^*$.
- Define $\phi : \mathbb{R} \to GL(2, \mathbb{R})$ by $\phi(x) = \begin{pmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{pmatrix}$. This is a homomorphism where $\ker\phi = 2\pi\mathbb{Z}$ and the image is the special orthogonal group $SO(2)$. Hence, $\mathbb{R}/2\pi\mathbb{Z} \cong SO(2)$.

Proof. Define a map

$$\Psi : G/\ker\phi \to \text{im } \phi$$

by $\Psi(g\ker\phi) := \phi(g)$. We verify the following properties:

1. Ψ **is well-defined and injective:** Let $a, b \in G$. Then:

$$\begin{aligned} a\ker\phi = b\ker\phi &\iff a^{-1}b \in \ker\phi \\ &\iff \phi(a^{-1}b) = e_H \\ &\iff \phi(a)^{-1}\phi(b) = e_H \\ &\iff \phi(a) = \phi(b) \\ &\iff \Psi(a\ker\phi) = \Psi(b\ker\phi). \end{aligned}$$

   Reading this from left to right shows Ψ is well-defined; reading from right to left shows Ψ is injective.

2. Ψ **is a homomorphism:** For any $a, b \in G$:

$$\begin{aligned} \Psi((a\ker\phi)(b\ker\phi)) &= \Psi(ab\ker\phi) \\ &= \phi(ab) \\ &= \phi(a)\phi(b) \\ &= \Psi(a\ker\phi)\Psi(b\ker\phi). \end{aligned}$$

3. Ψ **is surjective:** By definition, $\text{im}\ \phi = \{\phi(g) \mid g \in G\}$. Since $\Psi(g\ker\phi) = \phi(g)$, every element in the image of ϕ has a preimage under Ψ.

Since Ψ is a bijective homomorphism, $G/\ker\phi \cong \text{im}\ \phi$. $\square$

We end this section with the notion of simple groups. It will play a big role in the Galois theory course (MAT 5210).

Definition 2.87 (Simple Group)**.** A group G is called **simple** if G has no normal subgroups other than the trivial subgroup $\{e\}$ and the group G itself.

Remark 2.88. The concept of simple groups is analogous to prime numbers in arithmetic. If a group G is **not** simple, there exists a proper normal subgroup $N \triangleleft G$. This allows us to "decompose" G into two smaller groups: the normal subgroup N and the quotient group G/N, which can then be studied individually.

However, it is important to note that if G is not simple with $N \triangleleft G$, it is **not** necessarily isomorphic to the direct product $N \times (G/N)$ (Exercise: Find one such example of N and G).

Example 2.89.

- The symmetric group S_n is **not** simple for $n \geq 3$ because the alternating group A_n is a proper normal subgroup ($A_n \triangleleft S_n$). We can analyze S_n via A_n and the quotient $S_n/A_n \cong \mathbb{Z}_2$, but $S_n \ncong A_n \times \mathbb{Z}_2$ for $n \geq 3$.
- The alternating group A_4 is **not** simple because the Klein four-group

$$K = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$$

is a normal subgroup of A_4.

- The alternating group A_n is simple for all $n \geq 5$ (Homework). This result is a cornerstone of Galois Theory, as it implies that there is no general formula using radicals to solve polynomial equations of degree 5 or higher.

2.11 Fundamental Theorem of Finite Abelian Groups

We will conclude our study of groups by giving a classification of all finite abelian groups up to isomorphism. Here are some initial observations:

(1) $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$ is abelian.

(2) G and H are abelian if and only if $G \times H$ is abelian.

(3) If $\gcd(m,n) = 1$, then $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$.

Theorem 2.90. *All finite abelian groups are of the form $G \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}}$ where $p_1 \leq \cdots \leq p_k$ are primes and $a_i \in \mathbb{N} \setminus \{0\}$.*

The proof consists of two main steps:

- **Step 1:** If $|G| = q_1^{a_1} \ldots q_r^{a_r}$ for distinct primes q_i, then $G \cong G_1 \times \cdots \times G_r$ where $|G_i| = q_i^{a_i}$.
- **Step 2:** If $|G| = q^b$, then $G \cong \mathbb{Z}_{q^{b_1}} \times \cdots \times \mathbb{Z}_{q^{b_r}}$ with $\sum b_j = b$.

Proof of Step 1 : We deal with the case when $r = 2$, and the general case follows by inductively from this case.

Suppose $|G| = mn$ where $\gcd(m,n) = 1$ are coprime (e.g. $m = p_1^{a_1}$ and $n = p_2^{a_2}$). We want to show $G \cong G_1 \times G_2$ with $|G_1| = m$ and $|G_2| = n$. Let

$$G^m := \{g \mid g^m = e\}, \quad G^n := \{g \mid g^n = e\},$$

then it is obvious that G^m and G^n are subgroups of G (exercise).

- **Claim 1 -** $G^m \cap G^n = \{e\}$:

  Suppose $x \in G^m \cap G^n$. Then $x^m = x^n = e$. Since $\gcd(m,n) = 1$, there exist $\alpha, \beta \in \mathbb{Z}$ such that $\alpha m + \beta n = 1$ by Bezout's Theorem. Then

  $$x^1 = x^{\alpha m + \beta n} = (x^m)^\alpha (x^n)^\beta = e \cdot e = e.$$

- **Claim 2 -** $\phi : G^m \times G^n \to G$ given by $\phi(x,y) := xy$ is an isomorphism:

  Note that

  $$\phi((x,y) * (x',y')) = \phi(xx', yy') = xx'yy' = xyx'y' = \phi(x,y)\phi(x',y')$$

  is a homomorphism, where the third equality uses that fact that G is abelian.

For injectivity, if $\phi(x,y) = xy = e$, then $x = y^{-1} \in G^m \cap G^n = \{e\}$ by Claim 1, so $\ker\phi = \{(e,e)\}$. For surjectivity, for any $g \in G$,

$$g = g^{\alpha m+\beta n} = g^{\beta n} g^{\alpha m}.$$

But $x := g^{\beta n}$ satisfies $x^m = g^{\beta mn} = g^{\beta|G|} = e$ and similarly $y := g^{\alpha m}$ satisfies $y^n = e$. Therefore, $x \in G^m$ and $y \in G^n$, with

$$\phi(x,y) = xy = g^{\beta n} g^{\alpha m} = g.$$

- **Claim 3 -** $|G^m| = m$ and $|G^n| = n$:

  We first show that $\gcd(|G^m|, n) = 1$: Suppose on contrary that there exists a prime factor $q|n$ such that $q \mid |G^m|$. By Cauchy's Theorem (proved in Homework), there exists $x \in G^m$ of order q. By Bezout's Theorem, $\gamma q + \delta m = 1$ for some $\gamma, \delta \in \mathbb{Z}$. Then

  $$x = x^{\gamma q+\delta m} = (x^q)^\gamma (x^m)^\delta = e \cdot e = e,$$

  contradicting $\mathrm{ord}(x) = q$. Thus $|G^m|$ is coprime with all prime factors of n. Analogously, $|G^n|$ is coprime with all prime factors of m.
  Since $G^m \times G^n \cong G$, we have

  $$|G^m| \times |G^n| = mn \quad \Rightarrow \quad \begin{cases} |G^m| \mid mn \\ |G^n| \mid mn \end{cases} \quad \Rightarrow \quad \begin{cases} |G^m| \mid m \\ |G^n| \mid n \end{cases},$$

  where the last $\Rightarrow$ follows from the coprime properties. This forces $|G^m| = m$, $|G^n| = n$ to be the largest possible values.

Proof of Step 2 : Suppose $|G| = p^a$. By Lagrange's Theorem, for all $x \in G$:

$$\mathrm{ord}(x) = p^i \quad \text{for some } 0 \le i \le a.$$

Let $\mu \in G$ be an element with maximum order $\mathrm{ord}(\mu) = p^m$ in G.

Proposition 2.91. *Let G be an abelian group with $|G| = p^a$. Suppose $\mu \in G$ has maximum order $\mathrm{ord}(\mu) = p^m$ in G, then*

$$G \cong \langle \mu \rangle \times K \cong \mathbb{Z}_{p^m} \times K.$$

Assuming the validity of the proposition, then one can reduce the study of G into that of K, since K is also abelian with $|K| = |G|/|\langle\mu\rangle| = p^{a-m}$. Then **Step 2** follows by induction on power of p.

Proof of Proposition. We proceed by induction on $|G| = p^a$. For the base case $a = 1$, $|G| = p$ implies $G \cong \mathbb{Z}_p$, and the result holds with $K = \{e\}$. Assume the proposition holds for all abelian groups of order p^r with $r < a$.

Now let $|G| = p^a$ and let $\mathrm{ord}(\mu) = p^m$ be maximal. If $m = a$, then $\langle \mu \rangle = G$ and the result is trivial. Otherwise, if $m < a$, let $\nu \in G$ be a non-identity element not in $\langle \mu \rangle$ with the smallest possible order p^k.

- **Claim 4 -** $\mathrm{ord}(\nu) = p$:
  Since $\mathrm{ord}(\nu^p) < p^k$, by the minimality of p^k, we must have $\nu^p \in \langle \mu \rangle$, so $\nu^p = \mu^i$ for some $i \in \mathbb{Z}$. Then:
  $$(\mu^i)^{p^{m-1}} = (\nu^p)^{p^{m-1}} = \nu^{p^m} = e$$
  since every element in G has order $p^i \mid p^m$. This implies $\mathrm{ord}(\mu^i) \mid p^{m-1}$, so $|\langle \mu^i \rangle| < p^m$, which means $\gcd(i, p^m) > 1$. Thus, $i = pj$ for some $j \in \mathbb{Z}$, and we have:
  $$\nu^p = \mu^{pj}.$$
  Let $c := \nu\mu^{-j}$. Then $c \notin \langle \mu \rangle$ and:
  $$c^p = (\nu\mu^{-j})^p = \nu^p\mu^{-pj} = e.$$
  Since ν was chosen with the smallest order among elements not in $\langle \mu \rangle$, it follows that $\mathrm{ord}(\nu) = p$.
- **Claim 5 -** $\langle \mu \rangle \cap \langle \nu \rangle = \{e\}$:
  Let $\nu^l \in \langle \mu \rangle \cap \langle \nu \rangle$. Since $\mathrm{ord}(\nu) = p$, we have $0 \le l < p$. If $l \ne 0$, then $\gcd(l, p) = 1$. By Bezout's Theorem, there exist α, β such that $\alpha l + \beta p = 1$. Then:
  $$\nu = \nu^{\alpha l + \beta p} = (\nu^l)^\alpha (\nu^p)^\beta = (\nu^l)^\alpha \in \langle \mu \rangle$$
  which contradicts our choice of ν. Thus $l = 0$ and the intersection is trivial.
- **Claim 6 -** $\bar{\mu} \in \bar{G} := G/\langle \nu \rangle$ has order p^m:
  Clearly $\bar{\mu}^{p^m} = e_{\bar{G}}$. Suppose $\mathrm{ord}(\bar{\mu}) = p^u$ for some $u < m$. Then $\mu^{p^u}\langle \nu \rangle = e\langle \nu \rangle$, meaning $\mu^{p^u} \in \langle \nu \rangle$. By Claim 5:
  $$\mu^{p^u} \in \langle \mu \rangle \cap \langle \nu \rangle = \{e\}$$
  which contradicts $\mathrm{ord}(\mu) = p^m$. Thus, $\bar{\mu}$ retains the maximal order p^m in $\bar{G}$.
  Since
  $$|\bar{G}| = |G/\langle \nu \rangle| = |G|/|\langle \nu \rangle| = p^a/p = p^{a-1},$$
  and $\overline{\mu} \in \overline{G}$ has maximal order p^m, one can apply inductive hypothesis to obtain a subgroup $\bar{K} \le \bar{G}$ such that
  $$\bar{G} \cong \langle \bar{\mu} \rangle \times \bar{K}.$$
  Let $\pi : G \to \overline{G}$ be defined by $\pi(g) := g\langle \nu \rangle$ (Exercise: Show that π is a homomorphism), and $K := \pi^{-1}(\overline{K})$ (here we abuse notations and treat $\langle \overline{\mu} \rangle$, $\overline{K} \le \overline{G}$ as subgroups).

- **Claim 7 -** The map $\pi|_K : K \to \bar{K}$ has $\ker(\pi|_K) = \langle \nu \rangle$:

$$x \in \ker(\pi|_K) \iff \pi(x) = e_{\bar{G}} \iff x \in \langle \nu \rangle.$$

- **Claim 8 -** $\langle \mu \rangle \cap K = \{e\}$ in G:
  Let $\mu^i \in \langle \mu \rangle \cap K$. Then $\pi(\mu^i) = \bar{\mu}^i \in \langle \bar{\mu} \rangle \cap \bar{K}$. Since $\bar{G} \cong \langle \bar{\mu} \rangle \times \bar{K}$ is an external direct product, $\langle \bar{\mu} \rangle \cap \bar{K} = \{e_{\overline{G}}\}$ and:
$$\bar{\mu}^i = e_{\bar{G}} \implies \mu^i \in \langle \nu \rangle.$$
  By Claim 5, $\mu^i \in \langle \mu \rangle \cap \langle \nu \rangle = \{e\}$.

Finally, consider the homomorphism $\theta : \langle \mu \rangle \times K \to G$ defined by $\theta(\mu^i, k) := \mu^i k$. By Claim 8, θ is injective. By Claim 7 and the First Isomorphism Theorem, the order of K is:

$$|K| = p \cdot |\bar{K}| = p \cdot \frac{|\bar{G}|}{p^m} = p \cdot \frac{p^{a-1}}{p^m} = p^{a-m}.$$

Thus, the order of the product group is:

$$|\langle \mu \rangle \times K| = p^m \cdot p^{a-m} = p^a = |G|.$$

Therefore, θ is a bijection, and $G \cong \langle \mu \rangle \times K$. □

Example 2.92. Consider an abelian group G with $|G| = 4$. By Lagrange's Theorem, the possible orders for elements in G are 1, 2, and 4.

(i) If G contains an element m of order 4, then $G = \langle m \rangle \cong \mathbb{Z}_4$.

(ii) If every non-identity element of G has order 2, let $m \in G$ be an element of order 2. By the previous proposition, $G \cong \langle m \rangle \times K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Corollary 2.93. *If G is an abelian p-group with $|G| = p^a$, then*

$$G \cong \mathbb{Z}_{p^{b_1}} \times \mathbb{Z}_{p^{b_2}} \times \cdots \times \mathbb{Z}_{p^{b_\ell}}$$

where $\sum_{j=1}^{\ell} b_j = a$.

Proof. The proof follows by induction on $|G| = p^a$, using the proposition that $G \cong \mathbb{Z}_{p^m} \times K$ to repeatedly factor out cyclic components until the remaining group is trivial. □

Example 2.94. To classify all abelian groups of order $|G| = 360 = 2^3 \times 3^2 \times 5$:

- **By Step 1:** The group decomposes into:

$$G \cong G_8 \times G_9 \times G_5$$

  where $|G_8| = 2^3$, $|G_9| = 3^2$, and $|G_5| = 5$ (these are called **Sylow p-subgroups** ($p = 2, 3, 5$) of G).

- **By Step 2:** We find all possible structures for each p-group:
  - G_8 can be $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, or $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.
  - G_9 can be $\mathbb{Z}_9$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$.
  - G_5 can only be $\mathbb{Z}_5$.

Thus, the total number of non-isomorphic abelian groups of order 360 is $3 \times 2 \times 1 = 6$. The possible structures are:

$$G \cong \begin{Bmatrix} \mathbb{Z}_8 \\ \mathbb{Z}_4 \times \mathbb{Z}_2 \\ \mathbb{Z}_2^3 \end{Bmatrix} \times \begin{Bmatrix} \mathbb{Z}_9 \\ \mathbb{Z}_3 \times \mathbb{Z}_3 \end{Bmatrix} \times \mathbb{Z}_5$$

Corollary 2.95 (Smith Normal Form)**.** *Every finite abelian group G is isomorphic to a group of the form:*

$$G \cong \mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \cdots \times \mathbb{Z}_{d_k}$$

where each $d_i > 1$ is an integer and $d_i \mid d_{i+1}$ for all $i = 1, \ldots, k-1$.

Furthermore, this representation is unique: if $H \cong \mathbb{Z}_{e_1} \times \cdots \times \mathbb{Z}_{e_\ell}$ with $e_i \mid e_{i+1}$, then $G \cong H$ if and only if $k = \ell$ and $d_i = e_i$ for all i.

Chapter 3

Rings

3.1 Basic Definition

The concept of a ring generalizes the arithmetic properties we observe in the integers, allowing us to study structures with both addition and multiplication.

Definition 3.1 (Ring)**.** A **ring** $(R, +, \cdot)$ is a set R equipped with two binary operations, addition $(+)$ and multiplication $(\cdot)$, such that:

1. $(R, +)$ is an **abelian group** with additive identity 0_R.
2. Multiplication is **associative**: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in R$.
3. The **distributive laws** hold for all $a, b, c \in R$:

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c$$

Example 3.2.

- The sets $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ with standard addition and multiplication are rings.
- **Gaussian Integers:** $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$. Addition and multiplication follow the rules of complex numbers.
- **Cyclotomic Integers:** Let $\zeta_n = e^{2\pi i/n}$. Then $\mathbb{Z}[\zeta_n] = \{a_0 + a_1\zeta_n + \cdots + a_{n-1}\zeta_n^{n-1} \mid a_j \in \mathbb{Z}\}$ is a ring. For $n = 4$, $\zeta_4 = i$, we recover the Gaussian integers.
- **Polynomial Rings:** $\mathbb{Z}[x]$ is the set of all polynomials with integer coefficients. Multiplication is defined by the Cauchy product:

$$\left(\sum a_p x^p\right)\left(\sum b_q x^q\right) = \sum_i \left(\sum_{p+q=i} a_p b_q\right) x^i$$

- **Matrix Rings:** $M_{n\times n}(\mathbb{Z})$, the set of $n \times n$ matrices with integer entries, is a ring under matrix addition and multiplication.
- **Multiple Integers:** $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ is a ring for any $n \in \mathbb{Z}$.

- **Non-example:** The set of odd integers $\{\dots,-3,-1,1,3,\dots\}$ is **not** a ring because it is not closed under addition (the sum of two odd integers is even).

Remark 3.3 (Motivation). The study of rings like $\mathbb{Z}[i]$ or $\mathbb{Z}[\zeta_n]$ is motivated by number theory.

- To find integer solutions to $x^2+y^2=z^2$, we can factor the left side in $\mathbb{Z}[i]$ as $(x+iy)(x-iy)=z^2$. This leads to the question: does the **Fundamental Theorem of Arithmetic** hold in $\mathbb{Z}[i]$? (i.e., is there unique factorization into primes?)
- In Fermat's Last Theorem, we want to study integer solutions of

$$x^n+y^n=z^n,$$

  In the ring $\mathbb{Z}[\zeta_n]$, one can factor the sum as $x^n+y^n=\prod_{k=0}^{n-1}(x+\zeta_n^k y)$.

While all rings satisfy the basic axioms, we often focus on rings with additional properties.

Definition 3.4 (Unital and Commutative Rings)**.** Let R be a ring.

1. R is **unital** if there exists a multiplicative identity $1_R \in R$ such that $1_R \cdot r = r \cdot 1_R = r$ for all $r \in R$.
2. If R is unital, an element $u \in R$ is called a **unit** if it has a multiplicative inverse; that is, there exists $u^{-1} \in R$ such that $uu^{-1}=u^{-1}u=1_R$. The set of all units is denoted $U(R)$.
3. R is **commutative** if $a \cdot b = b \cdot a$ for all $a, b \in R$.

Example 3.5.

- $\mathbb{Z}_n$ is a commutative unital ring. Its units are $U(\mathbb{Z}_n)=\{[a]\in\mathbb{Z}_n \mid \gcd(a,n)=1\}$.
- $M_{n\times n}(\mathbb{R})$ is unital but **not** commutative for $n \geq 2$. Its units are $U(M_{n\times n}(\mathbb{R}))=GL(n,\mathbb{R})$.
- $(\mathbb{Z}_n,+,\cdot)$ is a commutative and unital ring with the multiplication identity $1_R=[1]$. The units of $\mathbb{Z}_n$ are $U(\mathbb{Z})=\mathbb{Z}_n^*=\{a \mid gcd(a,n)=1\}$
- $R=\mathbb{Z}$ is commutative unital, with $U(R)=\{\pm1\}$.
- $R=\mathbb{Z}[i]$ is commutative unital, with $U(R)=\{\pm1,\pm i\}$.
- For $R=\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, all nonzero elements are units.
- $n\mathbb{Z}$ for $n>1$ is a commutative ring that is **not** unital.

Remark 3.6.

- The additive identity $0_R \in R$ is unique in R. If R is unital, then the multiplication identity $1_R \in R$ is also unique.
- We write $-r \in R$ to be the additive inverse of $r \in R$, i.e.: $(-r)+r=r+(-r)=0_R$
- If $r \in U(R)$ is a unit in a (commutative) unital ring, then we write r^{-1} to be the multiplication inverse of r, i.e.: $r^{-1}r=rr^{-1}=1_R$
- For $n \in \mathbb{N}$ and $r \in R$, we write $n \cdot r := r+\cdots+r$, and $-n\cdot r := (-r)+\cdots+(-r)$ (n terms in both sums).

- For $a, b \in R$, we write
$$a \mid b$$
if there exists $c \in R$ such that $ac = b$.

Proposition 3.7. *In any unital ring R:*

1. $0_R \cdot r = r \cdot 0_R = 0_R$ *for all* $r \in R$.
2. $(-1_R) \cdot r = -r$ *(if R is unital).*
3. $(-r) \cdot (-s) = r \cdot s$.

Proof. (1) Since $0_R = 0_R + 0_R$, we have $0_R \cdot r = (0_R + 0_R) \cdot r = 0_R \cdot r + 0_R \cdot r$. By the cancellation law in the additive group $(R, +)$, we obtain $0_R \cdot r = 0_R$.

(2) $0_R = (1_R + (-1_R)) \cdot r = 1_R \cdot r + (-1_R) \cdot r = r + (-1_R) \cdot r$. Thus, $(-1_R) \cdot r$ is the additive inverse of r, so $(-1_R) \cdot r = -r$.

(3) Exercise. □

Definition 3.8 (Subring)**.** A subset $S \leq R$ is a **subring** if S itself is a ring under the operations of R.

Proposition 3.9 (Subring Test)**.** *A non-empty subset $S \subseteq R$ is a subring if and only if for all $a, b \in S$:*

1. $a - b \in S$
2. $a \cdot b \in S$

Example 3.10.
- $\mathbb{Z}[i] \leq \mathbb{C}$ is a subring.
- $n\mathbb{Z} \leq \mathbb{Z}$ is a subring.
- $\{\cdots, -3, -1, 1, 3, \cdots\} \subset \mathbb{Z}$ is **NOT** a subring.

Definition 3.11 (Field)**.** A **field** F is a commutative unital ring in which every non-zero element is a unit. That is,
$$U(F) = F \setminus \{0_F\}.$$

Example 3.12.
- $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ are fields.
- $\mathbb{Z}$ is **not** a field (the only units are ± 1).
- $\mathbb{Z}_n$ is a field if and only if n is prime. If p is prime, we often denote this field as $\mathbb{F}_p$.
- $M_{n \times n}(\mathbb{R})$ is **not** a field (it is not commutative, and not every non-zero matrix is invertible).

3.2 Ring Homomorphisms

Just as group homomorphisms preserve the group structure, ring homomorphisms are maps between rings that preserve both addition and multiplication.

Definition 3.13 (Ring Homomorphism)**.** Let R and S be rings. A map $\phi : R \to S$ is a **ring homomorphism** if for all $a, b \in R$:

1. $\phi(a +_R b) = \phi(a) +_S \phi(b)$
2. $\phi(a \cdot_R b) = \phi(a) \cdot_S \phi(b)$

Furthermore:

- If R and S are unital, ϕ is **unital** if $\phi(1_R) = 1_S$.
- If ϕ is bijective, it is called a **ring isomorphism**, denoted $R \cong S$.

Example 3.14.

- Consider $\phi : \mathbb{Z} \to \mathbb{Z}$ given by $\phi(n) := 2n$.
    1. ϕ is a group homomorphism on $(\mathbb{Z}, +)$.
    2. However, ϕ is **not** a ring homomorphism because for $a, b \neq 0$:

$$\phi(a \cdot b) = 2ab \neq 4ab = (2a) \cdot (2b) = \phi(a) \cdot \phi(b)$$

- In $\mathbb{Z}_{10}$, the map $\phi(n) = 2n$ is not a ring homomorphism, but $\phi(n) := 5n$ **is** a ring homomorphism.
    - Addition: $5(a + b) \equiv 5a + 5b \pmod{10}$.
    - Multiplication: $\phi(ab) = 5ab$ and $\phi(a)\phi(b) = 25ab$. Since $25 \equiv 5 \pmod{10}$, $\phi(ab) = \phi(a)\phi(b)$.
- The natural projection $\pi : \mathbb{Z} \to \mathbb{Z}_m$ where $\pi(n) = [n]_m$ is a ring homomorphism.
- **Evaluation Homomorphism:** Let R be a commutative unital ring. Define $\phi : R[x] \to R$ by $\phi(p(x)) := p(1_R)$. This map evaluates a polynomial at the multiplicative identity.
- The inclusion maps $\mathbb{Z} \hookrightarrow \mathbb{Q}$, $\mathbb{Q} \hookrightarrow \mathbb{R}$, and $\mathbb{R} \hookrightarrow \mathbb{C}$ are ring homomorphisms.

Proposition 3.15. *Let $\phi : R \to S$ be a ring homomorphism.*

1. $\phi(0_R) = 0_S$.
2. $\phi(-a) = -\phi(a)$ *for all* $a \in R$.
3. *If ϕ is unital and $a \in U(R)$, then $\phi(a) \in U(S)$ with $\phi(a)^{-1} = \phi(a^{-1})$.*
4. *If $\phi : R \to S$ is an isomorphism, then $\phi^{-1} : S \to R$ is also a ring isomorphism.*

Proof.

1. $\phi(r) = \phi(0_R + r) = \phi(0_R) + \phi(r)$. By the uniqueness of the additive identity in S, $\phi(0_R) = 0_S$.
2. This follows from the properties of group homomorphisms under addition.
3. $\phi(a^{-1}) \cdot \phi(a) = \phi(a^{-1}a) = \phi(1_R) = 1_S$. Similarly, $\phi(a)\phi(a^{-1}) = 1_S$. Thus, $\phi(a)$ is a unit with inverse $\phi(a^{-1})$.
4. Since ϕ is a bijection, ϕ^{-1} exists as a set-theoretic map. Let $\alpha, \beta \in S$. Then there exist

unique $a, b \in R$ such that $\phi(a) = \alpha$ and $\phi(b) = \beta$. Then:

$$\phi^{-1}(\alpha \cdot \beta) = \phi^{-1}(\phi(a) \cdot \phi(b)) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(\alpha) \cdot \phi^{-1}(\beta)$$

The proof for addition follows the same logic.

□

Proposition 3.16. *Let $\phi : R \to S$ be a ring homomorphism. Then* $\ker \phi := \{r \in R \mid \phi(r) = 0_S\}$ *is a subring of R, and im $\phi := \{\phi(r) \mid r \in R\}$ is a subring of S.*

Proof. Let $a, b \in \ker \phi$, so $\phi(a) = \phi(b) = 0_S$. We check the subring criteria for $\ker \phi$:

- $\phi(a + b) = \phi(a) + \phi(b) = 0_S + 0_S = 0_S$, so $a + b \in \ker \phi$.
- $\phi(-a) = -\phi(a) = -0_S = 0_S$, so $-a \in \ker \phi$.
- $\phi(a \cdot b) = \phi(a) \cdot \phi(b) = 0_S \cdot 0_S = 0_S$, so $ab \in \ker \phi$.

The proof for im ϕ is similar. □

3.3 Ideals

The concept of an ideal is the ring-theoretic analogue of a normal subgroup. Indeed, we are using the same notation $\triangleleft$ to denote an ideal of a ring:

Definition 3.17 (Ideal)**.** Let R be a ring. A subset $I \triangleleft R$ is an **ideal** if:

1. I is an additive subgroup of $(R, +)$.
2. For all $x \in R$ and $i \in I$, we have $xi \in I$ and $ix \in I$.

Remark 3.18. If $I \triangleleft R$, then I is automatically a subring of R. This is because for any $a, b \in I$, the second condition (absorption) implies that $ab \in I$ since $a \in R$ and $b \in I$.

However, not all subrings are ideals. For instance, $\mathbb{Z} \leq \mathbb{Q}$ is a subring but **not** an ideal, since $2 \in \mathbb{Z}$ and $\frac{1}{3} \in \mathbb{Q}$, but $2 \cdot \frac{1}{3} = \frac{2}{3} \notin \mathbb{Z}$.

Example 3.19.

- For $R = \mathbb{Z}$, the subset $I = n\mathbb{Z}$ is an ideal. For any $na \in I$ and $m \in \mathbb{Z}$, $(na)m = n(am) \in I$.
- Let $R = \mathbb{Z}[x]$ and $I = \{p(x) \in R \mid p(0) = 0\}$, the set of polynomials with no constant term. For any $p(x) \in I$ and $q(x) \in R$, the product $p(x)q(x)$ satisfies $p(0)q(0) = 0 \cdot q(0) = 0$, so $p(x)q(x) \in I$.
- Let $R = \mathbb{Z}[x]$ and $I = \{p(x) \in R \mid p(0) \in 2\mathbb{Z}\}$, the set of polynomials where the constant term is an even integer. This is an ideal of $\mathbb{Z}[x]$.

Question:

For any ring R, how do we construct an ideal I?

Definition 3.20 (Ideal Generated by V)**.** Let R be a ring and $V \subseteq R$ be a subset. The **ideal generated by** V, denoted $\langle V \rangle$, is the smallest ideal in R containing all elements of V.

Example 3.21.
- Let $R = \mathbb{Z}$ and $V = \{n\}$. Any ideal I containing n must contain all its additive multiples nk for $k \in \mathbb{Z}$ because I is a subgroup. Thus $n\mathbb{Z} \subseteq I$. Since $n\mathbb{Z}$ is itself an ideal, we have $\langle n \rangle = n\mathbb{Z}$.
- In the general case, let R be a unital commutative ring and $V = \{a_1, \ldots, a_n\}$. Then:

$$\langle a_1, \ldots, a_n \rangle = \{r_1 a_1 + \cdots + r_n a_n \mid r_i \in R\}$$

  To verify this:

  1. This set S contains each a_i (e.g., $a_1 = 1_R \cdot a_1 + 0_R \cdot a_2 + \ldots$).
  2. S is an ideal: the sum of two such linear combinations is another linear combination, and multiplying by $r \in R$ distributes over the terms.
  3. Any ideal I containing $\{a_1, \ldots, a_n\}$ must contain $r_i a_i$ by absorption, and their sum by the subgroup property. Thus $S \subseteq I$.
- Let $R = \mathbb{Z}[x]$. The ideal $\langle 2, x \rangle = \{2p(x) + xq(x) \mid p(x), q(x) \in R\}$ is exactly the set of polynomials with an even constant term.

Lemma 3.22. *Suppose $I_1, \ldots, I_k \lhd R$. Then:*

1. *$I_1 + \cdots + I_k := \{i_1 + \cdots + i_k \mid i_r \in I_r\}$ is an ideal of R.*
2. *$\bigcap_{i=1}^{k} I_i$ is an ideal of R.*

Proof.
1. Let $x = (i_1 + \cdots + i_k)$ and $y = (i'_1 + \cdots + i'_k)$ be elements of the sum. Then $x - y = (i_1 - i'_1) + \cdots + (i_k - i'_k)$. Since each I_r is an additive subgroup, $i_r - i'_r \in I_r$, so $x - y \in I_1 + \cdots + I_k$. For any $r \in R$, $r(i_1 + \cdots + i_k) = ri_1 + \cdots + ri_k$. Since $ri_j \in I_j$, the sum is in $I_1 + \cdots + I_k$.
2. Let $x, y \in \bigcap I_i$. Then for all i, $x, y \in I_i$. Since each I_i is an ideal, $x - y \in I_i$ and $rx \in I_i$ for all i. Therefore, $x - y \in \bigcap I_i$ and $rx \in \bigcap I_i$.

□

Example 3.23. In $R = \mathbb{Z}$, let $I_r = \langle a_r \rangle = a_r\mathbb{Z}$ for $a_r \in \mathbb{N}$. Then:

- $I_1 + \cdots + I_k = \langle \gcd(a_1, \ldots, a_k) \rangle = \gcd(a_1, \ldots, a_k)\mathbb{Z}$
- $I_1 \cap \cdots \cap I_k = \langle \operatorname{lcm}(a_1, \ldots, a_k) \rangle = \operatorname{lcm}(a_1, \ldots, a_k)\mathbb{Z}$

Indeed, one can show that if $I \lhd \mathbb{Z}$ is an ideal, then $I = \langle i \rangle$ for some $i \in \mathbb{Z}$ (Homework).

Definition 3.24 (Principal Ideal)**.** Let R be a unital commutative ring. An ideal $I \lhd R$ is called **principal** if $I = \langle a \rangle = \{ar \mid r \in R\}$ for some $a \in R$.

For instance, all ideals in $\mathbb{Z}$ are principal. However, the ideal $\langle 2, x \rangle \lhd \mathbb{Z}[x]$ is not principal - one cannot find any $p(x) \in \mathbb{Z}[x]$ such that $\langle 2, x \rangle = \langle p(x) \rangle$ (Homework).

Proposition 3.25. *Let $\phi : R \to S$ be a ring homomorphism. Then* $\ker\phi \lhd R$.

Proof. From previous results, we know that $\ker\phi$ is a subring of R. It remains to show the absorption property: for all $r \in R$ and $i \in \ker\phi$, $ri \in \ker\phi$ and $ir \in \ker\phi$. Indeed, $\phi(ri) = \phi(r)\phi(i) = \phi(r)\cdot 0_S = 0_S$. Hence, $ri \in \ker\phi$. Similarly, $\phi(ir) = \phi(i)\phi(r) = 0_S \cdot \phi(r) = 0_S$, so $ir \in \ker\phi$. □

3.4 Quotient Ring

Definition 3.26 (Quotient Ring)**.** Let R be a ring and $I \lhd R$ an ideal. The **quotient ring** R/I is the set of additive cosets $R/I := \{r + I \mid r \in R\}$ equipped with the operations:

$$(r_1 + I) + (r_2 + I) := (r_1 + r_2) + I$$
$$(r_1 + I)\cdot(r_2 + I) := (r_1 \cdot r_2) + I$$

We need to check the above definition is well-defined, and satisfies the axioms of a ring.

- **Multiplication is well-defined:** Suppose $r_1 + I = r_1' + I$ and $r_2 + I = r_2' + I$. This implies $r_1' = r_1 + i_1$ and $r_2' = r_2 + i_2$ for some $i_1, i_2 \in I$. Then:

  $$r_1'r_2' = (r_1 + i_1)(r_2 + i_2) = r_1r_2 + r_1i_2 + i_1r_2 + i_1i_2$$

  Since I is an ideal, r_1i_2, i_1r_2, and i_1i_2 are all in I. Thus $r_1'r_2' - r_1r_2 \in I$, which means $r_1'r_2' + I = r_1r_2 + I$.

- **Addition is well-defined:** This follows directly from the fact that $(I, +)$ is a normal subgroup of $(R, +)$.

- **Ring Axioms:** Associativity and Distributivity in R/I are inherited from the properties of R. For example:

  $$(r_1+I)((r_2+I)+(r_3+I)) = r_1(r_2+r_3)+I = (r_1r_2+r_1r_3)+I = (r_1+I)(r_2+I)+(r_1+I)(r_3+I)$$

It is easy to see that the following holds for quotient rings:

- If R is commutative, then R/I is commutative.
- If R is unital, then R/I is unital with identity $1_{R/I} = 1_R + I$.

We will leave it to the reader to verify these statements.

Example 3.27.

- Consider $\mathbb{R}[x]/\langle x^2 - 1\rangle$. Let $\overline{p(x)} = p(x) + \langle x^2 - 1\rangle$. In this ring,

  $$\overline{x-1}\cdot\overline{x+1} = \overline{x^2-1} = \overline{0}.$$

  So there exists two nonzero elements such that its product is zero (these elements are called **zerodivisors**, which will be discussed in greater detail in Section 3.6 below). This occurs because $x^2 - 1$ is reducible in $\mathbb{R}[x]$.

- In $\mathbb{R}[x]/\langle x^2+1\rangle$, we have $\overline{x}\cdot\overline{x}=\overline{x^2}$. Since $x^2\equiv -1 \pmod{x^2+1}$, we have $\overline{x}^2=\overline{-1}$. This construction behaves like the complex numbers, where $\overline{x}$ plays the role of 'square root of -1'.
- Consider $\mathbb{Z}[x]/\langle 2,x\rangle$. Any polynomial $p(x)=a_nx^n+\cdots+a_1x+a_0$ is congruent to its constant term a_0 modulo the ideal $\langle 2,x\rangle$. Furthermore, $a_0\equiv 0$ or $1 \pmod 2$. Thus,

$$\mathbb{Z}[x]/\langle 2,x\rangle\cong\mathbb{Z}_2.$$

Theorem 3.28 (First Isomorphism Theorem of Rings)**.** *Let $\Phi: R\to S$ be a ring homomorphism. Then the map*

$$\psi: R/\ker\Phi\to \operatorname{im}\,\Phi$$

defined by $\psi(r+\ker\Phi):=\Phi(r)$ is a ring isomorphism.

Proof.
- **Well-defined and Bijective:** From the First Isomorphism Theorem for Groups, we know ψ is a well-defined bijection for the additive structure.
- **Homomorphism:** We check the multiplicative property:

$$\psi((r+\ker\Phi)(r'+\ker\Phi))=\psi(rr'+\ker\Phi)=\Phi(rr')=\Phi(r)\Phi(r')=\psi(r+\ker\Phi)\psi(r'+\ker\Phi).$$

Since ψ preserves both addition and multiplication and is a bijection, it is a ring isomorphism.

□

Example 3.29.
- Define $\Phi:\mathbb{R}[x]\to\mathbb{C}$ by $\Phi(p(x))=p(i)$. This is a surjective homomorphism. The kernel consists of polynomials $p(x)$ such that $p(i)=0$. By the Factor Theorem,

$$(x-i)\mid p(x).$$

Since $p(x)\in\mathbb{R}[x]$ has real coefficients, the complex roots come in pairs, so one also has $(x+i)\mid p(x)$. Thus $(x^2+1)\mid p(x)$, meaning $\ker\Phi=\langle x^2+1\rangle$. By the First Isomorphism Theorem:

$$\mathbb{R}[x]/\langle x^2+1\rangle\cong\mathbb{C}$$

- Define $\Phi:\mathbb{Z}[x]\to\mathbb{Z}_2$ by $\Phi(p(x))=p(0) \pmod 2$. The kernel is the set of polynomials with an even constant term, which is the ideal $\langle 2,x\rangle$. By the First Isomorphism Theorem:

$$\mathbb{Z}[x]/\langle 2,x\rangle\cong\mathbb{Z}_2$$

3.5 Product Ring and Chinese Remainder Theorem

Definition 3.30 (Product Ring)**.** Given rings R_1 and R_2, the Cartesian product $R_1\times R_2$ is a ring under component-wise operations:

$$(r_1,r_2)+(s_1,s_2)=(r_1+s_1,r_2+s_2)$$

$$(r_1, r_2) \cdot (s_1, s_2) = (r_1 \cdot s_1, r_2 \cdot s_2).$$

We can generalize the above definition to $\prod_{i=1}^{n} R_i = R_1 \times R_2 \times \cdots \times R_n$.

Definition 3.31. Let R be a unital commutative ring. We say $I_1, I_2 \triangleleft R$ **coprime** if $I_1 + I_2 = R$.

Example 3.32. Let $R = \mathbb{Z}$ and $I_1 = \langle m \rangle = m\mathbb{Z}$, $I_2 = \langle n \rangle = n\mathbb{Z}$. We have seen before that $I_1 + I_2 := \{mp + nq \mid p, q \in \mathbb{Z}\} = \langle gcd(m, n) \rangle$. Therefore,

$$I_1, I_2 \text{ are coprime} \Leftrightarrow I_1 + I_2 = \mathbb{Z} \Leftrightarrow \langle \gcd(m, n) \rangle = \mathbb{Z} \Leftrightarrow \gcd(m, n) = 1 \text{ are coprime as integers.}$$

The above example generalizes our understanding of "coprime" from $\mathbb{Z}$ to any R: Two elements r_1, r_2 are coprime in R means $\langle r_1 \rangle + \langle r_2 \rangle = R$.

Theorem 3.33 (Chinese Remainder Theorem)**.** *Let R be a commutative unital ring, and let $I_1, \ldots, I_k \triangleleft R$ be ideals such that I_i and I_j are pairwise coprime for $i \neq j$. Then we have a ring isomorphism*

$$\phi : R/(I_1 \cap \cdots \cap I_k) \to R/I_1 \times \cdots \times R/I_k$$

defined by $\phi(r + I_1 \cap \cdots \cap I_k) := (r + I_1, \ldots, r + I_k)$.

Proof. Let $\Phi : R \to R/I_1 \times \cdots \times R/I_k$ be the ring homomorphism defined by

$$\Phi(r) := (r + I_1, \ldots, r + I_k).$$

By the First Isomorphism Theorem, we need to show:

(i) $\ker \Phi = I_1 \cap \cdots \cap I_k$, and

(ii) $\operatorname{im} \Phi = R/I_1 \times \cdots \times R/I_k$.

Part (i): We have

$$\begin{aligned} r \in \ker \Phi &\Leftrightarrow \Phi(r) = (0 + I_1, \ldots, 0 + I_k) \\ &\Leftrightarrow r + I_\ell = 0 + I_\ell \text{ for all } \ell = 1, \ldots, k \\ &\Leftrightarrow r \in I_\ell \text{ for all } \ell = 1, \ldots, k \\ &\Leftrightarrow r \in I_1 \cap \cdots \cap I_k. \end{aligned}$$

Part (ii): Fix $j \in \{1, \ldots, k\}$. Since R is unital and I_j and I_i are coprime for all $i \neq j$, we have $I_i + I_j = R$. Thus there exist $z_i \in I_j$ and $w_i \in I_i$ such that

$$z_i + w_i = 1 \quad \text{for all } i \neq j.$$

Consider the identity $1 = (1 - \prod_{i \neq j} w_i) + \prod_{i \neq j} w_i$. Define

$$x_j := 1 - \prod_{i \neq j} w_i = 1 - \prod_{i \neq j} (1 - z_i).$$

Expanding the product, we see that x_j is a sum of products of the z_i's with no constant term, hence $x_j \in I_j$. Also, define

$$y_j := \prod_{i \neq j} w_i \in \bigcap_{i \neq j} I_i,$$

since each $w_i \in I_i$ and ideals are closed under multiplication.

Therefore, for each fixed j, we have

$$1 = x_j + y_j, \quad \text{where } x_j \in I_j \text{ and } y_j \in \bigcap_{i \neq j} I_i.$$

We now verify that Φ is surjective. Let $(u_1 + I_1, \ldots, u_k + I_k) \in R/I_1 \times \cdots \times R/I_k$ be an arbitrary element.

Claim: $\Phi(u_1 y_1 + \cdots + u_k y_k) = (u_1 + I_1, \ldots, u_k + I_k)$.

Proof of Claim: We have

$$\Phi(u_1 y_1 + \cdots + u_k y_k) = (u_1 y_1 + \cdots + u_k y_k + I_1, \ldots, u_1 y_1 + \cdots + u_k y_k + I_k).$$

For the ℓ-th component, note that $y_j \in I_\ell$ for all $j \neq \ell$ (by definition of y_j), so

$$\begin{aligned} u_1 y_1 + \cdots + u_k y_k + I_\ell &= u_\ell y_\ell + I_\ell \\ &= u_\ell (1 - x_\ell) + I_\ell \\ &= u_\ell - u_\ell x_\ell + I_\ell \\ &= u_\ell + I_\ell, \end{aligned}$$

where the last equality holds because $x_\ell \in I_\ell$.

Thus Φ is surjective, completing the proof. □

Corollary 3.34. *Let $p_1, \cdots, p_n$ be distinct prime numbers. Then*

$$\mathbb{Z}/\langle p_1^{a_1} \cdots p_n^{a_n} \rangle \cong \mathbb{Z}/\langle p_1^{a_1} \rangle \times \cdots \times \mathbb{Z}/\langle p_n^{a_n} \rangle.$$

Therefore, for each $b_i \in \mathbb{Z}/\langle p_i^{a_i} \rangle \cong \mathbb{Z}_{p_i^{a_i}}$, there exists $x \in \mathbb{Z}$ such that $\Phi(x) = (b_1 + \langle p_1^{a_1} \rangle, \cdots, b_n + \langle p_n^{a_n} \rangle)$, i.e.

$$x \equiv b_i \pmod{\mathbb{Z}_{p_i^{a_i}}}$$

for all i.

3.6 Integral Domains

In many rings, the product of two non-zero elements can be zero. Integral domains are a class of rings where this "pathological" behavior does not occur, much like in the integers.

Definition 3.35 (Zerodivisor)**.** Let R be a ring. A non-zero element $r \in R$ is called a **zerodivisor** if there exists a non-zero element $s \in R$ such that $r \cdot s = 0$ or $s \cdot r = 0$.

Example 3.36.

- In the ring $R = \mathbb{Z}_6$, the element $[2]$ is a zerodivisor because $[2]\cdot[3] = [6] = [0]$, and both $[2], [3] \neq [0]$.
- Let R be unital commutative ring, and $R[x]$ be its polynomial ring, and let $p(x) = a(x)b(x) \in R[x]$ be a reducible polynomial. Then

$$\overline{a(x)} := a(x) + \langle p(x) \rangle, \quad \overline{b(x)} := b(x) + \langle p(x) \rangle$$

  are zerodivisors of the quotient ring $R[x]/\langle p(x) \rangle$ (c.f. Example 3.27).
- Finally, if R_1, R_2 are not zero rings, then the product ring $R_1 \times R_2$ has zerodivisors. Namely,

$$(r_1, 0_{R_2}) \cdot (0_{R_1}, r_2) = (0_{R_1}, 0_{R_2}) = 0_{R_1 \times R_2}$$

  for $r_1 \neq 0$, $r_2 \neq 0$. So they are both zerodivisors.

As one noticed in the construction of ideal $\langle a_1, \ldots, a_r \rangle \lhd R$ generated by $a_1, \ldots, a_r \in R$ and the Chinese Remainder Theorem, unital commutative rings have a much nicer structure than general rings. Therefore, we separate the case when R is unital commutative in the definition below:

Definition 3.37 (Integral Domain)**.** A ring R is a **domain** if it has no zerodivisors. A *unital, commutative* ring with no zerodivisors is called an **integral domain** (ID).

Example 3.38.

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, and the Gaussian integers $\mathbb{Z}[i]$ are all integral domains.
- $\mathbb{Z}_n$ is an integral domain if and only if n is prime. If n is composite, say $n = ab$ with $1 < a, b < n$, then $[a][b] = [0]$ in $\mathbb{Z}_n$, so it has zerodivisors.
- If R is an integral domain, then the polynomial ring $R[x]$ is also an integral domain.

Proposition 3.39 (Cancellation Property)**.** *Let R be a (unital, commutative) ring. Then R is a(n integral) domain if and only if for all $a, b, c \in R$ with $c \neq 0$:*

$$ca = cb \implies a = b$$

Proof. ($\Rightarrow$) Suppose R is an integral domain and $ca = cb$ with $c \neq 0$. Then $ca - cb = 0$, which implies $c(a - b) = 0$. Since R has no zerodivisors and $c \neq 0$, we must have $a - b = 0$, hence $a = b$.

($\Leftarrow$) Suppose the cancellation property holds. Let $c, x \in R$ such that $cx = 0$ and $c \neq 0$. We can write this as $cx = c \cdot 0$. By cancellation, $x = 0$. Thus, R has no zerodivisors. □

Remark 3.40. The cancellation property is the primary reason integral domains are 'well-behaved'. It allows us to solve algebraic equations similarly to how we do in the integers or fields.

Hierarchy of Rings:
One of the main goals of ring theory is to understand the following inclusions:

$$\begin{aligned}\text{Fields} &\subsetneq \text{Euclidean Domains (ED)}\\ &\subsetneq \text{Principal Ideal Domains (PID)}\\ &\subsetneq \text{Unique Factorization Domain (UFD)}\\ &\subsetneq \text{Integral Domains (ID)}\end{aligned}$$

Indeed, one can easily see the coarsest inclusion holds:

Proposition 3.41. *If R is a field, then R is an integral domain.*

Proof. Since R is a field, it is a commutative ring with unit 1_R. Suppose on contrary that there exist $a, b \in R$ such that $ab = 0_R$ with $a, b \neq 0_R$. Since R is a field and $a \neq 0_R$, the element a has a multiplicative inverse a^{-1}. Multiplying both sides of the equation $ab = 0$ by a^{-1} gives

$$a^{-1}(ab) = a^{-1}0_R.$$

By associativity,

$$(a^{-1}a)b = 0_R.$$

Since $a^{-1}a = 1_R$, we obtain

$$1_R \cdot b = 0_R,$$

which implies $b = 0_R$, contradicting the assumption that $ab = 0_R$ with $a, b \neq 0_R$. Therefore, R has no zero divisors, and R is an integral domain. □

In the case when R is a finite ring, the converse holds:

Lemma 3.42. *Let R be a finite domain. Then R is unital.*

Proof. Let $a \in R$ be a non-zero element. Consider the set of powers $\{a, a^2, a^3, \dots\}$. Since R is finite, there must be repetitions: $a^m = a^n$ for some $m > n > 0$. By the cancellation property, $a^{m-n}a^n = 1 \cdot a^n$, so a^{m-n} acts as the identity for powers of a.

Specifically, for any $x \in R$, consider xa^n. We have $xa^m = xa^n a^{m-n}$. Since $a^m = a^n$, this implies $xa^n = (xa^{m-n})a^n$. By cancellation of a^n, we get $x = xa^{m-n}$. Thus a^{m-n} is the multiplicative identity 1_R. □

Proposition 3.43. *Every finite commutative domain R is a field.*

Proof. Let R be a finite commutative domain, and let $R \setminus \{0_R\} = \{r_1, r_2, \dots, r_m\}$. By the Lemma above, we have seen that R is unital. To further show R is a field, we must show every non-zero $a \in R$ has a multiplicative inverse.

Consider the set $S = \{ar_1, ar_2, \ldots, ar_m\}$. If $ar_i = ar_j$, then by the cancellation property $r_i = r_j$, so $i = j$. Thus, all m elements in S are distinct and non-zero. Since $S \subseteq R \setminus \{0_R\}$ and $|S| = |R \setminus \{0_R\}|$, it must be that $S = R \setminus \{0_R\}$.

Since the identity 1_R is in $R \setminus \{0\}$, there must exist some r_k such that $ar_k = 1_R$. Thus a is a unit, and R is a field. □

For the rest of this chapter, we will further focus on the case when R **is an (infinite) integral domain**, and study the intermediate rings in the above inclusions.

3.7 Prime and Maximal Ideals

As stated in the previous section, one would like to study various arithmetic properties for integral domain R.

To facilitate for generalization, we would first define these properties in terms of **ideals**, and then derive the equivalent conditions in terms of **elements**.

Definition 3.44 (Divides and Associates)**.** Let $a, b \in R$, where R is an integral domain. We say that a **divides** b (written $a \mid b$) if

$$\langle b \rangle \subseteq \langle a \rangle.$$

(or equivalently, $\quad b \in \langle a \rangle \quad \Leftrightarrow \quad b = qa$ for some $q \in R$.).

We say that a and b are **associates**, written $a \sim b$, if $a \mid b$ and $b \mid a$. Or equivalently,

$$\langle a \rangle = \langle b \rangle.$$

To understand these definitions purely in terms of elements, one has:

Lemma 3.45. *Let R be an integral domain and let $a, b \in R$. Then*

$$a \sim b \textit{ if and only if there exists } u \in U(R) \textit{ such that } a = ub.$$

Proof. ($\Leftarrow$) Suppose $a = ub$ for some unit $u \in U(R)$. Then $a \in \langle b \rangle$, so $\langle a \rangle \subseteq \langle b \rangle$. Since u is a unit, u^{-1} exists, and

$$b = u^{-1}a,$$

so $b \in \langle a \rangle$, hence $\langle b \rangle \subseteq \langle a \rangle$ and therefore $\langle a \rangle = \langle b \rangle$. In other words, $a \sim b$.

($\Rightarrow$) Suppose $a \sim b$, i.e. $\langle a \rangle = \langle b \rangle$. Then $a \in \langle b \rangle$, so there exists $p \in R$ such that $a = pb$. Similarly, since $b \in \langle a \rangle$, there exists $q \in R$ such that $b = qa$.

Substituting gives

$$a = p(qa) = (pq)a.$$

Since R is an integral domain and $a \neq 0$, we may cancel a to obtain

$$1 = pq.$$

Thus p is a unit (with inverse q), and

$$a = pb$$

with $p \in U(R)$. Hence $a = ub$ for some unit u. $\square$

Now we define what 'prime' really means for R. Rather than studying whether an element $r \in R$ is prime directly, we define what it means for an ideal $I \triangleleft R$ to be prime, and then define what it means for $p \in R$ to be prime. As an example, recall that we defined

$$I_1,\, I_2 \text{ is coprime if } I_1 + I_2 = R \qquad \text{(coprime ideals)}$$

And we define:

Definition 3.46 (Coprime elements)**.** Let R be a unital commutative ring. Two elements a, b are **coprime** if the principal ideals $\langle a \rangle$, $\langle b \rangle \triangleleft R$ are coprime, i,e,

$$\langle a \rangle + \langle b \rangle = R.$$

To better comprehend the above definition, the above condition is equivalent to saying:

$$a, b \in R \text{ are coprime } \iff \text{ there exists } p, q \in R \text{ such that } ap + bq = 1 \text{ (Exercise).}$$

Upshot: We defined what it means for $a, b \in R$ to be coprime without defining 'common factors'!

Note that in the prototype case when $R = \mathbb{Z}$, the above statement and Bezout's theorem imply that

$$a, b \in \mathbb{Z} \text{ are coprime } \iff \gcd(a, b) = 1.$$

which gives us back our usual definition of what coprime means in $\mathbb{Z}$.

Definition 3.47 (Prime and Maximal Ideals)**.** Let R be an integral domain. We say a *proper* ideal (i.e. $I \neq R$) $I \triangleleft R$ is

1. **prime** if for all $a, b \in R$ such that $ab \in R$, then we must have $a \in I$ or $b \in I$;
2. **maximal** if for all ideals $J \triangleleft R$ s.t. $I \subset J \subset R$, then $J = I$ or $J = R$.

Definition 3.48 (Prime Element)**.** Let R be a unital commutative ring. An element $p \in R$ is called **prime** if the ideal $\langle p \rangle$ is a prime ideal.

Equivalently, p is prime if and only if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

for all $a, b \in R$.

Example 3.49.
- Back into our prototype $R = \mathbb{Z}$. It is obviously true that if $p \in \mathbb{Z}$ is a prime number, then $p \mid ab \;\Leftrightarrow\; p \mid a \textit{ or } p \mid b$ and vice versa.

  (**Exercise:** $\langle n \rangle$ is a maximal ideal $\Leftrightarrow$ n is prime $\Leftrightarrow$ $\langle n \rangle$ is prime ideal)

- Let $R = \mathbb{Z}[x]$, and $I = \langle x \rangle$ be all polynomials with 0 constant term is a prime ideal – take

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m, \quad q(x) = b_0 + b_1 x + \cdots + b_n x^n$$

  Then

$$p(x)q(x) \in I \;\Leftrightarrow\; a_0 b_0 = 0 \;\Leftrightarrow a_0 = 0 \textit{ or } b_0 = 0 \Leftrightarrow\; p(x) \in I \textit{ or } q(x) \in I$$

  However, I is **NOT** maximal, since

$$I \subsetneq \langle 2, x \rangle \subsetneq R.$$

Proposition 3.50. *Let R be an integral domain, and let $I \lhd R$. Then:*

1. *I is prime $\Longleftrightarrow$ R/I is an integral domain.*
2. *I is maximal $\Longleftrightarrow$ R/I is a field.*

Example 3.51. Consider the ideal $\langle x \rangle \subset \mathbb{Z}[x]$. Then

$$\mathbb{Z}[x]/\langle x \rangle \cong \mathbb{Z}.$$

Since $\mathbb{Z}$ is an integral domain, it follows that $\langle x \rangle$ is prime. However, $\mathbb{Z}$ is *not* a field, hence $\langle x \rangle$ is not maximal.

Proof. 1. Let $a + I, b + I \in R/I$. Then $(a + I)(b + I) = 0 + I \Longleftrightarrow ab \in I$.

($\Rightarrow$) Suppose I is prime. Suppose $a + I, b + I$ be such that $(a + I)(b + I) = 0 + I$. By the first sentence of the proof, this implies

$$ab \in I \Longrightarrow a \in I \text{ or } b \in I.$$

Hence

$$a + I = 0 \text{ or } b + I = 0.$$

Therefore R/I has no zero divisors and is an integral domain.

($\Leftarrow$) Conversely, suppose R/I is an integral domain. Consider $ab \in I$. Then the first sentence of the proof says

$$(a + I)(b + I) = 0 + I.$$

Since R/I has no zero divisors, this implies

$$a + I = 0 + I \text{ or } b + I = 0 + I,$$

hence $a \in I$ or $b \in I$. Thus I is prime.

2. Suppose I is maximal and let J be an ideal such that

$$I \subseteq J \subseteq R.$$

Then $J = I$ or $J = R$.

By the Correspondence Theorem (Homework), ideals J with $I \subseteq J \subseteq R$ correspond bijectively to ideals of R/I via

$$J \longmapsto J/I.$$

Hence the only ideals of R/I are $\{0\}$ and R/I.

Recall (from Homework) that a ring $\mathbb{F}$ is a field if and only if its only ideals are $\{0\}$ and $\mathbb{F}$. Therefore R/I is a field. Conversely, if R/I is a field, then its only ideals are $\{0\}$ and R/I. By correspondence theorem again, the only ideals of R containing I are I and R. Hence I is maximal.

□

Corollary 3.52. *Let R be an integral domain. If $I \triangleleft R$ is maximal, then I is prime.*

Proof. If I is maximal, then R/I is a field. Since every field is an integral domain, R/I is an integral domain. Hence I is prime. □

Indeed, one can proof the above corollary directly without using quotient rings. We leave it as an exercise.

3.8 Principal Ideal Domains

Recall that for any integral domain (indeed for unital commutative rings as well),

$$I \text{ maximal } \Longrightarrow I \text{ prime}.$$

However, the converse is *false* in general – for example, $\langle x \rangle \subset \mathbb{Z}[x]$ is prime but not maximal.

A natural goal is to find classes of integral domains R where prime ideals are "as close as possible" to maximal ideals. One particularly important class is when *every* ideal is generated by one element.

Definition 3.53 (Principal Ideal Domain)**.** Let R be an integral domain. We say that R is a **principal ideal domain (PID)** if every ideal $I \triangleleft R$ is principal, i.e. there exists $a \in R$ such that

$$I = \langle a \rangle = \{ar \mid r \in R\}.$$

Example 3.54.

1. **All fields $\mathbb{F}$ are PIDs** – By Homework, the only ideals in $\mathbb{F}$ are $\{0_{\mathbb{F}}\} = \langle 0_{\mathbb{F}} \rangle$ and $\mathbb{F} = \langle 1_{\mathbb{F}} \rangle$.
2. $\mathbb{Z}$ **is a PID** – Let $I \triangleleft \mathbb{Z}$ be a nonzero ideal. Choose $d > 0$ to be the smallest positive integer in I.
   Claim: $I = \langle d \rangle$.
   Clearly $\langle d \rangle \subseteq I$ since $d \in I$. For the other inclusion, let $n \in I$. By the division algorithm in $\mathbb{Z}$, write $n = qd + r$ with $0 \leq r < d$. Then $r = n - qd \in I$. By minimality of d, we must have $r = 0$, hence $n \in \langle d \rangle$. Therefore $I \subseteq \langle d \rangle$, so $I = \langle d \rangle$.
3. **If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is a PID** – Let $I \triangleleft \mathbb{F}[x]$ be a nonzero ideal. Choose a nonzero polynomial $f(x) \in I$ of *minimal degree.* (Multiplying by a unit in $\mathbb{F}$ if necessary, we may assume f is monic.)
   Claim: $I = \langle f(x) \rangle$.
   Clearly $\langle f \rangle \subseteq I$. For the other inclusion, let $g(x) \in I$. By the division algorithm in $\mathbb{F}[x]$, write
   $$g(x) = q(x)f(x) + r(x), \qquad \deg r < \deg f.$$
   Then $r(x) = g(x) - q(x)f(x) \in I$. By minimality of $\deg f$, we must have $r(x) = 0$. Hence $g(x) \in \langle f \rangle$, so $I \subseteq \langle f \rangle$ and therefore $I = \langle f \rangle$.
4. $\mathbb{Z}[x]$ **is not a PID** – In particular, the ideal $J = \langle 2, x \rangle \triangleleft \mathbb{Z}[x]$ is not principal (Homework).

Remark 3.55. The proofs that $\mathbb{Z}$ and $\mathbb{F}[x]$ are PIDs are the same pattern: choose an element of *minimal size* (smallest positive integer / smallest degree polynomial) and use the division algorithm to show it generates the entire ideal.

Indeed, these rings belong to a more restrictive class of Integral Domains, called **Euclidean domains (EDs)**. We will study EDs in full detail later, and will show that all EDs are PIDs.

In a PID, prime and maximal ideals are very close. Note that since a PID is an integral domain, the zero ideal $\langle 0 \rangle$ is always prime (because $R/\langle 0 \rangle \cong R$ is an integral domain). Therefore "prime $\Leftrightarrow$ maximal" cannot hold for *all* ideals unless R is a field. The correct statement is for *nonzero* prime ideals.

Proposition 3.56. *Let R be a PID and let $I \triangleleft R$ be a nonzero proper ideal. Then*

$$I \text{ is prime } \iff I \text{ is maximal.}$$

Equivalently, for $0 \neq p \in R$ nonunit,

$$\langle p \rangle \text{ is prime } \iff \langle p \rangle \text{ is maximal.}$$

Proof. ($\Leftarrow$) Maximal $\Rightarrow$ prime holds in every unital commutative ring.

($\Rightarrow$) Assume I is a nonzero prime ideal. Since R is a PID, $I = \langle p \rangle$ for some $0 \neq p \in R$. To show

$\langle p \rangle$ is maximal, let J be any ideal with

$$\langle p \rangle \subseteq J \subseteq R.$$

Again using that R is a PID, write $J = \langle a \rangle$ for some $a \in R$. The inclusion $\langle p \rangle \subseteq \langle a \rangle$ means $p \in \langle a \rangle$, i.e. $p = ab$ for some $b \in R$.

Now we use the fact that $\langle p \rangle$ is prime ideal, or equivalently $p \in R$ is prime element. One has

$$p = ab \;\Rightarrow\; p \mid ab \;\Rightarrow\; p \mid a \;\text{ or }\; p \mid b.$$

Suppose first of all $p \mid a$. Then $a = pq$ for some $q \in R$. Hence

$$p = ab = pqb \;\Rightarrow\; 1 = qb,$$

i.e. $b \in U(R)$. Consequently

$$p = ab,\; b \in U(R) \;\Rightarrow\; p \sim a \;\Rightarrow\; \langle p \rangle = \langle a \rangle = J$$

On the other hand, suppose $p \mid b$. Then one has $a \in U(R)$ similarly, and hence $J = \langle a \rangle = R$. Either way, there is no ideal J lying strictly between $I = \langle p \rangle$ and R, so I is maximal. □

3.9 Irreducible Elements

Recall that in a PID, we have shown that nonzero prime ideals are maximal. A further goal is to understand how elements of an integral domain factor. The key notion is that of an *irreducible element*, which serves as the "building block" for factorization.

Definition 3.57 (Irreducible Element)**.** Let R be a unital commutative ring. An element $a \in R$ is **irreducible** if $\langle a \rangle$ is maximal among all proper principal ideals of R, i.e.,

$$\text{whenever } \langle a \rangle \subseteq \langle b \rangle \subseteq R \text{ for some } b \in R, \text{ we have } \langle a \rangle = \langle b \rangle \text{ or } \langle b \rangle = R.$$

As before, we give a characterization of irreducibility in terms of elements of R.

Lemma 3.58. *Let R be an integral domain, and let $a \in R$ be a nonzero nonunit. Then*

$$a \textit{ is irreducible} \iff \textit{whenever } a = xy, \textit{ we have } a \sim x \textit{ or } a \sim y.$$

(Recall that $a \sim x$ means $a = ux$ for some unit $u \in U(R)$.)

Proof. ($\Rightarrow$): Suppose a is irreducible and $a = xy$. Then $a \in \langle x \rangle$, so $\langle a \rangle \subseteq \langle x \rangle$. By irreducibility, either $\langle a \rangle = \langle x \rangle$ or $\langle x \rangle = R$.

- If $\langle a \rangle = \langle x \rangle$, then $a \sim x$.

- If $\langle x \rangle = R$, then x is a unit. Since $a = xy$ and x is a unit, we have $a \sim y$.

($\Leftarrow$): Suppose that whenever $a = xy$ we have $a \sim x$ or $a \sim y$. Let $\langle a \rangle \subseteq \langle b \rangle$ for some $b \in R$. Then $a = bb'$ for some $b' \in R$, so by hypothesis, $a \sim b$ or $a \sim b'$.

- If $a \sim b$, then $\langle a \rangle = \langle b \rangle$.
- If $a \sim b'$, write $a = ub'$ for some unit u. Then from $a = bb'$ and $b' = u^{-1}a$, we get $a = b \cdot u^{-1}a$, so $bu^{-1} = 1$ by cancellation in R, meaning b is a unit and $\langle b \rangle = R$. $\square$

Remark 3.59. As an immediate consequence of the lemma, $a \in R$ is irreducible if and only if there do *not* exist nonunits $b, c \in R$ such that $a = bc$. In other words, irreducible elements are precisely the "atoms" of R: they cannot be factored into strictly smaller pieces. This makes irreducible elements the natural building blocks for factorization in R.

The following proposition shows that prime elements are always irreducible.

Proposition 3.60. *Let R be an integral domain. If $0 \neq a \in R$ is prime, then a is irreducible.*

Proof. Suppose $a = xy$. Then $a \mid xy$, and since a is prime, $a \mid x$ or $a \mid y$. Without loss of generality, assume $a \mid x$, so $x = aa'$ for some $a' \in R$. Then $a = xy = aa'y$, and cancelling a (which is valid since R is an integral domain and $a \neq 0$) gives $a'y = 1$, so y is a unit. Hence $a \sim x$. $\square$

Remark 3.61. The converse is *false* in general. For example, $3 \in \mathbb{Z}[\sqrt{-5}]$ is irreducible but not prime. However, we will see below that in a PID — and more generally in a UFD — the two notions coincide for nonzero nonunits.

In a PID, the interplay between maximal ideals, prime ideals, and irreducible elements gives the following clean picture.

Proposition 3.62. *Let R be a PID and let $0 \neq a \in R$ be a nonunit. Then*

$$\langle a \rangle \text{ is maximal} \iff \langle a \rangle \text{ is prime} \iff a \text{ is irreducible} \iff a \text{ is prime.}$$

Proof. The equivalence $\langle a \rangle$ maximal $\Leftrightarrow$ $\langle a \rangle$ prime was established in the previous section. The implication prime $\Rightarrow$ irreducible holds in any integral domain by the proposition above.

It remains to show that irreducible $\Rightarrow$ prime in a PID. Let $a \in R$ be irreducible. By definition, $\langle a \rangle$ is maximal among proper principal ideals. Since every ideal in a PID is principal, $\langle a \rangle$ is in fact a maximal ideal, hence a prime ideal, so a is prime. $\square$

To conclude the four conditions — maximal ideal, prime ideal, prime element, irreducible element — are all equivalent (for nonzero nonunits) for PIDs. This is a very special property: for instance, in $\mathbb{Z}[x]$ (which is *not* a PID), the element x generates a prime ideal $\langle x \rangle$ that is *not* maximal; and in $\mathbb{Z}[\sqrt{-5}]$ (which is not even a UFD), the element 3 is irreducible but *not* prime.

3.10 Unique Factorization Domain

We now make precise the notion that every element can be written as a finite product of irreducibles.

Definition 3.63 (Factorization Domain)**.** Let R be an integral domain. We say R is a **factorization domain** if for every nonzero nonunit $x \in R$, there exist irreducible elements $x_1, \ldots, x_r \in R$ such that

$$x = x_1 x_2 \cdots x_r.$$

To prove that PIDs are factorization domains, we first establish a chain condition.

Definition 3.64 (ACCP)**.** Let R be an integral domain. We say R satisfies the **ascending chain condition on principal ideals** (ACCP) if every ascending chain of principal ideals

$$\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \langle a_3 \rangle \subseteq \cdots$$

stabilizes, i.e., there exists $N \in \mathbb{N}$ such that $\langle a_n \rangle = \langle a_N \rangle$ for all $n \geq N$.

Lemma 3.65. *Every PID satisfies the ACCP.*

Proof. Let $\langle a_1 \rangle \subseteq \langle a_2 \rangle \subseteq \cdots$ be an ascending chain of principal ideals in a PID R. Set $I := \bigcup_{n=1}^{\infty} \langle a_n \rangle$. One verifies that I is an ideal of R (see Homework), so since R is a PID, $I = \langle r \rangle$ for some $r \in R$. Since $r \in I$, we have $r \in \langle a_M \rangle$ for some $M \in \mathbb{N}$. Then for all $n \geq M$,

$$I = \langle r \rangle \subseteq \langle a_M \rangle \subseteq \langle a_n \rangle \subseteq I,$$

so $\langle a_n \rangle = I$ for all $n \geq M$, and the chain stabilizes. $\square$

Lemma 3.66. *If R satisfies the ACCP, then R is a factorization domain.*

Proof. Let

$$F := \{x \in R \mid x \sim x_1 \cdots x_r,\ x_i \text{ irreducibles}, r \in \mathbb{N} \cup \{0\}\}$$

We want to show that $R = F$. Note that

(a) $1 \in F$. More generally, all units are in F $(r = 0)$.

(b) All irreducible elements $x \in R$ are in F $(r = 1)$.

(c) F is closed under multiplication (i.e. if $a, b \in F$, then $ab \in F$).

Suppose on contrary there exists $x_0 \in R \backslash F$. Then by (b) $x_0 = y_0 z_0$ is reducible, where y_0, z_0 are not units. Therefore,

$$x_0 \nsim y_0 \text{ and } x_0 \nsim z_0 \iff \langle x_0 \rangle \neq \langle y_0 \rangle, \langle z_0 \rangle$$

By (c), either $y_0 \in R \backslash F$ or $z_0 \in R \backslash F$ (or both). Without loss of generality, assume $x_1 = y_0 \in R \backslash F$. Therefore,

$$x_0 = x_1 z_0 \implies x_0 \in \langle x_1 \rangle \implies \langle x_0 \rangle \subsetneq \langle x_1 \rangle \quad (\text{and } \langle x_1 \rangle \neq R).$$

Since x_1 not unit, we continue same argument on x_1 and get

$$\langle x_0 \rangle \subsetneq \langle x_1 \rangle \subsetneq \langle x_2 \rangle \subsetneq \cdots$$

contradicting (ACCP) which says there exists $N \in \mathbb{N}$ such that

$$\langle x_n \rangle = \langle x_{n+1} \rangle = \langle x_{n+2} \rangle = \cdots$$

This is a contradiction, and consequently there is no such $x_0 \in R \setminus F$, In other words, $R = F$. $\square$

Theorem 3.67. *Every PID is a factorization domain.*

Proof. Every PID satisfies the ACCP by the first lemma, and every ring satisfying the ACCP is a factorization domain by the second lemma. $\square$

We have shown that in a PID, every nonzero nonunit factors into irreducibles. We now address uniqueness.

Proposition 3.68. *Let R be an integral domain and suppose*

$$x_1 x_2 \cdots x_r \sim y_1 y_2 \cdots y_n,$$

where $u \in U(R)$ is a unit, $x_1, \ldots, x_r$ are primes, and $y_1, \ldots, y_n$ are irreducible nonunits. Then $r = n$ and there exists a permutation $\sigma \in S_r$ such that $x_i \sim y_{\sigma(i)}$ for all i.

Proof. We proceed by induction on r.

Base case ($r = 0$): The left-hand side equals a unit, so $y_1 \cdots y_n \sim 1$. This forces each y_j to be a unit, contradicting the assumption that y_j are nonunits. Hence $n = 0$.

Inductive step: Assume the result holds for all products of fewer than r primes. Consider

$$u \cdot x_1 \cdots x_r = y_1 \cdots y_n.$$

Since x_r is prime and $x_r \mid y_1 \cdots y_n$, we have $x_r \mid y_j$ for some j. Write $y_j = x_r \cdot a$ for some $a \in R$. Since y_j is irreducible and x_r is a nonunit, a must be a unit, so $y_j \sim x_r$. Canceling x_r from both sides (valid since R is an integral domain) gives

$$u \cdot x_1 \cdots x_{r-1} = y_1 \cdots \widehat{y_j} \cdots y_n \iff x_1 \cdots x_{r-1} \sim y_1 \cdots \widehat{y_j} \cdots y_n$$

where $\widehat{y_j}$ denotes that y_j is omitted. By the inductive hypothesis, $r - 1 = n - 1$ (so $r = n$) and there exists $\sigma \in S_{r-1}$ matching $x_i \sim y_{\sigma(i)}$ for $i = 1, \ldots, r-1$. Extending σ by setting $\sigma(r) = j$ yields the desired permutation in S_r. $\square$

Corollary 3.69. *Let R be a PID. Then every nonzero nonunit $x \in R$ can be written as a finite product of irreducibles (= primes), and this factorization is unique up to order and associates.*

This generalizes the Fundamental Theorem of Arithmetic (the case $R = \mathbb{Z}$) to all PIDs.

Definition 3.70 (Unique Factorization Domain)**.** Let R be an integral domain. We say R is a **unique factorization domain** (UFD) if every nonzero nonunit $x \in R$ can be written as a finite product of irreducibles, and this factorization is unique up to order and associates.

The corollary above shows that every PID is a UFD. In a UFD, the notions of prime and irreducible also coincide.

Proposition 3.71. *Let R be a UFD. Then $p \in R$ is prime if and only if p is irreducible.*

Proof. Prime implies irreducible in any integral domain. Conversely, suppose p is irreducible and $p \mid ab$, so $pc = ab$ for some $c \in R$. Write $a = a_1 \cdots a_s$, $b = b_1 \cdots b_t$, and $c = c_1 \cdots c_u$ as products of irreducibles. Then

$$p \cdot c_1 \cdots c_u = a_1 \cdots a_s \cdot b_1 \cdots b_t.$$

By uniqueness of factorization, p must be an associate of some a_i or some b_j. In the former case $p \mid a$, and in the latter $p \mid b$. Hence p is prime. □

Summary. We have established the following strict chain of inclusions among classes of integral domains:

$$\text{Fields} \subsetneq \text{PIDs} \subsetneq \text{UFDs} \subsetneq \text{Integral Domains}.$$

Examples separating these classes:

- $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD, since irreducible does not imply prime (Homework), while irreducible and prime are the same for UFDs;
- $\mathbb{Z}[x]$ is a UFD but not a PID - we have seen it is not a PID, and will prove that it is a UFD in Section 3.12 below.

3.11 Euclidean Domain

We now establish the connection between Euclidean domains and principal ideal domains.

Definition 3.72 (Euclidean Function)**.** Let R be an integral domain. A function

$$N : R \setminus \{0\} \to \mathbb{N} \cup \{0\}$$

is called a **Euclidean function** on R if for all $a, b \in R$ with $b \neq 0$, either

(i) $b \mid a$; or

(ii) there exist $q, r \in R$ with $r \neq 0$ such that $a = bq + r$ and $N(r) < N(b)$.

Example 3.73. We have already seen the following examples of Euclidean functions:

- $R = \mathbb{Z}$, $N(m) := |m|$. Then (ii) is the division algorithm for integers.

- $R = \mathbb{F}[x]$ (polynomials over a field), $N(p(x)) := \deg(p)$. Then (ii) is the division algorithm for polynomials.

Definition 3.74 (Euclidean Domain)**.** An integral domain R is a **Euclidean domain (ED)** if there exists at least one Euclidean function on R.

If a Euclidean function N satisfies $N(ab) = N(a)N(b)$ for all nonzero $a, b \in R$, we say N is **multiplicative**.

Remark 3.75. The Euclidean function $N(a) = |a|$ on $\mathbb{Z}$ is multiplicative. However, the degree function $N(p) = \deg(p)$ on $\mathbb{F}[x]$ is not multiplicative. To make it multiplicative, one can take $N'(p) := 2^{\deg(p)}$ which is still a (multiplicative) Euclidean function on $\mathbb{F}[x]$.

Theorem 3.76. *Let R be an integral domain.*

(a) If R is a field, then R is a Euclidean domain.

(b) If R is a Euclidean domain, then R is a principal ideal domain.

Proof. (a): Let R be a field and define $N : R \setminus \{0\} \to \mathbb{N} \cup \{0\}$ by $N(a) := 1$ for all $a \in R \setminus \{0\}$. For any $a, b \in R$ with $b \neq 0$, since R is a field, b is a unit, so $b \mid a$. Thus condition (i) of the Euclidean function is satisfied, making N a Euclidean function on R.

(b): Let N be a Euclidean function on R, and let I be a nonzero ideal of R. Choose $a \in I \setminus \{0\}$ such that $N(a)$ is minimal among all nonzero elements in I (such an element exists since N takes values in $\mathbb{N} \cup \{0\}$ and I is nonzero). We claim that $I = \langle a \rangle$.

For any $i \in I$, apply the definition of Euclidean function to the pair (i, a). Either $a \mid i$, or there exist $q, r \in R$ with $r \neq 0$ such that $i = aq + r$ with $N(r) < N(a)$.

In the first case, $i \in \langle a \rangle$.

In the second case, we have $r = i - aq$. Since $i, a \in I$ and I is an ideal, we have $r \in I$. Moreover, $r \neq 0$ and $N(r) < N(a)$, contradicting the minimality of $N(a)$. Therefore, this case cannot occur.

Thus for all $i \in I$, we have $a \mid i$, which means $i \in \langle a \rangle$. Therefore $I \subseteq \langle a \rangle$. Since $a \in I$, we also have $\langle a \rangle \subseteq I$. Hence $I = \langle a \rangle$, so R is a principal ideal domain. □

Summary. We have established the following strict chain of inclusions among classes of integral domains:

$$\text{Fields} \subsetneq \text{EDs} \subsetneq \text{PIDs} \subsetneq \text{UFDs} \subsetneq \text{Integral Domains}.$$

It is harder to get an example of a PID but not an ED:

Example 3.77 (PID but not ED)**.** An example of a principal ideal domain that is not a Euclidean domain is more subtle. We present the ring

$$R = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right] = \left\{ a + b\frac{1+\sqrt{-19}}{2} \;\middle|\; a, b \in \mathbb{Z} \right\}.$$

The reason why R is a PID is one of the main content of MAT 5210 – Advanced Abstract Algebra I, based on the fact that the class number of the *number field* $\mathbb{Q}(\sqrt{-19})$ is 1.

As for why R is not a Euclidean domain, it is left as an Homework exercise.

Proposition 3.78. *The Gaussian integers* $\mathbb{Z}[i] := \{a + bi \mid a, b \in \mathbb{Z}\}$ *form a Euclidean domain with Euclidean function*

$$N(a + bi) = a^2 + b^2.$$

Proof. Let $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$. Consider $\frac{\alpha}{\beta} = c_1 + c_2 i$ where $c_1, c_2 \in \mathbb{Q}$.

Let $n_1, n_2 \in \mathbb{Z}$ be integers closest to c_1 and c_2 respectively, so that $|n_j - c_j| \leq \frac{1}{2}$ for $j = 1, 2$. Set $q := n_1 + n_2 i \in \mathbb{Z}[i]$. Then

$$\frac{\alpha}{\beta} - q = (c_1 - n_1) + (c_2 - n_2)i.$$

Multiplying by β gives

$$\alpha - \beta q = [(c_1 - n_1) + (c_2 - n_2)i]\beta.$$

Set $r := \alpha - \beta q$. If $r = 0$, then $\beta \mid \alpha$. Otherwise, $r \neq 0$ and

$$\begin{aligned} N(r) &= N[(c_1 - n_1) + (c_2 - n_2)i] \cdot N(\beta) \\ &= [(c_1 - n_1)^2 + (c_2 - n_2)^2] \cdot N(\beta) \\ &\leq \left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right] N(\beta) \\ &= \frac{1}{2} N(\beta) < N(\beta). \end{aligned}$$

Thus the Euclidean property holds, and $\mathbb{Z}[i]$ is a Euclidean domain. □

Classification of Gaussian Primes. Since $\mathbb{Z}[i]$ is a ED (and hence a PID), irreducible elements are precisely prime elements. We now classify the primes in $\mathbb{Z}[i]$.

Lemma 3.79. *The units in* $\mathbb{Z}[i]$ *are* $U(\mathbb{Z}[i]) = \{\pm 1, \pm i\}$. *An element* $\alpha = a + bi \in \mathbb{Z}[i]$ *is a unit if and only if* $N(\alpha) = 1$.

Lemma 3.80. *If* $\beta \in \mathbb{Z}[i]$ *satisfies* $N(\beta) = p$ *where* p *is a prime integer, then* β *is prime in* $\mathbb{Z}[i]$.

Proof. Suppose $\beta = \gamma\delta$ for some $\gamma, \delta \in \mathbb{Z}[i]$. Then

$$p = N(\beta) = N(\gamma)N(\delta).$$

Since p is prime, either $N(\gamma) = 1$ or $N(\delta) = 1$. Thus γ or δ is a unit, so β is irreducible. Since $\mathbb{Z}[i]$ is a UFD, β is prime. □

Lemma 3.81. *Let* p *be a prime integer in* $\mathbb{Z}$. *Then either*

(i) p *is prime in* $\mathbb{Z}[i]$, *or*

(ii) $p = (a+bi)(a-bi)$ *where* $a+bi$ *and* $a-bi$ *are primes in* $\mathbb{Z}[i]$ *with* $a, b \neq 0$.

Proof. Suppose p is not prime in $\mathbb{Z}[i]$. Then $p = \gamma\delta$ for nonunits $\gamma, \delta \in \mathbb{Z}[i]$. Taking norms,

$$p^2 = N(p) = N(\gamma)N(\delta).$$

Thus $N(\gamma) = N(\delta) = p$. By the previous lemma, both γ and δ are prime in $\mathbb{Z}[i]$.

Since $p \in \mathbb{Z}$ is real, $\bar{p} = p$. If $p = \gamma\delta$, then $p = \overline{\gamma\delta} = \bar{\gamma}\bar{\delta}$. Write $\gamma = a + bi$. Then $\bar{\gamma} = a - bi$, and by uniqueness of factorization in the UFD $\mathbb{Z}[i]$, we have $\bar{\gamma} \sim \delta$ (up to units and reordering). Since $N(\gamma) = N(\bar{\gamma}) = p$, we have $p = \gamma\bar{\gamma} = (a+bi)(a-bi)$. $\square$

Lemma 3.82. *Let* $a+bi \in \mathbb{Z}[i]$ *with* $a, b \neq 0$. *Then* $a+bi$ *is prime in* $\mathbb{Z}[i]$ *if and only if* $a^2+b^2 = p$ *for some prime integer* p.

Proof. ($\Leftarrow$) If $a^2 + b^2 = p$ is prime, then $N(a+bi) = p$, so by the previous lemma, $a+bi$ is prime.

($\Rightarrow$) Suppose $a+bi$ is prime in $\mathbb{Z}[i]$ with $a, b \neq 0$. Then $\overline{a+bi} = a - bi$ is also prime. Consider the factorization of $(a+bi)(a-bi) = a^2+b^2$ into primes in $\mathbb{Z}$. Write $a^2+b^2 = p_1 \cdots p_r$ where each p_i is prime in $\mathbb{Z}$. This gives a factorization in $\mathbb{Z}[i]$ into irreducibles. By uniqueness of factorization, we have $r \leq 2$.

If $r = 2$, then $(a+bi)(a-bi) = p_1p_2$. Both p_1 and p_2 are irreducible (hence prime) in $\mathbb{Z}[i]$. By uniqueness, we have $a+bi \sim p_i$ for some i, meaning $a+bi = u \cdot p_i$ for a unit $u \in \{\pm 1, \pm i\}$. But then $a+bi = \pm p_i$ or $a+bi = \pm ip_i$, contradicting $a, b \neq 0$.

Therefore $r = 1$, so $a^2+b^2 = p_1$ is prime in $\mathbb{Z}$. $\square$

Theorem 3.83. *Let* p *be a prime integer. Then* p *is prime in* $\mathbb{Z}[i]$ *if and only if* $p \equiv 3 \pmod 4$.

Proof. We prove the contrapositive in both directions.

($\Rightarrow$) Suppose $p \not\equiv 3 \pmod 4$. If $p = 2$, then $2 = (1+i)(1-i)$, so 2 is not prime in $\mathbb{Z}[i]$. If p is an odd prime with $p \equiv 1 \pmod 4$, then by Fermat's theorem on sums of two squares (or Lagrange's lemma), there exist integers a, b with $a^2+b^2 = p$. Thus $p = (a+bi)(a-bi)$ with $a, b \neq 0$, so p is not prime in $\mathbb{Z}[i]$.

($\Leftarrow$) Suppose p is not prime in $\mathbb{Z}[i]$. If $p = 2$, then $2 = (1+i)(1-i)$ with $1+i, 1-i$ prime (since $N(1+i) = 2$), so $2 \equiv 2 \pmod 4 \not\equiv 3 \pmod 4$. If p is odd and not prime in $\mathbb{Z}[i]$, then by the previous lemma, $p = a^2+b^2$ for some integers a, b. Now $a^2+b^2 \equiv 0, 1,$ or $2 \pmod 4$ depending on the parities of a and b. Thus $p \not\equiv 3 \pmod 4$. $\square$

Corollary 3.84. *The primes in* $\mathbb{Z}[i]$ *are precisely the elements of the following forms:*

(i) $p \in \mathbb{Z}$ *with* $p \equiv 3 \pmod 4$ *(which remain prime);*

(ii) $1+i$ *and* $1-i$ *(which split the prime* 2*);*

(iii) $a+bi \in \mathbb{Z}[i]$ *with* $a, b \neq 0$ *and* $a^2+b^2 = p$ *where* p *is a prime in* $\mathbb{Z}$ *with* $p \equiv 1 \pmod 4$ *(and their associates).*

Corollary 3.85 (Fermat's Theorem on Sums of Two Squares)**.** *If p is a prime integer with $p \equiv 1 \pmod 4$, then there exist integers a, b such that $p = a^2 + b^2$.*

Proof. If $p \equiv 1 \pmod 4$, then p is not prime in $\mathbb{Z}[i]$ by the previous theorem. Therefore, $p = (a+bi)(a-bi)$ for some $a+bi$ prime in $\mathbb{Z}[i]$ with $a, b \neq 0$. Taking norms, $p^2 = N(a+bi) \cdot N(a-bi) = (a^2+b^2)^2$, so $p = a^2 + b^2$. □

Example 3.86. We have $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $17 = 1^2 + 4^2$, and $29 = 2^2 + 5^2$.

3.12 Polynomial Rings

In this section, we study rings of polynomials $R[x]$. A main reason why polynomial rings are important is the following:

Theorem 3.87. *Let $\mathbb{F}$ be a field and suppose $f(x) \in \mathbb{F}[x]$ is irreducible. Then $\mathbb{F}[x]/\langle f(x)\rangle$ is a field.*

Proof. Since $F[x]$ is a Euclidean domain, it is a principal ideal domain. The ideal $I = \langle f(x)\rangle$ is prime (= maximal) because $f(x)$ is irreducible (= prime). Therefore, $I = \langle f(x)\rangle$ is a maximal ideal, and the result follows from the general theory of maximal ideals. □

As a consequence, there is an injective homomorphism of fields

$$\phi : F \to F[x]/\langle f(x)\rangle =: K$$

defined by $\phi(a) := a + \langle f(x)\rangle$. This allows us to "extend" F to a larger field.

In **Galois Theory**, one tries to understand roots of (irreducible) polynomials $f(x) \in F[x]$. Let $\bar{x} := x + \langle f(x)\rangle \in K$. Then in K, we have

$$f(\bar{x}) = f(x) + \langle f(x)\rangle = 0_K.$$

So $\bar{x}$ is a root of $f(x)$ in K! So the above construction gives a way to construct roots of any (irreducible) polynomials.

Under such perspective, it is important to determine when a polynomial is irreducible. In this section, we present methods for checking whether $f(x) \in \mathbb{Q}[x]$ (or $\mathbb{Z}[x]$) is irreducible. Although $\mathbb{Z}$ is **not** a field, it is useful for understanding irreducibility of polynomials $f(x) \in \mathbb{Q}[x]$ with integer coefficients.

Lemma 3.88. *Let $\phi : R \to S$ be a unital ring homomorphism (e.g., $\mathbb{Z} \hookrightarrow \mathbb{Q}$ or $\mathbb{Z} \twoheadrightarrow \mathbb{Z}_p$). Then the map $\tilde{\phi} : R[x] \to S[x]$ defined by*

$$\tilde{\phi}(a_0 + a_1 x + \cdots + a_n x^n) := \phi(a_0) + \phi(a_1)x + \cdots + \phi(a_n)x^n$$

is a unital ring homomorphism.

Theorem 3.89 (Reduction Modulo Prime Test)**.** *Let $f \in \mathbb{Z}[x]$ be monic and let p be a prime integer. If $\tilde{\phi}(f) \in \mathbb{Z}_p[x]$ is irreducible, then f is irreducible in $\mathbb{Z}[x]$.*

Proof. Suppose $f = gh$ for some $g, h \in \mathbb{Z}[x]$. We must show that g or h is a unit.

Applying $\tilde{\phi}$, we have $\tilde{f} = \tilde{g} \cdot \tilde{h}$ in $\mathbb{Z}_p[x]$.

Since f is monic, both g and h must be monic as well (up to units ± 1). Therefore, $\tilde{f}$, $\tilde{g}$, and $\tilde{h}$ are all monic with $\deg(\tilde{g}) = \deg(g)$ and $\deg(\tilde{h}) = \deg(h)$.

Since $\tilde{f}$ is irreducible, either $\deg(\tilde{g}) = 0$ or $\deg(\tilde{h}) = 0$. Therefore, either $\deg(g) = 0$ or $\deg(h) = 0$, which means g or h is a constant polynomial. Since f is monic with degree greater than 0, this constant must be ± 1, making g or h a unit in $\mathbb{Z}[x]$. Thus f is irreducible. □

Example 3.90. Consider $f(x) = x^3 + 3x + 7$ and reduce modulo $p = 2$. We get $\tilde{f}(x) = x^3 + x + 1 \in \mathbb{Z}_2[x]$.

To show $\tilde{f}$ is irreducible in $\mathbb{Z}_2[x]$, note that if $\tilde{f}$ factors, then it must have a linear factor, and hence either $\tilde{f}(0) = 0$ or $\tilde{f}(1) = 0$ in $\mathbb{Z}_2$ (by the factor theorem). We compute:

$$\tilde{f}(0) = 1 \neq 0 \pmod 2, \quad \tilde{f}(1) = 1 + 1 + 1 = 1 \neq 0 \pmod 2.$$

Therefore, $\tilde{f}$ has no linear factors in $\mathbb{Z}_2[x]$, so it is irreducible. Thus $f(x) = x^3 + 3x + 7$ is irreducible in $\mathbb{Z}[x]$.

Theorem 3.91 (Eisenstein's Criterion)**.** *Let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ and suppose p is a prime integer such that:*

(1) $p \mid a_i$ for all $0 \leq i < n$;

(2) $p \nmid a_n$;

(3) $p^2 \nmid a_0$.

Moreover, suppose $\gcd(a_n, \ldots, a_0) = 1$ *(i.e., f is primitive). Then $f(x)$ is irreducible in $\mathbb{Z}[x]$.*

Proof. Suppose $f = gh$ for some $g, h \in \mathbb{Z}[x]$. We must show that g or h is a unit.

Consider the reduction map $\tilde{\phi} : \mathbb{Z}[x] \to \mathbb{Z}_p[x]$. By condition (1), we have

$$\tilde{f}(x) = \overline{a_n} x^n$$

where $\overline{a_n} \neq 0$ in $\mathbb{Z}_p$ by condition (2).

From $\tilde{f} = \tilde{g} \cdot \tilde{h}$ and $\tilde{f} \sim x^n$ in $\mathbb{Z}_p[x]$ (up to nonzero scalar multiple), we must have $\tilde{g} \sim x^i$ and $\tilde{h} \sim x^{n-i}$ for some $0 \leq i \leq n$.

Now suppose $0 < i < n$. Then the constant term of $\tilde{g}$ is 0 in $\mathbb{Z}_p$, which means p divides the constant term of g. Similarly, p divides the constant term of h. Therefore, p^2 divides the constant term of $gh = f$, contradicting condition (3).

Thus either $i = 0$ or $i = n$, which means either $\tilde{g}$ or $\tilde{h}$ is a constant in $\mathbb{Z}_p[x]$.

Since $\deg(g) + \deg(h) = \deg(f) = \deg(\tilde{f}) = \deg(\tilde{g}) + \deg(\tilde{h})$ and $\deg(\tilde{g}) \leq \deg(g)$, $\deg(\tilde{h}) \leq \deg(h)$, we have $\deg(\tilde{g}) = \deg(g)$ and $\deg(\tilde{h}) = \deg(h)$.

Therefore, either $\deg(g) = 0$ or $\deg(h) = 0$, so g or h is a constant polynomial. Since f is primitive, this constant must be ± 1, making g or h a unit. Thus f is irreducible. □

Example 3.92. Consider $g(x) = x^4 + 1$. Define $f(x) = g(x+1) = (x+1)^4 + 1$. Expanding:

$$f(x) = x^4 + 4x^3 + 6x^2 + 4x + 2.$$

Taking $p = 2$, we verify Eisenstein's criterion:

1. $2 \mid 4, 6, 4, 2$ (all non-leading coefficients);
2. $2 \nmid 1$ (the leading coefficient);
3. $2^2 = 4 \nmid 2$ (the constant term).

Therefore, $f(x)$ is irreducible in $\mathbb{Z}[x]$, and hence $g(x) = x^4 + 1$ is also irreducible in $\mathbb{Z}[x]$.

Definition 3.93 (Primitive Polynomial)**.** A polynomial $f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ is called **primitive** if $\gcd(a_n, \ldots, a_0) = 1$.

Now we address irreducibility in $\mathbb{Q}[x]$.

Theorem 3.94 (Gauss' Lemma)**.** *A nonconstant polynomial $f \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ if and only if f is primitive and irreducible in $\mathbb{Q}[x]$.*

Proof. ($\Leftarrow$) Suppose $f \in \mathbb{Z}[x]$ is primitive, irreducible in $\mathbb{Q}[x]$, and $f = gh$ for some $g, h \in \mathbb{Z}[x]$. We must show that g or h is a unit in $\mathbb{Z}[x]$.

Since $g, h \in \mathbb{Z}[x] \subseteq \mathbb{Q}[x]$ and f is irreducible in $\mathbb{Q}[x]$, either g or h must be a unit in $\mathbb{Q}[x]$, i.e., a nonzero constant. Say $g = c \in \mathbb{Q}$.

Since $g \in \mathbb{Z}[x]$, we have $c \in \mathbb{Z}$. Moreover, since f is primitive, gcd(coefficients of f) $= 1$. If $c \neq \pm 1$, then c divides all coefficients of $f = gh = c \cdot h$, contradicting primitivity of f. Therefore, $c = \pm 1$, so g is a unit in $\mathbb{Z}[x]$.

($\Rightarrow$) Suppose f is irreducible in $\mathbb{Z}[x]$.

First, we show f is primitive. If not, then gcd(coefficients of f) $= d \geq 2$. We can write $f = d \cdot f'$ where $f' \in \mathbb{Z}[x]$ and $\deg(f') = \deg(f) > 0$. Since neither d nor f' is a unit in $\mathbb{Z}[x]$, this contradicts irreducibility of f. Thus f is primitive.

Next, we show f is irreducible in $\mathbb{Q}[x]$. Suppose $f = gh$ for some $g, h \in \mathbb{Q}[x]$. We must show that g or h is a unit in $\mathbb{Q}[x]$.

Let $\lambda, \mu \in \mathbb{N}$ be the smallest positive integers such that λg and μh have integer coefficients. Define $g' := \lambda g \in \mathbb{Z}[x]$ and $h' := \mu h \in \mathbb{Z}[x]$. Then

$$f = gh = \frac{g'}{\lambda} \cdot \frac{h'}{\mu},$$

so $\lambda \mu f = g'h'$.

Claim: $\lambda = 1$.

Proof of claim: Suppose for contradiction that $\lambda > 1$, so there exists a prime $p \mid \lambda$. Then $p \mid \lambda\mu f$, so all coefficients of $\lambda\mu f \in \mathbb{Z}[x]$ are divisible by p. In particular, $p \mid g'h'$ in $\mathbb{Z}[x]$.

Consider the reduction map $\tilde{\phi} : \mathbb{Z}[x] \to \mathbb{Z}_p[x]$. We have $\tilde{g'} \cdot \tilde{h'} \equiv 0$ in $\mathbb{Z}_p[x]$. Since $\mathbb{Z}_p[x]$ is an integral domain, either $\tilde{g'} \equiv 0$ or $\tilde{h'} \equiv 0$ in $\mathbb{Z}_p[x]$. This means either $p \mid g'$ or $p \mid h'$.

If $p \mid g'$, then all coefficients of $g' = \lambda g$ are divisible by p. But then $(\lambda/p)g$ has integer coefficients and is a scalar multiple of g, contradicting the minimality of λ.

Similarly, if $p \mid h'$, then $(\mu/p)h$ has integer coefficients, contradicting the minimality of μ.

Therefore, $\lambda = 1$, and $g' = g \in \mathbb{Z}[x]$ and $h' = \mu h \in \mathbb{Z}[x]$.

By the same argument applied to μ, we also have $\mu = 1$, so $h \in \mathbb{Z}[x]$.

Now $f = gh$ with $g, h \in \mathbb{Z}[x]$. Since f is irreducible in $\mathbb{Z}[x]$, either g or h is a unit in $\mathbb{Z}[x]$, which means either g or h is $\pm 1 \in \mathbb{Q}[x]$. Thus f is irreducible in $\mathbb{Q}[x]$. $\square$

Theorem 3.95. *$\mathbb{Z}[x]$ is a unique factorization domain, though not a principal ideal domain (by homework).*

Proof. First, we show that every primitive polynomial $f \in \mathbb{Z}[x]$ with $\deg(f) \geq 1$ can be factored into primitive irreducible polynomials. We proceed by induction on $\deg(f)$.
Base case: If $\deg(f) = 1$, then f is irreducible.
Inductive step: Assume the claim holds for all primitive polynomials of degree less than k. Let f be primitive with $\deg(f) = k > 1$. Either:

- f is irreducible; or
- $f = h_1 h_2$ for some $h_1, h_2 \in \mathbb{Z}[x]$ with $1 \leq \deg(h_i) < k$.

In the second case, since f is primitive, h_1 and h_2 must also be primitive (otherwise their gcd would divide f's coefficients, contradicting primitivity). By induction, both h_1 and h_2 factor into primitive irreducibles, so f does as well.

Now, to show factorization is unique, suppose $f = g_1 \cdots g_n = h_1 \cdots h_m$ where $g_i, h_j \in \mathbb{Z}[x]$ are primitive and irreducible.

By Gauss' Lemma, g_i and h_j are also irreducible in $\mathbb{Q}[x]$. Since $\mathbb{Q}[x]$ is a principal ideal domain (hence a UFD), we have $n = m$ and there exists a permutation $\sigma \in S_n$ such that g_i and $h_{\sigma(i)}$ are associates in $\mathbb{Q}[x]$. That is, $g_i = c_i h_{\sigma(i)}$ for some nonzero $c_i \in \mathbb{Q}$.

Since $g_i, h_{\sigma(i)} \in \mathbb{Z}[x]$ and both are primitive, we must have $c_i \in \{\pm 1\}$ (a rational number that makes integer polynomials remain primitive). Therefore, $g_i \sim h_{\sigma(i)}$ in $\mathbb{Z}[x]$.

Thus factorization into primitive irreducibles is unique up to order and associates.

For a general polynomial $f \in \mathbb{Z}[x]$, write $f = d \cdot f'$ where $d \in \mathbb{Z}$ and f' is primitive. Factor d uniquely in $\mathbb{Z}$ and factor f' uniquely into primitive irreducibles in $\mathbb{Z}[x]$. The factorization of f is therefore unique, making $\mathbb{Z}[x]$ a UFD. $\square$

Remark 3.96. More generally, if R is a unique factorization domain, then $R[x]$ is also a unique factorization domain. The proof uses the field of fractions $F = \mathrm{Frac}(R)$ and the fact that $F[x]$ is a principal ideal domain (hence a UFD).

Chapter 4

Fields

Recall: F is a field if F is an integral domain (unital) and for all $x \in F \setminus \{0\}$, there exists a multiplicative inverse $x^{-1} \in F \setminus \{0\}$.

In general, there are 3 kinds of fields:

1. **Number field** (a subfield of $\mathbb{C}$)
   (e.g., $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Q}(i)$, $\ldots$)
2. **Finite field** (a field F with $|F| < \infty$)
   (e.g., $\mathbb{Z}_p = \mathbb{F}_p$ for p prime. $\mathbb{F}_2[x]/\langle x^3 + x + 1\rangle$ has $8 = 2^3$ elements.)
3. **Function fields**
   (e.g., $\mathbb{C}(x) = \mathrm{Frac}(\mathbb{C}[x]) = \{f(x)/g(x) \mid f(x), g(x) \in \mathbb{C}[x], g(x) \neq 0\}$)

We'll only focus on (1) or (2).

4.1 Field Extension and Degree

Motivation: Given a field F, construct a "bigger" field K that "contains" F.

For example, let $F = \mathbb{F}_2$ and $K = \mathbb{F}_2[x]/\langle x^3 + x + 1\rangle$. Then we have an injective unital ring (or field) homomorphism $i : F \to K$ where $i(0) = 0 + \langle x^3 + x + 1\rangle$ and $i(1) = 1 + \langle x^3 + x + 1\rangle$. So we say K "contains" F.

Definition 4.1. Let F and K be fields. We say F is a **subfield** of K or K is a **field extension** of F if there is an injective unital field homomorphism $i : F \to K$.

In this case, K is a vector space over F. There is a scalar multiplication map $\cdot : F \times K \to K$ defined by $\alpha \cdot x = i(\alpha)x$ for $\alpha \in F, x \in K$, that satisfies:

- $0 \cdot x = 0$
- $1 \cdot x = x$
- $\alpha \cdot (x + y) = (\alpha \cdot x) + (\alpha \cdot y)$
- $\ldots$

The **degree** of extension is defined as:

$$[K : F] := \dim_F(K) \quad \text{(as a vector space over } F\text{)}$$

Example 4.2. 1. $F = \mathbb{R}$, $K = \mathbb{C}$, with the inclusion $i : \mathbb{R} \hookrightarrow \mathbb{C}$.
Then $[\mathbb{C} : \mathbb{R}] = 2$, since $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\} = \text{span}_{\mathbb{R}}\{1, i\}$.
Thus, $\dim_{\mathbb{R}}(\mathbb{C}) = 2$ (since $\{1, i\}$ is linearly independent over $\mathbb{R}$).

2. $F = \mathbb{Q}$, $K = \mathbb{R}$, with the inclusion $i : \mathbb{Q} \hookrightarrow \mathbb{R}$.
To find $[\mathbb{R} : \mathbb{Q}]$, consider that $\mathbb{R} = \text{span}_{\mathbb{Q}}\{1, \sqrt{2}, \sqrt{3}, e, \pi, \dots\}$.
The set of transcendental numbers like e, π is infinite and they are linearly independent over $\mathbb{Q}$. Thus, $[\mathbb{R} : \mathbb{Q}] = \infty$.

4.2 Splitting Extension

In field theory, we want to understand roots of polynomials $p(x) \in F[x]$. More precisely, we would like to construct $E : F$ such that E contains some (or all) roots of $F[x]$. The first step is the following:

Theorem 4.3 (Kronecker)**.** *Let F be a field, and $p(x) \in F[x]$ be an irreducible polynomial of degree m. Then $K = F[x]/\langle p(x)\rangle$ is a field extension of F such that $[K : F] = m$. Moreover, there exists an element $\alpha \in K$ such that:*

- $K = F[\alpha] := \{a_n\alpha^n + \cdots + a_1\alpha + a_0 \mid a_i \in F, n \in \mathbb{N}\}$.
- $p(\alpha) = 0$ *in* K.

Proof. Since $p(x) = b_m x^m + \cdots + b_1 x + b_0 \in F[x]$ is irreducible, the ideal $\langle p(x)\rangle$ is a maximal ideal in $F[x]$. This implies that $K = F[x]/\langle p(x)\rangle$ is a field.

Let's study the dimension, $\dim_F(K)$. Consider the element $\alpha := x + \langle p(x)\rangle \in K$. Then in K,

$$\begin{aligned}
p(\alpha) &= b_m\alpha^m + \cdots + b_1\alpha + b_0 \\
&= b_m(x + \langle p(x)\rangle)^m + \cdots + b_1(x + \langle p(x)\rangle) + b_0(1 + \langle p(x)\rangle) \\
&= (b_m x^m + \cdots + b_1 x + b_0) + \langle p(x)\rangle \\
&= p(x) + \langle p(x)\rangle = 0 + \langle p(x)\rangle = 0_K
\end{aligned}$$

So, the second part of the theorem is proved. Moreover, this also implies that

$$\alpha^m = -\frac{1}{b_m}(b_{m-1}\alpha^{m-1} + \cdots + b_0)$$

i.e. the set $\{1, \alpha, \dots, \alpha^m\}$ is linearly dependent in K. More precisely, α^m is a linear combination of $\{1, \alpha, \dots, \alpha^{m-1}\}$. Similarly, for any $n \geq m$, α^n is also a linear combination of $\{1, \alpha, \dots, \alpha^{m-1}\}$.

Claim: The set $\{1, \alpha, \dots, \alpha^{m-1}\}$ is a basis of K over F.

Spanning set: Take any element $k \in K$. By definition, $k = (c_l x^l + \cdots + c_1 x + c_0) + \langle p(x) \rangle$ for some $c_i \in F$. This is equal to $c_l \alpha^l + \cdots + c_1 \alpha + c_0$. Since all powers α^j for $j \geq m$ can be reduced to a linear combination of $\{1, \alpha, \dots, \alpha^{m-1}\}$, any element $k \in K$ can be written as a linear combination of these basis elements. Thus, $K = \mathrm{Span}_F\{1, \alpha, \dots, \alpha^{m-1}\}$.

Linearly independent: Suppose $d_{m-1}\alpha^{m-1} + \cdots + d_1\alpha + d_0 = 0_K$ for some $d_i \in F$, not all zero. This is equivalent to $(d_{m-1}x^{m-1} + \cdots + d_1 x + d_0) + \langle p(x) \rangle = 0_K$. Let $g(x) = d_{m-1}x^{m-1} + \cdots + d_0$. The equation means $g(x) \in \langle p(x) \rangle$. This implies that $p(x)$ divides $g(x)$, i.e., $g(x) = p(x) \cdot r(x)$ for some $r(x) \in F[x]$. But this is a contradiction, because $\deg(g(x)) \leq m - 1$ while $\deg(p(x)) = m$. The degree of a non-zero polynomial $g(x)$ cannot be less than the degree of a non-zero polynomial $p(x)$ that divides it. Thus, $g(x)$ must be the zero polynomial, which means all d_i must be zero. The set is linearly independent.

Since $\{1, \alpha, \dots, \alpha^{m-1}\}$ is a basis for K over F, the dimension is m. Therefore, $[K : F] = m$. $\square$

Example 4.4. Let $F = \mathbb{Q}$, $p(x) = x^3 - 2 \in \mathbb{Q}[x]$ (irreducible by Eisenstein's criterion). Then $[K : \mathbb{Q}] = 3$ for $K = \mathbb{Q}[x]/\langle x^3 - 2 \rangle$. Let $\alpha = x + \langle x^3 - 2 \rangle$. Then $K = \mathrm{Span}_{\mathbb{Q}}\{1, \alpha, \alpha^2\} = \{a_2\alpha^2 + a_1\alpha + a_0 \mid a_i \in \mathbb{Q}\}$. We know $\alpha^3 - 2 = 0$, so $\alpha^3 = 2$. K is a field that contains $\mathbb{Q}$ and a cube root of 2. We also know how to do arithmetic in K, for instance:

$$\begin{aligned}
(2\alpha^2 + \tfrac{3}{4}) \cdot (\alpha - 6) &= 2\alpha^3 - 12\alpha^2 + \tfrac{3}{4}\alpha - \tfrac{18}{4} \\
&= 2(2) - 12\alpha^2 + \tfrac{3}{4}\alpha - \tfrac{9}{2} \\
&= 4 - 12\alpha^2 + \tfrac{3}{4}\alpha - \tfrac{9}{2} \\
&= -12\alpha^2 + \tfrac{3}{4}\alpha - \tfrac{1}{2}
\end{aligned}$$

Exercise: What is $\alpha^{-1} \in K$?

In the above example, under the extension $K : \mathbb{Q}$, one can treat the polynomial $p(x) = x^3 - 2 \in \mathbb{Q}[x] \subseteq K[x]$ under the extended field. By Kronecker's theorem, $p(x) \in K[x]$ has a root:

$$p(x) = (x - \alpha)(x^2 + \alpha x + \alpha^2) \in K[x]$$

However, K does not contain all the roots of $p(x)$.

Definition 4.5. Let F be a field, and $p(x) \in F[x]$. We say a field extension $E : F$ splits $p(x)$ if $p(x) = a(x - a_1) \cdots (x - a_n)$.

Proposition 4.6. *For any field F and any polynomial $p(x) \in F[x]$, there exists a field extension $E : F$ such that E splits $p(x)$.*

Proof. Prove by induction on the degree of $p(x)$ (for any base field F). If $p(x) \in F$ is of degree 1, we are done. By induction hypothesis, assume the theorem holds for all polynomials of degree $< k$ over any field. Consider a polynomial $p(x)$ of degree k. Let $f(x)$ be an irreducible factor of $p(x)$. Then take $K = F[x]/\langle f(x) \rangle$ so that $a := x + \langle f(x) \rangle \in K$ is a root of $f(x)$ and thus of $p(x)$. So

$p(x) = (x-a)g(x)$ in $K[x]$, where $\deg(g(x)) = k-1$. By the induction hypothesis, there exists an extension $E : K$ such that $g(x)$ splits in $E[x]$, say $g(x) = c(x-a_2)...(x-a_k)$. So we have $p(x) = c(x-a)(x-a_2)...(x-a_k)$ in $E[x]$, and we are done. □

Now we know for any $p(x) \in F[x]$, there is a field extension $E : F$ such that E contains all the roots of $p(x)$. We want to find 'the smallest' extension that contains all the roots of $p(x)$.

Definition 4.7. Let $E : F$ be a field extension, and $e_1, \ldots, e_n \in E$. Write $F(e_1, \ldots, e_n)$ for the smallest field extension of F containing $e_1, \ldots, e_n$. In other words,

$$F(e_1, \ldots, e_n) := \bigcap_{K \leq E,\ e_1, \ldots, e_n \in K} K$$

Proposition 4.8. $F(e_1, e_2, \ldots, e_n) = (F(e_1))(e_2, \ldots, e_n) = \cdots = ((F(e_1)(e_2)) \ldots (e_{n-1}))(e_n)$.

Proof. Easy. □

Definition 4.9. Suppose $E : F$ is a field extension such that E splits a polynomial $p(x) \in F[x] \subseteq E[x]$ with roots $a_1, \ldots, a_n \in E$. Then a **splitting field** of $p(x)$ over F is the smallest subfield $F(a_1, \ldots, a_n)$ of E containing all the roots a_i of $p(x)$ in E.

Example 4.10.

1. Let $p(x) = x^2 + 3x + 3 \in \mathbb{Q}[x]$. Then for $E = \mathbb{Q}[x]/\langle p(x) \rangle$, the polynomial $p(x)$ splits automatically into $(x-a)(x-b)$. So $\mathbb{Q}(a, b) = \mathbb{Q}[x]/\langle p(x) \rangle$ is a splitting field.

   On the other hand, $p(x)$ splits in $\mathbb{C}$ and the roots are $-3/2 \pm i\sqrt{3}/2$. So $\mathbb{Q}(i\sqrt{3})$ is also a splitting field of $p(x)$.

2. Let $p(x) = x^3 - 2 \in \mathbb{Q}[x]$. Then $F = \mathbb{Q}[x]/\langle x^3 - 2 \rangle$ has a root $\alpha \in E$. We can form $E = F[x]/\langle x^2 + \alpha x + \alpha^2 \rangle$. The field E is a splitting field, which we can write as $\mathbb{Q}(a, b, c)$ where a, b, c are the roots.

   On the other hand, working in $\mathbb{C}$, the roots are $2^{1/3}, 2^{1/3}\omega, 2^{1/3}\omega^2$ (where $\omega = e^{i2\pi/3}$). Then $\mathbb{Q}(2^{1/3}, 2^{1/3}\omega, 2^{1/3}\omega^2) = \mathbb{Q}(2^{1/3}, \omega)$ is also a splitting field of $p(x)$.

Theorem 4.11. *(will prove in MAT5210) All splitting fields of the same polynomial are isomorphic.*

4.3 Simple Extension

In the previous section, we begin with a irreducible polynomial $p(x) \in F[x]$, and extend to E so that E contains some (or all) roots of $p(x)$. In this section, we change our perspective a little bit – we begin with a field extension $E : F$, and see which elements in E are roots of a polynomial $p(x) \in F[x]$. Such $E : F$ always exists by the virtue of the previous section.

Definition 4.12. Let F be a field and L be a field extension of F (e.g., $F = \mathbb{Q} \subseteq L = \mathbb{C}$). An element $\alpha \in L$ is

- **algebraic** if there exists a non-zero polynomial $p(x) \in F[x]$ such that $p(\alpha) = 0$ (in L).
- Otherwise, α is called **transcendental**.

Example 4.13 (Elements over $\mathbb{Q}$)**.**

- $i = \sqrt{-1}$ is algebraic over $\mathbb{Q}$ (its polynomial is $p(x) = x^2 + 1$).
- $\sqrt[4]{2}, \sqrt{3}$ are algebraic over $\mathbb{Q}$ (their polynomials are $p(x) = x^4 - 2$ and $x^2 - 3$, respectively).
- $\sqrt{2} + \sqrt{3}$ is algebraic over $\mathbb{Q}$ (HW12).
- e, π are transcendental over $\mathbb{Q}$ (there is NO polynomial with $\mathbb{Q}$-coefficients having e or π as a root).

Theorem 4.14. *If α is transcendental over F, then $F[\alpha] \cong F[x]$ as rings (but not as fields).*

As for β algebraic:

Proposition 4.15. *Let $\beta \in L$ be algebraic over F. Consider the set $I = \{f(x) \in F[x] \mid f(\beta) = 0\}$. Then:*

1. *I is a non-zero ideal in $F[x]$.*
2. *$I = \langle p(x) \rangle$ for some unique monic polynomial $p(x) \in F[x]$. (This is because $F[x]$ is a PID).*
3. *$p(x)$ is the unique monic polynomial of smallest degree in I.*
4. *$p(x)$ is irreducible.*

*This polynomial $p(x)$ is called the **minimal polynomial** of β over F.*

Proof. (a) Let's show $p(x)$ has the smallest degree. By the division algorithm, suppose on the contrary there exists $g(x) \in I$ with $\deg(g) < \deg(p)$. Since $I = \langle p(x) \rangle$, $g(x)$ must be a multiple of $p(x)$, which is impossible unless $g(x) = 0$. Contradiction.

(b) Let's show $p(x)$ is irreducible. Suppose on the contrary that $p(x) = h_1(x)h_2(x)$ with $\deg(h_1) > 0$ and $\deg(h_2) > 0$. Then $p(\beta) = h_1(\beta)h_2(\beta) = 0$ in the field L. This implies that either $h_1(\beta) = 0$ or $h_2(\beta) = 0$. This means either $h_1(x) \in I$ or $h_2(x) \in I$. But $\deg(h_1) < \deg(p)$ and $\deg(h_2) < \deg(p)$, which contradicts the fact that $p(x)$ is a polynomial of smallest degree in I. Thus, $p(x)$ must be irreducible. $\square$

Definition 4.16. Let F be a field, $\beta \in L$ (a field extension of F). The polynomial $m_\beta(x) := p(x)$ appearing in Proposition 4.15 above is called the **minimal polynomial** of β over F.

Example 4.17. Minimal polynomials over $\mathbb{Q}$:

- For $\beta = i$, the minimal polynomial is $m_\beta(x) = x^2 + 1$.
- For $\beta = \sqrt[4]{2}$, the minimal polynomial is $m_\beta(x) = x^4 - 2$.
- For $\beta = e^{2\pi i/5}$, the minimal polynomial is $m_\beta(x) = x^4 + x^3 + x^2 + x + 1$.
  (Since $\beta^5 = 1$, and $x^5 - 1 = (x - 1)p(x)$).

Theorem 4.18. *Let $L : F$ be a field extension, and $\beta \in L$ be algebraic. Then the smallest field extension $F(\beta) : F$ containing β is equal to the polynomial ring $F[\beta]$, which is isomorphic to $F[x]/\langle m_\beta(x)\rangle$.*

Proof. Consider the ring homomorphism $\phi : F[x] \to F[\beta]$ defined by evaluation at β:

$$\phi(a_n x^n + \cdots + a_0) = a_n \beta^n + \cdots + a_0$$

The kernel of this map is precisely the ideal $I = \{f(x) \in F[x] \mid f(\beta) = 0\}$, which we know is equal to $\langle m_\beta(x)\rangle$. So the result follows by the First Isomorphism Theorem for rings. Since $\langle m_\beta(x)\rangle$ is a maximal ideal, $F[x]/\langle m_\beta(x)\rangle$ is a field, and therefore $F[\beta]$ is also a field. $\square$

Definition 4.19. The field extension $F(\beta) := F[\beta]$ of F is called the **simple extension** of F by β.

Corollary 4.20. *If $\alpha, \beta \in L$ are algebraic over F and are roots of the same irreducible polynomial $p(x) \in F[x]$, then $F(\alpha) \cong F(\beta)$.*

Proof. $m_\alpha(x) = m_\beta(x) = p(x)$. So $F(\alpha) \cong F[x]/\langle p(x)\rangle \cong F(\beta)$. $\square$

Let $E : F$ be a field extension, and $\alpha_1, \ldots, \alpha_k \in E$ be algebraic over F. Then the field extension $F(\alpha_1, \ldots, \alpha_k)$ can be constructed inductively as follows: Define a sequence of fields by $F_0 = F$ and

$$F_{i+1} := F_i(\alpha_{i+1}) \cong \frac{F_i[x]}{\langle m^{F_i}_{\alpha_{i+1}}(x)\rangle} \quad \text{for } 0 \le i \le k-1.$$

where $m^{F_i}_{\alpha_{i+1}}(x) \in F_i[x]$ is the minimal polynomial of the element α_{i+1} over the field F_i. Then $F_k = F(\alpha_1, \ldots, \alpha_k)$.

4.4 Algebraic Extension

We have seen that a simple extension $F(\beta) : F$ contains an algebraic element β. What about the other elements in $F(\beta) = \mathrm{Span}_F\{1, \beta, \ldots, \beta^{\deg(m_\beta(x))-1}\}$? Are they algebraic? More generally, how about the non-simple extension $F(\beta_1, \ldots, \beta_k)$?

Definition 4.21. An extension $E : F$ is called **algebraic** if every element $\alpha \in E$ is algebraic over F.

Theorem 4.22. *If $[E : F] < \infty$ (i.e., it is a finite extension), then E is an algebraic extension of F.*

Proof. Suppose $[E : F] = n$. Then for any $\alpha \in E$, consider the set $\{1, \alpha, \alpha^2, \ldots, \alpha^n\}$. This is a set of $n+1$ elements in an n-dimensional vector space, so it must be linearly dependent over F. This implies there exist scalars $c_0, \ldots, c_n \in F$, not all zero, such that $c_0 \cdot 1 + c_1\alpha + \cdots + c_n\alpha^n = 0$. Thus, α is a root of the non-zero polynomial $p(x) = c_n x^n + \cdots + c_0 \in F[x]$, which means α is algebraic over F. Since α was arbitrary, the extension is algebraic. $\square$

Corollary 4.23. *Let $E : F$ be a field extension, and $\beta_1, \ldots, \beta_k \in E$ are algebraic over F. Then the extension $F(\beta_1, \ldots, \beta_k) : F$ is algebraic.*

Proof. For the special case when $k = 1$, one has $[F(\beta) : F] = \deg(m_\beta(x))$ which is finite. The proof of the general case is given in Corollary 4.27 below. □

Theorem 4.24 (Primitive Element Theorem)**.** *Let F be a field with characteristic $char(F) = 0$. Let $a, b \in K$ be algebraic over F in some extension $K : F$. Then there exists an element $c \in F(a, b)$ such that $F(c) = F(a, b)$. (Such an element c is called a primitive element for the extension).*

Proof. Let $p(x)$ be the minimal polynomial of a over F. Let $q(x)$ be the minimal polynomial of b over F.

Take a field extension $L : K$ such that $p(x)$ and $q(x)$ split completely in $L[x]$. Suppose $a_1, \ldots, a_m \in L$ are the roots of $p(x)$, with $a_1 = a$. Suppose $b_1, \ldots, b_n \in L$ are the roots of $q(x)$, with $b_1 = b$.

Since $\text{char}(F) = 0$, the field F is infinite. This allows us to choose an element $d \in F$ such that

$$d \neq \frac{a_i - a_j}{b_k - b_l}$$

for all i, j and for all $k \neq l$. In particular, we choose $d \in F$ such that for any i and any $j > 1$:

$$d \neq \frac{a_i - a}{b - b_j}$$

(Note that the elements on the right-hand side may not be in F, but there are only a finite number of such values, so we can always find a $d \in F$ that avoids them).

Consider the element $c := a + db$. Then, it is clear that $F(c) \subseteq F(a, b)$ since $a, b \in F(a, b)$ and $d \in F \subseteq F(a, b)$.

To show the other inclusion, $F(a, b) \subseteq F(c)$, it is enough to show that $b \in F(c)$, because if $b \in F(c)$, then $a = c - db \in F(c)$ as well.

Consider the polynomials $r(x) = p(c - dx)$ and $q(x)$, both viewed as polynomials in $F(c)[x]$. We evaluate both polynomials at b:

- $r(b) = p(c - db) = p(a) = 0$.
- $q(b) = 0$.

Let $m(x) \in F(c)[x]$ be the minimal polynomial of b over the field $F(c)$. Since b is a root of both $r(x)$ and $q(x)$ (which are in $F(c)[x]$), it must be that $m(x)$ divides both $r(x)$ and $q(x)$ in $F(c)[x]$.

$$m(x) \mid r(x) \quad \text{and} \quad m(x) \mid q(x)$$

Now let's study the polynomial $m(x) \in F(c)[x] \subseteq L[x]$.

Since $m(x) \mid q(x)$ and $q(x)$ splits in $L[x]$, the roots of $m(x)$ must be a subset of the roots of $q(x)$, which are $\{b = b_1, b_2, \ldots, b_n\}$.

Consider any root b_j of $m(x)$. Since $m(x)$ also divides $r(x)$, this b_j must also be a root of $r(x)$. So, for such a b_j, we must have $r(b_j) = p(c - db_j) = 0$. This implies that $c - db_j$ must be one of the roots of $p(x)$, say a_i.

$$c - db_j = a_i$$

Substituting $c = a + db$:

$$a + db - db_j = a_i$$

$$d(b - b_j) = a_i - a$$

If $j \neq 1$, then $b \neq b_j$. We can write:

$$d = \frac{a_i - a}{b - b_j}$$

But this contradicts our initial choice of d, which was chosen specifically to not be equal to any of these values. Therefore, the only possibility is that $j = 1$, which means $b_j = b_1 = b$, and hence

$$m(x) = x - b$$

in $F(c)[x]$, i.e. $b \in F(c)$ and the theorem is proved. □

4.5 Tower Law

Theorem 4.25 (Tower Law)**.** *Let $F \subseteq K \subseteq L$ be field extensions. (e.g., $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$). Then:*

$$[L : F] = [L : K][K : F]$$

Proof. Let $\{e_1, \ldots, e_n\}$ be a basis of K over F. This means $[K : F] = n$. Let $\{f_1, \ldots, f_m\}$ be a basis of L over K. This means $[L : K] = m$.

Claim: The set $\mathcal{B} = \{e_i f_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$ is a basis of L over F. If this claim is true, then the dimension of L over F is $n \cdot m$, which proves the theorem. We need to show that $\mathcal{B}$ is a spanning set and is linearly independent over F.

Spanning set: Let x be any element in L. Since $\{f_j\}$ is a basis of L over K, we can write x as a linear combination:

$$x = \sum_{j=1}^{m} \mu_j f_j, \quad \text{where each } \mu_j \in K$$

Now, for each coefficient $\mu_j \in K$, we can express it as a linear combination of the basis elements of K over F. For each j, we have:

$$\mu_j = \sum_{i=1}^{n} \nu_{ij} e_i, \quad \text{where each } \nu_{ij} \in F$$

Substituting this back into the expression for x:

$$x = \sum_{j=1}^{m} \left(\sum_{i=1}^{n} \nu_{ij} e_i \right) f_j = \sum_{j=1}^{m} \sum_{i=1}^{n} \nu_{ij} (e_i f_j)$$

Since every element $x \in L$ can be written as a linear combination of the elements $\{e_i f_j\}$ with coefficients $\nu_{ij} \in F$, this set spans L over F.

Linear Independence: Suppose we have a linear combination of the elements $\{e_i f_j\}$ that equals zero, with coefficients β_{ij} from the field F:

$$\sum_{i=1}^{n} \sum_{j=1}^{m} \beta_{ij} e_i f_j = 0, \quad \text{where } \beta_{ij} \in F$$

We can regroup the terms by factoring out the f_j:

$$\sum_{j=1}^{m} \left(\sum_{i=1}^{n} \beta_{ij} e_i \right) f_j = 0$$

For each j, the inner sum $\sum_{i=1}^{n} \beta_{ij} e_i$ is a linear combination of elements of the basis $\{e_i\}$ with coefficients in F. This means the term in the parenthesis is an element of the field K. So, we have a linear combination of the basis elements $\{f_j\}$ with coefficients from K that equals zero. By the linear independence of the set $\{f_1, \ldots, f_m\}$ over K, all the coefficients must be zero:

$$\sum_{i=1}^{n} \beta_{ij} e_i = 0 \quad \text{for all } j = 1, \ldots, m$$

Now, for each j, we have a linear combination of the basis elements $\{e_i\}$ over F that equals zero. By the linear independence of the set $\{e_1, \ldots, e_n\}$ over F, all the coefficients must be zero:

$$\beta_{ij} = 0 \quad \text{for all } i, j$$

This shows that the set $\{e_i f_j\}$ is linearly independent over F, which completes the proof of the Tower Law. $\square$

Example 4.26. Let $F = \mathbb{F}_2$ (which is the same as $\mathbb{Z}_2$). Let $L = \mathbb{F}_2[x]/\langle x^3+x+1 \rangle$. So, $|L| = 2^3 = 8$ and $[L : F] = 3$.

Are there any proper subfields K with $\mathbb{F}_2 \subset K \subset L$? Suppose such a subfield K exists. Then by the Tower Law, we must have:

$$[L : \mathbb{F}_2] = [L : K][K : \mathbb{F}_2]$$

Substituting the known value:

$$3 = [L : K][K : \mathbb{F}_2]$$

Since 3 is a prime number and the degrees of extensions are integers greater than or equal to 1, the only possibilities are:

1. $[L : K] = 1$ and $[K : \mathbb{F}_2] = 3$. This implies $K = L$.
2. $[L : K] = 3$ and $[K : \mathbb{F}_2] = 1$. This implies $K = \mathbb{F}_2$.

Therefore, there are no proper intermediate subfields between L and $\mathbb{F}_2$.

Corollary 4.27. *Suppose we have a tower of fields $F \subseteq E \subseteq K$. If $K : E$ is an algebraic extension and $E : F$ is an algebraic extension, then $K : F$ is also an algebraic extension.*

Proof. Let $a \in K$. Since $K : E$ is algebraic, a is a root of some polynomial $p(x) = b_n x^n + \cdots + b_1 x + b_0 \in E[x]$.

Since $E : F$ is algebraic, each coefficient $b_i \in E$ is algebraic over F. Consider the tower of fields:

$$F \subseteq F_0 := F(b_0) \subseteq F_1 := F_0(b_1) \subseteq \cdots \subseteq F_n := F_{n-1}(b_n) = F(b_0, \ldots, b_n).$$

Since each b_i is algebraic over F, it is also algebraic over F_{i-1}. Therefore, each extension in this tower, $F_i : F_{i-1}$, is a finite extension. By the Tower Law, $[F_n : F]$ is finite.

$$[F_n : F] = [F_n : F_{n-1}] \ldots [F_1 : F_0][F_0 : F] < \infty.$$

Now, the polynomial $p(x)$ has its coefficients in F_n, so $p(x) \in F_n[x]$. Since a is a root of $p(x)$, the extension $F_n(a) : F_n$ is finite, with $[F_n(a) : F_n] \leq \deg(p) = n$.

Consider the degree of the extension $F_n(a)$ over F:

$$[F_n(a) : F] = [F_n(a) : F_n][F_n : F].$$

This product is finite. Since $F(a)$ is a subfield of $F_n(a)$, we have $[F(a) : F] \leq [F_n(a) : F] < \infty$. Because $F(a) : F$ is a finite extension, it must be an algebraic extension. This implies that the element a is algebraic over F. Since a was an arbitrary element of K, the extension $K : F$ is algebraic. $\square$

Corollary 4.28. *The set of all algebraic elements in an extension $E : F$ forms a subfield of E.*

Proof. Let $a, b \in E$ be algebraic over F. We need to show that $a + b$, $a - b$, ab, and a/b (for $b \neq 0$) are also algebraic over F. Consider the field $F(a, b)$. By the Tower Law,

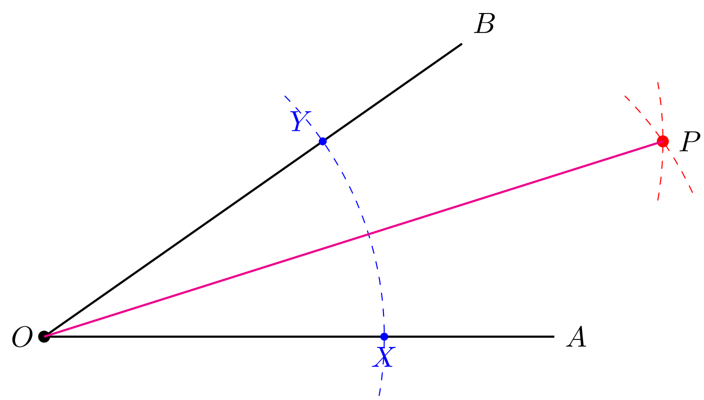
$$[F(a, b) : F] = [F(a, b) : F(a)][F(a) : F].$$

Since a is algebraic over F, $[F(a) : F]$ is finite. Since b is algebraic over F, it is also algebraic over the larger field $F(a)$. Thus, $[F(a, b) : F(a)] = [F(a)(b) : F(a)]$ is also finite. Therefore, $[F(a, b) : F]$ is finite. Since $F(a, b)$ is a finite extension of F, it is an algebraic extension. Any element of $F(a, b)$ is algebraic over F. This includes $a + b, a - b, ab$, and a/b. $\square$

4.6 Ruler-And-Compass Constructions

As a final application of the Tower Law, we present two problems raised by the ancient Greek more than 2000 years ago – What can we do with a ruler (with no markings) and a compass?

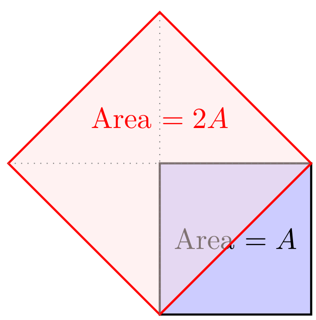
- **Bisect an angle**: One can bisect an angle by the following steps:



Construction Steps:
1. Arc from O marks X and Y at equal distances.
2. Arc from X into the angle interior.
3. Arc from Y (same radius) intersects at P.
4. Line OP is the angle bisector.

Question: Can we **tri-sect** any angle?

- **Double a square**: It is easy to draw a square whose area is twice as big as the original one:



Question: Can we **double the volume of a cube**?

To address these ancient Greek problems, we translate geometric constructions into the language of field theory.

Definition 4.29. A real number α is **constructible** if we can construct a line segment of length $|\alpha|$ using only a ruler and compass, starting from a segment of length 1.

Theorem 4.30. *If $\alpha \in \mathbb{R}$ is constructible, then the degree of the extension $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ must be a power of* 2.

Sketch. Every point created by a ruler and compass is the intersection of two lines, a line and a circle, or two circles. Algebraically, these intersections involve solving linear or quadratic equations. Thus, every new point lies in a field K_{i+1} obtained by a quadratic extension of the previous field K_i:

$$\mathbb{Q} = K_0 \subset K_1 \subset K_2 \subset \cdots \subset K_m$$

where $[K_{i+1} : K_i] = 2$ for all i. By the **Tower Law**:

$$[K_m : \mathbb{Q}] = [K_m : K_{m-1}] \dots [K_1 : K_0] = 2^m$$

If α is constructed, then $\alpha \in K_m$. Since $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq K_m$, the Tower Law again tells us that $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ must divide 2^m. Therefore, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^k$ for some $k \leq m$. $\square$

4.6.1 Application 1: Trisecting the Angle

We showed earlier that bisecting an angle is always possible. However, the Greeks could not find a way to divide an arbitrary angle into three equal parts. We show this is impossible by focusing on a specific angle: 60°.

Proposition 4.31. *It is impossible to trisect an angle of* 60° *using a ruler and compass.*

Proof. Trisecting 60° is equivalent to constructing the angle 20°. Constructing an angle θ is equivalent to constructing the number $\cos\theta$ on the x-axis. Recall the triple-angle formula for cosine:

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$$

Let $\theta = 20°$, so $3\theta = 60°$. Since $\cos 60° = 1/2$, let $\alpha = \cos 20°$. We have:

$$\frac{1}{2} = 4\alpha^3 - 3\alpha \implies 8\alpha^3 - 6\alpha - 1 = 0$$

To simplify, let $u = 2\alpha$. Substituting $\alpha = u/2$:

$$8\left(\frac{u}{2}\right)^3 - 6\left(\frac{u}{2}\right) - 1 = 0 \implies u^3 - 3u - 1 = 0$$

Let $f(u) = u^3 - 3u - 1$. By the Rational Root Theorem, the only possible rational roots are ± 1. Checking them: $f(1) = -3 \neq 0$ and $f(-1) = 1 \neq 0$. Since $f(u)$ is a cubic polynomial with no rational roots, it is irreducible over $\mathbb{Q}$. Thus:

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(u) : \mathbb{Q}] = 3$$

Because 3 is not a power of 2, $\cos 20°$ is not constructible. Therefore, the angle 60° cannot be trisected. $\square$

This does not mean *no* angle can be trisected (for example, a $90°$ angle can be trisected because $30°$ is constructible). It means there is no **general** method that works for any given angle.

4.6.2 Application 2: Doubling the Cube

The problem of doubling the cube (the Delian problem) asks if we can construct a cube with volume $2V$, given a cube of volume V.

Suppose the original cube has side length 1. Its volume is $1^3 = 1$. The side length x of a cube with volume 2 must satisfy $x^3 = 2$, so $x = \sqrt[3]{2}$.

Proposition 4.32. *It is impossible to double the volume of a cube using a ruler and compass.*

Proof. Let $x = \sqrt[3]{2}$. The polynomial $p(t) = t^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's Criterion (with $p = 2$). Since $p(t)$ is the minimal polynomial of x over $\mathbb{Q}$, we have:

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = \deg(t^3 - 2) = 3$$

Since 3 is not a power of 2, the number $\sqrt[3]{2}$ is not constructible. Thus, the side of the doubled cube cannot be constructed. $\square$